+1 Routine

A routine for the identifying important ideas worth remembering.

After reading a text, watching a movie, listening to a lecture, or being presented with new information or ideas in some manner, a group of learners does the following:

Recall In 2-3 minutes and working individually, each learner generates a list of key ideas that he or she recalls from the presentation that he/she feels is important to hang onto. Learners do this from memory rather than reviewing notes or material.

Add (+) 1 Learners pass their papers to the right. Taking 1-2 minutes, each student reads through the list in front of him/her and adds one new thing to the list. The addition might be an elaboration (adding a detail), a new point (adding something that was missing), or a connection (adding a relationship between ideas). REPEAT this process at least two times.

Act Return the papers back to the original owner. Learners read through and review all the additions that have been made on their sheets. At the same time they may add any ideas they have picked up from reading other's sheets that they thought were worthwhile.

PURPOSE

What kind of thinking does this routine encourage?
The routine helps students make connections between new ideas and prior knowledge. It also encourages them to take stock of ongoing questions, puzzles and difficulties as they reflect on what they are learning. How is the artwork or object connected to something you know about?

APPLICATION

When and where can I use it?
As learners we often encounter new ideas, information, and content. At upper levels, students often take notes on this material but too often may do so in a superficial, unthinking manner by merely writing down everything for possible review later.

This routine could be used as an alternative to traditional note taking. The benefit of post-lecture note taking is that students are required to identify key ideas, an important processing move, which is often easier to do after material is presented than during its presentation. At the same time this routine harnesses the power of the group to enhance everyone's notes while providing individuals with a written record to hang onto for future reference.

LAUNCH

What are some tips for starting and using this routine?
When presenting students with new information, in whatever form, tell them that you will be trying out a new note taking routine that will help them identify key ideas from the lesson/material. Instead of taking notes, ask learners to listen, engage, and participate fully in the lesson/lecture.

You may want to assure them that they will have access to your slides or lecture notes so that students are worried about missing important information. Explaining the psychology behind this routine can help students learn how to learn. The routine is not simply an activity you are asking to students to do, but a tool designed to help us use out brains more effectively for learning. You may wish to tell students that research has shown it is important to engage memory right away when something is learned rather than waiting to try and memorize things for a test later.

At the end of the lesson allow 10-15 minutes for the routine. Timing may be influenced by the complexity, richness and depth of the material that was presented as well as the age of students. Debrief this note taking method and discuss how to make it better next time.

This thinking routine was developed as part of the PZ Connect project at Project Zero at the Harvard Graduate School of Education. Explore the full PZ Thinking Routine Toolbox at *pz.harvard.edu/thinking-routines*.

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project zero THINKING ROUTINES

3-2-1 Bridge

Explain how your new responses connect to or changed from your initial responses.

PURPOSE

What kind of thinking does this routine encourage?
This routine helps students understand their own process of learning by considering their conceptions of a topic before and after a learning experience and how their conceptions changed.

APPLICATION

When and where can I use it?
This routine works well when the topic of the learning experience is one in which students have some prior knowledge. If the topic is something that students would not recognize, it would not be a good choice to use with this routine. Use the “Before Learning” part when you want students to remember their prior ideas, questions, and understandings about the topic. Then, near or at the end of the learning experience, invite students to complete the “After Learning” and “Bridge” parts of the routine.

LAUNCH

What are some tips for starting and using this routine?

When you introduce the topic or concept in the learning experience, share just a few details, remembering that the goal is to use the "Before Learning" part of the routine to activate students' prior knowledge. This will give you (and them) a chance to see their initial thinking.

Have the students write down 3 words or thoughts, 2 questions, and 1 metaphor or simile about the topic. Encourage them to capture their thoughts quickly, using words and questions that they think of first. Next, engage students in the learning experience. This might be reading an article, watching a video, examining an artifact, observing artwork, or engaging in an activity having to do with the topic. The learning experience should introduce new information or perspectives about the topic that will extend students' thinking in new directions.

After the learning experience, students complete the "After Learning" part of the routine. Then, students will do the "Bridge" part where they make connections between their initial and new thinking, explaining how and why their thinking shifted. Explain to students that their initial thinking is not right or wrong; it is just a starting point. New experiences take our thinking in new directions.

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project zero THINKING ROUTINES

Beauty and Truth

A routine for exploring the complex interaction between beauty and truth.

- Can you find **beauty** in this [image, story]?
- Can you find **truth** in this [image, story]?
- How might beauty **reveal** truth?
- How might beauty **conceal** truth?

PURPOSE

What kind of thinking does this routine encourage?
This routine fosters students' critical analysis of information (news, images, stories), revealing the nuances and complexities of an issue, event, or concept and how it is represented.

APPLICATION

When and where can I use it?
This routine aims to slow down students' thinking, inviting them to reflect on the ways works of literature, art, and journalism, for example, use beauty to engage us and motivate us to learn more about an issue and seek truth. The routine also invites a critical analysis of the ways in which beauty can mislead.

LAUNCH

What are some tips for starting and using this routine?
This routine invites you and your students into a broad and deep conversation about an image or story. Allow time for individual students to share ideas of beauty and truth—constructs they are unlikely to have explored explicitly in their past learning experiences. (When focusing on photographs with this routine, be alert for the misconception, held by many students, that photographs by their very nature reveal truth.) In Questions 3 and 4, the terms "beauty" and "truth" can be inverted.

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project zero THINKING ROUTINES

Beginning, Middle, End

A routine for harnessing the power of narrative to observe connections, patterns, and meaning.

Choose one of these questions:

- If this artwork is the **beginning** of a story, what might happen next?
- If this artwork is the **middle** of a story, what might happen before? What might be about to happen?
- If this artwork is the **end** of a story, what might the story be?

PURPOSE

What kind of thinking does this routine encourage?
This routine uses the power of narrative to help students make observations and encourages them to use their imagination to elaborate on and extend their ideas. Its emphasis on storytelling also encourages students to look for connections, patterns, and meanings.

APPLICATION

When and where can I use it?
The routine works with any kind of visual art that stays still in time—such as painting or sculpture. Use "Beginning, Middle, or End" when you want students to develop their writing or storytelling skills.

LAUNCH

What are some tips for starting and using this routine?
You can use the questions in the routine in the open-ended way they are written. Or, if you are connecting the artwork to a topic in the curriculum, you can link the questions to the topic. For example, if you are studying population density, you can ask students to keep the topic in mind when they imagine their stories. The routine is especially useful as a writing activity. To really deepen students' writing, you can use the "Looking: Ten Times Two" routine with the same artwork prior to using this routine as a way of helping students generate descriptive language to use in their stories.

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project zero THINKING ROUTINES

Chalk Talk

A routine for silently considering ideas, questions, or problems and responding to others.

Looking at the topical question written on the chart paper:

- What ideas come to mind when you consider this idea, question, or problem?
- What connections can you make to the others' responses?
- What questions arise as you think about the ideas and consider the responses and comments of others?

PURPOSE

What kind of thinking does this routine encourage?
This routine asks learners to consider ideas, questions, or problems by silently responding in writing both to the prompt and the thoughts of others. This "silent conversation" provides learners with time to follow through thoughts without interruption by choosing when they are ready to consider other points of view and make comments.

APPLICATION

When and where can I use it?
This routine highlights the notion of building understanding in a collaborative way through putting forward ideas, questioning one another, and developing the ideas further. Since individuals are asked to sign their comments, there is a degree of an anonymity that will free up some learners to take more risk and offer ideas. This activity can also be used for reflection.

LAUNCH

What are some tips for starting and using this routine?

1. *Set Up*
   - Write (above) prompt on a large sheet of chart paper and place on tables around room. Have markers available. Decide if you wish to assign students to groups or allow them to move freely around the room.

2. *Present the "Chalk Talk" Prompt*
   - Invite learners to think about their reactions to the prompt and record the ideas and questions. Encourage learners to read and to add to each other's responses with additional comments and questions.

3. *Circulate*
   - Provide time for learners to circulate around the "Chalk Talk" paper, reading and adding to the prompts and responses as they build. If working in groups you may want them to stay with one recording sheet for 5 min. to allow conversation to develop. Groups can then rotate en masse to another group's paper, silently reading what is written there, and adding their reactions and questions to the paper.

4. *Facilitate*
   - May need to prompt the group about the types of responses they can make as they read: connecting ideas, elaborating on others' ideas, commenting on what others have written, asking others to respond with more detail, and so on.

5. *Share the Thinking*
   - If people have rotated as a group, allow them to return to their original starting places to read what others have written on "their" "Chalk Talk" paper. Ask the group what themes they noticed emerging. What did they see as common issues and reactions? What questions surprise them? Debrief the process itself, asking the group how their thinking developed during the "Chalk Talk" process.

It's important to monitor the time and keep group appraised of the time.

This routine is adapted from Hilton Smith of the Foxfire Fund.

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project zero THINKING ROUTINES

Circle of Viewpoints

A routine for exploring perspectives.

1. Brainstorm a list of difference perspectives.

2. Choose one perspective to explore, using these sentence-starters:

    - **I am thinking of** ... the topic ... **from the viewpoint of** ... the viewpoint you've chosen
    - **I think** ... describe the topic from your viewpoint. Be an actor—take on the character of your viewpoint.
    - **A question I have from this viewpoint is** ... ask a question from this viewpoint

PURPOSE

What kind of thinking does this routine encourage?
This routine helps students see and explore multiple perspectives. It helps them understand that different people can have different kinds of connections to the same thing, and that these different connections influence what people see and think.

APPLICATION

When and where can I use it?
The routine works well with topics and artworks that deal with complex issues. It also works well when students are having a hard time seeing other perspectives or when things seem like there are only two sides to an issue. The routine can be used to open discussions about dilemmas and other controversial issues.

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project zero THINKING ROUTINES

Circles of Action

A routine for organizing one's understanding of a topic through concept mapping.

What can I do to contribute...

- In my **inner circle** (of friends, family, the people I know)?
- In my **community** (my school, my neighborhood)?
- In the **world** (beyond my immediate environment)?

PURPOSE

What kind of thinking does this routine encourage?
This routine fosters students' disposition and motivation to take action in personal, local, and global spheres and prepares them to weigh courses of action and their consequences.

APPLICATION

When and where can I use it?
This routine can help students to develop their sensitivity to opportunities to influence their environments, their capacity to do so in effective, informed, and responsible ways, and their motivation to take actions over time.

When students learn about the world and feel inclined to make a difference, they can feel overwhelmed by the magnitude of issues like climate change, trans-national consumption, or human rights violations. A mismatch between calls for global citizenship and students' perceptions of their own capacity to act can stifle motivation to engage.

This routine helps students learn to recognize that actions can have multiple scopes (small-big) and occur in multiple spheres (personal-local-global). This routine can be used across disciplines (e.g., geography, science, literature, economics) and with a broad range of resources (e.g., films, narratives, photographs) typically addressing a conflict, problem, system, or design that can be improved through participation and engagement. This routine can also be used informally in daily school contexts and interactions in which students have the opportunity to change a situation or help to solve a problem (e.g., a conflict among friends, consumption patterns in the school or community, the integration of immigrant students).

This routine works best when students have a moderate understanding of the issue, are primed to care about it, and have a sense of urgency or see the need for a response. This routine can be particularly effective when students sense the need but are having difficulty considering viable paths for action.

LAUNCH

What are some tips for starting and using this routine?
You or your students might begin with a particular issue in mind, or you might involve students in selecting a topic through brainstorming issues that are concerning them and that they think require solutions.

This routine invites students to map possibilities for action, so prior to the discussion, prepare a way for you or the students to document their responses. Typically, the "map" involves three concentric circles. You might draw these concentric circles (typically labeled inner, community, and world) on the board (for a whole class discussion) or invite small groups of students to draw them on large pieces of paper.

Once students have had a chance (either as a whole group or in small groups) to brainstorm possible contributions at each of the levels (note that the order of questions can be inverted if necessary), you might invite the whole class to discuss: What are the barriers to students' capacity to take action at various levels? Drawing on a rich initial actions map, students may be invited to consider factors such as ethics, viability, personal interest, and potential impact as they decide what to do next.

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project zero THINKING ROUTINES

Claim, Support, Question

A routine for reasoning with evidence.

Drawing on your investigation, experience, prior knowledge, or reading:

- Make a claim about (or give an explanation for, or offer an interpretation of) the topic
- Identify support (things you see, feel, know) for your claim
- Ask a question related to your claim or the supports. What isn't explained?

PURPOSE

What kind of thinking does this routine encourage?
This routine supports reasoning by having students form claims, explanations, or interpretations and support them with evidence. Questioning claims helps students see reasoning as an ongoing process.

APPLICATION

When and where can I use it?
Statements of fact or belief are presented everywhere. This routine is designed to help students take notice of claims and thoughtfully examine them. Use the routine with any topic, work of art, piece of text, poem, etc. to invite careful reasoning and evaluation.

LAUNCH

What are some tips for starting and using this routine?
This routine can work well for individuals, small groups, and whole groups. Begin by modeling the routine in the whole group: Identify a **claim**, ask the group for evidence that ***supports*** the claim, then invite students to question the claim or any of the evidence that has been offered to support it.

As you process each step of the routine, document students' comments in a way that is accessible to all students. For example, you may want to write the claim on chart paper, then draw two columns, one for ***supports*** and one for *questions*. Once students have practiced the routine in the whole group, they can use the routine individually or in small groups, documenting their responses in a journal or in a way that can be displayed within their small groups.

You may want to return to the whole group discussion and invite individuals or small groups to share their **claims**, ***supports***, and *questions* with the whole group. Leave time for the other students to respond with additional ***supports*** for the claim or to raise additional *questions*. If there's time, reflect on the activity: What new thoughts do students have about the topic? Keep in mind that students are often in the habit of simply agreeing or disagreeing with claims. You might need to slow down and give students time to think as they generate ***supports*** and *questions* about a **claim**.

You may need to encourage deeper thinking by asking: What are some other questions you might want to ask about this claim? Can you think of reasons why this might be true? Why might it not be true? Thoughtful questions that challenge the plausibility of a claim often lead to a deeper understanding of the topic and the reasoning process.

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project zero THINKING ROUTINES

Color, Symbol, Image

A routine for distilling the essence of ideas non-verbally.

As you are reading, listening, or watching, make note of things that you find interesting, important, or insightful. When you finish:

- Choose a **color** that you feel best represents or captures the essence of a key idea
- Choose a **symbol** that you feel best represents or captures the essence of a key idea
- Choose an **image** that you feel best represents or captures the essence of a key idea

With a partner or group, first share your color and then share the item from your reading that it represents. Tell why you chose that color as a representation of that idea. Repeat the sharing process until every member of the group has shared his or her Color, Symbol, and Image.

PURPOSE

What kind of thinking does this routine encourage?
This routine asks students to identify and distill the essence of ideas from reading, watching, or listening in non-verbal ways by using a color, symbol, or image to represent the ideas.

APPLICATION

When and where can I use it?
This routine can be used to enhance comprehension of reading, watching, or listening. It can also be used as a reflection on previous events or learnings. It is helpful if students have had some previous experience with highlighting texts for important ideas, connections, or events. The synthesis happens as students select a color, symbol, and image to represent three important ideas. This routine also facilitates the discussion of a text or event as students share their colors, symbols, and images.

LAUNCH

What are some tips for starting and using this routine?
After the class has read a text, you might ask the class to identify some of the interesting, important, or insightful ideas from the text and list these on the board. Write CSI: Color, Symbol, Image on the board. Select one of the ideas the class has identified. Ask students, what color might they use to represent the essence of that idea. What color captures something about that idea, maybe it is the mood or tone. Select another idea and ask the class what symbol they could use to represent that idea. *You might define a symbol as a simple line representation or uncomplicated drawing, such as two crossed lines to denote an intersection of ideas, or a circle to represent wholeness or completeness.* Then pick another idea from the list and ask students what image they might use to represent that idea. *You might define an image as a visual image or metaphor that is more complex and fully developed than just a symbol.*

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project zero THINKING ROUTINES

Colors, Shapes, Lines

A routine for encouraging detailed observation of artwork.

Look at the artwork or object for a moment.

- What colors do you see?
- What shapes do you see?
- What lines do you see?

PURPOSE

What kind of thinking does this routine encourage?
This routine helps students make detailed observations by drawing their attention to the forms in an artwork or an object and giving them specific categories of things to look for

APPLICATION

When and where can I use it?
The routine can be used with any kind of visual art or non-art images or objects. Like the "Looking: Ten Times Two" routine, students can use this routine on its own, or as preparation for discussing an artwork using a different routine. It is especially useful before a writing activity because it helps students develop descriptive language.

LAUNCH

What are some tips for starting and using this routine?
You can begin using this routine by simply building a broad inventory with students: list colors, then shapes, then lines. You can have students write down their own lists first before sharing and discussing them with partners, small groups, or the whole class. Alternatively, you might use this routine to guide a whole group discussion, with you or a student serving as notetaker to capture the whole group's descriptions. Documentation can either be done by individual students or by a notetaker for the whole class. Such documentation provides a useful tool to support revisiting and reflecting on the group's initial observations at a later point during the lesson or unit of instruction. You might consider using this question as you facilitate a later reflection on their initial observations: What might students be able to add that they didn't notice in their initial observations?

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project zero THINKING ROUTINES

Compass Points

A routine for examining propositions.

1. E = Excited
What **excites** you about this idea or propositions? What's the upside?

2. W = Worrisome
What do you find **worrisome** about this idea or proposition? What's the downside?

3. N = Need to Know
What else do you **need to know** or find out about this idea or proposition? What additional information would help you to evaluate things?

4. S = Stance or Suggestion for Moving Forward
What is your current **stance** or opinion on the idea or proposition? How might you move forward in your evaluation of this idea or proposition?

PURPOSE

What kind of thinking does this routine encourage?
This routine helps students flesh out an idea or proposition and eventually evaluate it.

APPLICATION

When and where can I use it?
This routine works well to explore various sides and facets of a proposition or idea prior to taking a stand or expressing an opinion on it. For instance, the school may be considering the idea of a dress code, a teacher might present the class with idea of altering the room arrangement, a character in a book might be confronted with making a choice, a politician might be putting forth a new way of structuring taxes, and so on.

LAUNCH

What are some tips for starting and using this routine?
The routine needs to be modeled with the whole group initially with responses recorded for the entire class to see. This enables students to build on each other's ideas. You might record responses using the directions of a compass to provide a visual anchor. That is, draw a compass in the center of the board and then record responses corresponding the appropriate direction: E, W, N, or S. It is generally easiest for students to begin with what is exciting or positive about the idea or proposition and then move to worrisome and need to know. Students might be asked to write down their individual stance or suggestion for moving forward after the initial group discussion. You can also ask students to make an initial judgment or evaluation of the idea or proposition before doing the compass points and then ask them how their thinking has changed after discussion using the compass points routine.

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project zero THINKING ROUTINES

Connect, Extend, Challenge

A routine for drawing connections between new ideas and prior knowledge.

- How is the artwork or object **connected** to something you know about?
- What new ideas or impressions do you have that **extended** your thinking in new directions
- What is **challenging** or confusing? What do you wonder about?

PURPOSE

What kind of thinking does this routine encourage?
The routine helps students make connections between new ideas and prior knowledge. It also encourages them to take stock of ongoing questions, puzzles and difficulties as they reflect on what they are learning. How is the artwork or object connected to something you know about?

APPLICATION

When and where can I use it?
A natural place to use this routine is after students have experienced something new. The routine is broadly applicable: Use it after students have explored a work of art, or anything else newly introduced in he curriculum. Try it as a reflection during a lesson, after a longer project, or when completing a unit of study. Try using it after another routine!

LAUNCH

What are some tips for starting and using this routine?
This routine works well with the whole class, in small groups or individually. Keep a visible record of students' ideas. If you are working in a group, ask students to share some of their thoughts and collect a list of ideas in each of the three categories Or have students write their individual responses on post-it notes and add them to a class chart. Keep students' visible thinking alive over time: Continually add new ideas to the lists and revisit the ideas and questions on the chart as students' understanding around a topic develops.

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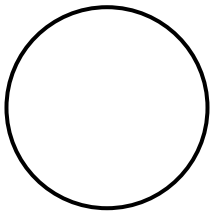
project zero THINKING ROUTINES

Creating Space for Learning

A routine to foster a disposition toward self-understanding, self-direction, and mindful and compassionate concentration.



Draw a circle on a blank piece of paper..

- **Breathe**
  Find a comfortable posture and take your time to breathe three times deeply.

- **Notice**
  What is on your mind and in your heart at this moment? Write or draw your inner throughts and feelings inside the circle.

- **Set Aside**
  What feelings and thoughts might you need to set aside for now to learn today? Gently "move" these to the margin of the page.

- **Keep at the Center**
  What feelings and thoughts might you want to keep or bring to your mind and heart as you prepare to learn today? Gently bring them to the center.

With these thoughts and feelings in mind, take a few final deep breaths and slowly turn to your learning.

PURPOSE

What kind of thinking does this routine encourage?
"Creating Space for Learning" is designed to foster a disposition toward self-understanding, self-direction and mindful, compassionate concentration. Through consistent use, students are likely to develop a greater awareness and capacity to talk about their thoughts and feelings. They are likely to develop greater agency in selfregulating their emotions and attention. In addition, students might experience a greater sense of well-being and an appreciation for full engagement in learning.

APPLICATION

When and where can I use it?

This routine can be used at the start of a school day, or before transitioning into a learning experience that would especially benefit from student presence and concentration. When preparing to use this routine, you may consider inviting students to stand, shake out their legs and arms, and stretch for a moment.

Afterwards, students can settle into a seated place, draw a circle in their notebook, and begin taking a few deep breathes. During the breathing and notice phase, students might find it helpful to close their eyes initially, before jotting down their initial notices. Facilitators may also choose to join students in taking deep breathes and checking in internally.

LAUNCH

What are some tips for starting and using this routine?

Throughout the routine, facilitators can create a calm environment through the use of relaxed tone of voice and unhurried pace. Providing learners with a few extra seconds before, during and after each step can deepen students' experience and add a sense of tranquility. When facilitators use this routine over time students are likely to became more comfortable with the process. Ideally facilitators encourage students to utilize these embodied self-reflective practices in their lives outside of school.

Sometimes facilitators can invite a few students to share a thought or feeling that will help them learn or that they set aside, thus creating an atmosphere of trust, inspiration and community. Depending on needs facilitators can experiment in modifying the routine for example focusing only on "breath—notice—keep at the center."

This thinking routine was developed as part of the e ID Global, Reimagining Migration project at Project Zero at the Harvard Graduate School of Education. Explore the full PZ Thinking Routine Toolbox at *pz.harvard.edu/thinking-routines*.

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project zero THINKING ROUTINES

Creative Comparisons

A routine for metaphorical thinking.

What do you see in the artwork? / What do you know about the topic?

- Compare: Choose a category from the list below or identify your own category.
- Imagine: If this topic / artwork was a kind of _____ (category), what would it be?
- Explain three ways that it compares.

PURPOSE

What kind of thinking does this routine encourage?

The routine encourages metaphorical thinking—central to the work of any artist and to creative thinking in any discipline. Metaphors provoke our imaginations to create comparisons between dissimilar things, often leading to deeper and richer understanding of each.

APPLICATION

When and where can I use it?

Creating metaphors help students understand unfamiliar subjects by linking it to what they already know. Use the routine when you want to help students make connections between disparate elements or ideas, or to stimulate new insights and solutions.

LAUNCH

What are some tips for starting and using this routine?

Teachers should provide a list of categories related to the artwork or the topic. Good category examples include those that have a wide variety of parts or types. Sample categories could include: musical instruments, plants, toys, cities, parts of the body, artworks of all kinds (for topics that aren't artworks), music (for any topic or artwork that isn't music).

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Creative Hunt

A routine for looking at parts, purposes, and audiences.

Key Prompts

- What's the main purpose here?
- What are the parts and their purposes?
- Which are especially smart or creative? Star them!
- Who is the audience for this?

PURPOSE

What kind of thinking does this routine encourage?
An important part of creativity is recognizing how creative things around us are. This is often inspiring. Because we are too used to things, we do not appreciate their creativity. It is also often practical: we see better the limitations of things and how they might be improved. It's also a good way of understanding things better, by looking into what they are for, how they work, and who their audiences are. Thus, this creativity routine has an understanding bonus.

APPLICATION

When and where can I use it?
This routine makes thinking visible by helping students to find the creative thinking behind ordinary things—doorknobs, pencils, newspapers, toys. It can also be applied to more important things and more abstract things, like forms of government, hospitals, or schools. The routine helps students to appreciate creativity and be more alert to creative opportunities.

The creativity hunt is a good way to awaken students to the creativity in ordinary objects around them. You can use it on everyday classroom objects, like a blackboard, a ballpoint pen, a paintbrush, or an article of clothing. You don't have to stick to concrete physical objects. You can use it on more abstract things, like the 24-hour day, recess, a sport, or a game.

Besides the things around us, you can easily use it to connect to the subject areas. Here are some tips about picking a good object:

- Pick something that comes from human beings and human creativity, like a telescope, a form of government, or a means of communication.
- Pick something relevant to the subject matter you are teaching. For instance, a cannon, a musket, or a military formation would be a good choice if you are teaching military history. A particular tax or policy might be a good choice if you are teaching about government. A sextant or telescope might be a good choice if you are teaching about science

- You can pick concrete things like a sextant or telescope but also abstract things like a particular tax or policy.
- Pick something that the students know enough about so that they can think some about what it's for, how it works, and who its audience is. For instance, you would not pick a telescope if students didn't know much about telescopes, but you might if you could bring one in and students could try it out and examine it. You would not pick an import duty if students had just heard the name but did not know how it worked, but if they had read and discussed it in general you might.

LAUNCH

What are some tips for starting and using this routine?
Here are some basic steps for starting the routine:

1. Identify something for students to think about—something ordinary like a ballpoint pen or larger and more abstract like a hospital. It is natural to pick something from a subject matter being taught.
2. Set up the target diagram and label the key elements: main purpose, parts and purposes, audience. Say something like this: "Let's look at this from a creative viewpoint. Creative things have jobs to do. They need to hit their target. So here is the target. Let's explore how this thing hits its target."
3. Lead students in filling out the target diagram. Let them suggest main purposes (sometimes there is more than one), particular parts and the purposes they serve, and who the audience is. Also, invite them to star (*) any part they think is particularly smart or creative. You can conduct this as a general conversation, but another good way is to ask students to fill out Post-Its individually or in small groups and stick them up on the diagram.
4. Sum up by looking for what's creative. Go over the *'s and invite more. Emphasize how this clever object hits its target

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project zero THINKING ROUTINES

Creative Question Starts

A routine to practice developing questions that provoke thinking and inquiry.

1. **Brainstorm** at least 12 questions about the topic, concept, artwork or object. Try some of the Creative Starts Questions.

2. **Review** your list, identify the most interesting questions, and select one to discuss.

3. R**eflect**: What new ideas do you have about the topic, concept, artwork, or object that you didn't have before?

Creative Question Starts:

- What colors do you see?
- Why...?
- What if...?
- What is the purpose of...?
- How would it be different if...?
- Suppose that...?
- What if we knew...?
- What would change if...?

PURPOSE

What kind of thinking does this routine encourage?
This routine provides practice developing questions that provoke thinking and inquiry. Brainstorming such questions helps students explore the complexity, depth, and multi-dimensionality of a topic.

APPLICATION

When and where can I use it?
Use this routine to expand and deepen students' thinking, to encourage their curiosity, and to increase their motivation to inquire. This routine works well throughout the study of a topic in a variety of ways: when introducing a new topic to help students get a sense of the topic's breadth; in the middle of studying a topic as a way to enliven students' curiosity; and, finally, near the end of studying a topic as a way of showing students how the knowledge they have gained helps them to ask more complex questions.

LAUNCH

What are some tips for starting and using this routine?

To begin, you might want to ask students what they think makes a good question. Then, when you introduce the routine, you can connect to their ideas about what makes good questions and explain that the routine is a tool for asking good questions that lead to greater insight. Initially, it may work best to do the routine together as an entire group. Once students are familiar with the routine, you can have them work in small groups or even solo.

In your planning, think about how you would like your students to engage at each step. For example, the students might do step 1 as a whole group, step 2 in pairs, and return to the whole group discussion in step 3.

Start the routine by providing a topic, concept, piece of artwork, or object. Ask them to use the Creative Question Starts to generate a list of questions about the topic. Creating the list is worthwhile since it gives students a sense of the breadth of a topic and sparks curiosity about it. Exploring the questions further can pique students' interest and can fluidly expand the discussion of the topic.

After the students finish generating questions, you may want to select one of the questions to investigate further, have a discussion about some of the questions, give students information to read about the topic, or explore some of the questions by imaginatively playing out possibilities. Additional ways students could explore or investigate their questions (and the topic) include writing a story or essay, drawing a picture, creating a play or dialogue, inventing a scenario, conducting an imaginary interview, or conducting a thought experiment.

The reflection step is an important component of the routine since it allows students to identify new insights and ideas about the topic, object, artwork, or concept. Consider documenting students' questions throughout their study of the topic and display the list in a visible place so that students can see how their questions about the topic evolve. If students are working individually with the routine, they could write their question lists in a journal. You also could create a collage out of students' individual lists and put it on display.

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project zero THINKING ROUTINES

Creative Questions

A routine for generating and transforming questions.

1. Pick an everyday object or topic and brainstorm a list of questions about it.

2. Look over the list and transform some of the questions into questions that challenge the imagination. Do this by transforming questions along the lines of:
   - What would it be like if...
   - How would it be different if...
   - What would change if...
   - How would it look differently if...
   - Suppose that...

3. Choose a question to imaginatively explore. Explore it by imaginatively playing out its possibilities. Do this by:
   - Writing a story or essay
   - Drawing a picture
   - Creating a play or dialogue
   - Inventing a scenario
   - Conducting an imaginary interview
   - Conducting a thought experiment

4. Reflect: What new ideas do you have about the topic, concept, or object that you didn't have before?

PURPOSE

What kind of thinking does this routine encourage?
Formulating and exploring an interesting question is often as important than finding a solution. This routine encourages students to create interesting questions and then imaginatively mess around with them for a while in order to explore their creative possibilities. It provides students with the opportunity to practice developing good questions that provoke thinking and inquiry into a topic.

APPLICATION

When and where can I use it?
Use Creative Questions to expand and deepen students' thinking, to encourage students' curiosity and to increase students motivation to inquire. This routine can be used when you are introducing a new topic to help students get a sense of its breadth. It can be used when you're in the middle of studying a topic as a way of enlivening students' curiosity. And it can be used when you are near the end of studying a topic to show students how the knowledge they have gained about the topic helps them to ask ever more interesting questions. This routine can also be used continuously throughout a topic to help the class keep a visible, evolving list of questions about the topic that can be added to at anytime.

LAUNCH

What are some tips for starting and using this routine?
Before using Creative Questions you might want to ask students what they think makes a good question. Then, when you show the Creative Questions, explain that this routine is a tool for asking good questions.

Start the routine by providing a topic, concept, or object—a painting, voting rights, a stethoscope, genetic engineering. Ask them to use the Creative Questions to generate a list of questions about the topic or object. Initially, it's best to work together as an entire group. Once students get the hang of the routine, you can have them work in small groups or even solo. After students finish generating questions, ask them to pick one of the questions to investigate further.

Encourage students to explore it by imaginatively playing out its possibilities. Writing a story or essay, drawing a picture, creating a play or dialogue, inventing a scenario, conducting an imaginary interview, or conducting a thought experiment are just some of the possible ways for students to find out about their questions. At the end of the exploration process, be sure to take time to reflect on new insights and ideas about the topic, object, or concept.

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project zero THINKING ROUTINES

Digital Habits Checkup

A protocol to help students rethink and revamp their digital habits.

- **Check**
  Check and take stock of your digital habits—all the tech habits that are part of your life right now. Habits can be positive, negative, or both. Brainstorm and list as many of your habits as you can. Draw/insert an emoji(s) next to each habit to represent how it usually makes you feel.

- **Choose**
  Choose one digital habit from the list you created that is something you want to change or try to do differently. Why is it important that you change this habit?

- **Challenge**
  Make a plan for how you would like to try a personal challenge to your habit for a set period of time (e.g., one or two weeks). Consider when you will do something different and what you can do instead while you're doing your challenge. (When you're trying to change a habit, it's not enough just to say what we won't do. We also have to come up with something to do instead!)

- **Boost**
  What are some ways you can give your new habit a "boost" and make it easier to do? If you're trying to break a bad habit, what are some things you can do so that you don't slip back into old ways?

- **Track**
  Keep track of your progress. Each day, answer two questions: How's it going with your challenge today? What made your challenge hard or easy today?

PURPOSE

What kind of thinking does this routine encourage?
Based on principles from habit science, the Digital Habits Checkup helps students be alert to and take stock of their digital habits so they can make positive changes that support well-being. This routine directs ongoing consideration to: the self and personal well-being. Using it repeatedly helps cultivate dispositions that support mindful digital engagement.

APPLICATION

When and where can I use it?
This activity can be used anywhere educators see a connection with their learning goals. For example, in:

- Advisory period, where students are learning SEL or character education skills
- Library or media class, where students are learning about digital citizenship and technology
- Health class, where students are learning about healthy and unhealthy lifestyle choices

This activity is flexible! Students can complete the Habits Check-up in a short period of time (e.g., 15 minutes). Or, you can extend the steps with more discussion and expand the activity so it's a special unit with follow-ups as students track and revamp their habits over a week or month.

LAUNCH

What are some tips for starting and using this routine?
To facilitate this activity, make a copy of the **Digital Habits Checkup: Student Handout** which provides scaffolding for each of the steps of the protocol. A tip for building interest before you dig in is to sharing this example related specifically to smart phone and computer "notifications."

To begin, have students turn up the volume on their phones and ask each student to document the notifications during a single class period.

The example provides a rich jumping-off point for conversations about the volume of digital disruptions. Or, start off by discussing persuasive design more generally and discuss how the features of tech (e.g., infinite scroll, notifications) are intentionally designed to capture and hold our attention (see The Center for Humane Technology's Persuasive Technology resource for reference).

In the first step **Check**, consider using a digital board like Linoit or Padlet to have students share digital habits and react to/build on their classmates' reflections.

For the **Choose** step, consider creating a list of the examples students are selecting for themselves. If students are having trouble getting started or crafting a statement about what behavior they want to do differently, consider providing them with the sentence starters "I want to start doing xx …" or "I want to stop or do less of xx …". These starters will model how it's helpful to include the intended or hoped for or changed action that the student is intending to take around the digital habit.

For the **Challenge** step, support students to name specific details of their challenge. What are they going to do (or not do) and when? Use a sentence starter like "I will …". Consider ending this step with a discussion of potential barriers or things that might get in the way of actually succeeding with their challenge. Propose the question:

- Could you commit to trying your challenge for three days (or one week or one month)? If not, why not?
- How could you revise your challenge so you pick something that feels both helpful and doable?

For the **Boost** step, encourage students to get specific about what they can do to be successful at sticking to their Challenge. What will they do instead of their current habit? What could they do to help themselves stay on track? Depending on the specific challenge, examples might include unfollowing certain accounts, setting screen time limits, using a media-use tracking app like Moment or ZenScreen, committing to tech-free wind down time, charging devices in not easy to reach places (outside a teen's room), setting device-free times in one's schedule, or adding a health or meditation app.

Let them know that the class or group will return to the "boost" step two to three days after they begin their challenge. They can do this individually or with a partner. They might find that their current plan is a good one, but they need to take a few extra boost steps to help themselves stick to the changed behaviors.

For the final **Track** step, encourage students to select a format or tool for how they plan to track their progress on their hoped for behavior change. Examples of such formats or tools include using a journal or daily log, writing a blog, or creating a documentary-style video diary. If time permits, use a few minutes each day to have them reflect with a partner on how it's going or to share as a group. You can use tracking as an opportunity to help students reflect and assess how their challenge is going and whether or not they need to revise their challenges as needed.

The key is to help them recognize that they have the power to identify digital habits they don't like and change them. Remind the group that sometimes starting with a very small goal is a good first step and way to find some initial success. Even just cutting back a negative habit or increasing a desirable habit a little bit can be a meaningful start.

This thinking routine was developed as part of the Digital Dilemmas project at Project Zero at the Harvard Graduate School of Education and colleagues at Common Sense Media. Explore the full PZ Thinking Routine Toolbox at *pz.harvard.edu/thinking-routines*.

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project zero THINKING ROUTINES

Does It Fit?

A routine for thinking creatively about options.

1. Fit Your Options to the **Ideal**
   - Identify what the ideal situation would look like and then evaluate each option against it.
   - Ask yourself: How well does each option fit with the ideal situation?

2. Fit Your Options to the **Criteria**
   - Identify the criteria or attributes that feel important for you to consider in this situation and then evaluate each option against those.
   - Ask yourself: How well does each option fit the criteria?

3. Fit Your Options to the **Situation**
   - Identify the realities and constraints of your situation, such as resources and time, and then evaluate each option against them.
   - Ask yourself: How well does each option fit the realities of the situation?

4. Fit Your Options to you **Personally**
   - Try out each option by running a "mental movie" in which you imagine yourself carrying out the option and try to get a sense of what it would feel like.
   - Ask yourself: Which option feels like the best fit for me?

PURPOSE

What kind of thinking does this routine encourage?
To help students more effectively flesh out and evaluate options, alternatives, and choices in a decision-making situation.

APPLICATION

When and where can I use it?
This routine is part of an overall decision-making process that begins with the generation of options, choices, or alternatives for solving the problem or satisfying the needs of a situation. Once options are identified, they need to be evaluated in order to make a choice. Use this routine whenever students need to make a thoughtful and reasoned decision. Examples could include: the choice of a final project; direction for an investigation; making a group or whole class decision on how to allocate time, money, or resources; electing a group leader or spokesman; choosing among possible classes, and so on.

LAUNCH

What are some tips for starting and using this routine?

The four different "fits" represent four distinct approaches to evaluating options rather than a multi-layered routine. The first part of the routine involves making a choice as to which of the "fits" is best for the situation at hand. Then, that particular "fit" is carried out. To make this initial choice, students need some practice and discussion of each of the "fits" to see in what kinds of situations each works best. You might introduce this routine by briefly discussing each one and then have the whole class try out one of "fits" to make a decision. Initially, you might choose which of the four "fits" is best for a given situation and then gradually involve students in this process once all the fits have been practiced on several occasions.

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project zero THINKING ROUTINES

Elaboration Game

A routine for encouraging close looking.

As a group, observe and describe several different sections of an artwork.

1. One person identifies a specific section of the artwork and describes what he or she sees.
    - Another person **elaborates** on the first person's observations by adding more detail about the section.
    - A third person **elaborates** further by adding yet more detail, and a fourth person adds yet more.
    - *Observers: Only describe what you see. Hold off giving your ideas about the art until the last step of the routine.*
2. After four people have described a section in detail, another person identifies a new section of the artwork and the process starts over.

PURPOSE

What kind of thinking does this routine encourage?
This routine encourages students to look carefully at details. It challenges them to develop verbal descriptions that are elaborate, nuanced, and imaginative. It also encourages them to distinguish between observations and interpretations by asking them to withhold their ideas about the artwork—their interpretations—until the end of the routine. This in turn strengthens students' ability to reason carefully because it gives them practice making sustained observations before jumping into judgment.

APPLICATION

When and where can I use it?

Use this routine with any kind of visual art that stays still in time, such as painting or sculpture. You can also use the routine with non-art objects, such as a microscope, an animal skeleton, or a plant. The Elaboration Game is an especially good way to launch a writing activity because it helps students develop a detailed descriptive vocabulary.

LAUNCH

What are some tips for starting and using this routine?

This is a great routine to launch a collaborative conversation about an artwork or object. With an object or sculpture you can ask students to view it from different sides or viewpoints. With an artwork or twodimensional piece, students have the opportunity to tackle smaller sections at a time. You could use a viewfinder or have the students use their hands to create a viewfinder to focus in on their section.

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Facts or Fiction

A routine to increase awareness of the many issues of truth and what to consider when exploring the truth of something.

Select a news item, an image, something that has happened to you, or an event in history and ask:

- What do you **perceive** is being conveyed by this?
- Who would **decide** to convey this message? Why?
- How else could this be **interpreted**?
- What do you **believe** is true in this? What makes you think that?

PURPOSE

What kind of thinking does this routine encourage?
Too often, information can be accepted at face value without any supporting evidence. This routine is designed to explore the complexity of the realm of truth. It encourages slow looking, perspective taking, seeking supporting evidence, and analysis to uncover the core and accuracy of information. Some questions that might emerge as students use the routine are: Is there supporting evidence? Are many facts and figures reported? Does it fit with what you have learned before? Never be afraid to ask questions.

APPLICATION

When and where can I use it?
The routine can be helpful to show students that images or information presented are not always truthful accounts of events. One suggestion for doing this is to select an image showing that if only part of it was visible, different 'truths' could be interpreted. Cut the image into separate pieces, show only a part of it at a time as if each piece was a stand-alone complete picture, facilitate a discussion on what can be seen and thoughts/interpretations of what is happening, and then show the complete image and compare with initial thoughts. Alternately, show only one of a sequence of images or read only part of an article initially, discuss it and then complete the sequence.

project zero THINKING ROUTINES

LAUNCH

What are some tips for starting and using this routine?

- The routine can help awaken students to the idea that facts or fiction within a story, image, report can be influenced by the agenda of the photographer, author and/or editor of the information. Identifying possible purposes of a message can provide insights into who is "sending" the message. Why do you think this image/story was communicated? Who would benefit from this "message" being in the public arena? What impact do you think it may have? After giving yourself, your students, or colleagues time to think about this, ask whether your original impressions changed. What other messages are being perceived? ***Circle of Viewpoints*** and ***Step Inside*** thinking routines could also assist in discussions throughout this routine and highlight different perspectives of the same information/image.

- Students often ask "What are the facts" and "Can there be more than one truth?" From a definitional standpoint, in the end there cannot be more than one truth on a particular point, e.g. either smoking doesn't cause cancer or it does (more precisely, heavy patterns of smoking increase the chances of cancer to such-and-such a degree). But there's a big catch to this: what's really true and what we entirely reasonably think is true at the moment are quite different things. What we now reasonably think is true sometimes turns out to be wrong later. Also, at a given point in time, different people may take different ideas as true. However, it is important for students to appreciate that what we reasonably take as true today may turn out to be wrong later; disagreements about what's true are normal and sorting out evidence is often a challenging and complex long-term endeavor.

FACILITATION

This activity is recommended for the following learner age ranges:

- Elementary School
  - 30 minutes to 1 hour
  - Heavily facilitated by peer/teacher/caregiver
- Middle School
  - 30 minutes to 1 hour
  - Lightly facilitated by peer/teacher/caregiver
- High School
  - 30 minutes to 1 hour
  - Lightly facilitated by peer/teacher/caregiver

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project zero THINKING ROUTINES

Feelings and Options

A routine that scaffolds perspective-taking and empathic problem-solving to help students navigate social dilemmas

Choose a dilemma that you want your students to analyze.

1. **Identify:** Who are the different people involved in the dilemma? What challenge are they facing?
2. **Feel:** What do you think each person in the dilemma is feeling? Why might the situation be hard or challenging for each of them?
3. **Imagine:** Imagine options for how the situation could be handled. Come up with as many ideas as possible. Highlight or circle which option might lead to the most positive outcome, where most people feel good or taken care of.
4. **Say:** Thinking more about the idea you chose for handling the situation, what could the people involved say?

PURPOSE

What kind of thinking does this routine encourage?
This routine supports reasoning by having students form claims, explanations, or interpretations and support them with evidence. Questioning claims helps students see reasoning as an ongoing process.

APPLICATION

When and where can I use it?
Feelings and Options is a thinking routine for engaging with social and emotional dilemmas. It's designed to support students to explore different perspectives and practice language for constructive and kind communication. By using this routine repeatedly, students can develop the sensitivity to recognize dilemmas and the dispositions to:

1. Explore and care about others' perspectives
2. Envision options and possible impacts before acting

This routine is designed to be paired with dilemma scenarios. You can create your own dilemma, or choose a dilemma from our digital dilemmas compilation.

Each step of Feelings and Options can be done by students individually, in pairs or small groups, or as a whole class discussion. Consider what might work best for your students.

This activity can be used anywhere educators see a connection to their learning goals. For example:

- advisory period, where students are learning SEL or character education skills
- library or media class, where students are learning about digital citizenship and technology
- health or wellness class, where students are learning about healthy relationships
- English language arts (ELA), connected to any reading that includes relational dilemmas or social-emotional conflict
- special topics classes, like religion, leadership, or student council

This activity is flexible! Students can complete the thinking routine in about 15 minutes. You can extend the steps with more discussion and expand the activity so it's a special unit with follow-ups over a week or month. We recommend that you use this thinking routine as just that: a repeated routine. This helps students internalize the steps and thinking we want to support outside of the classroom.

LAUNCH

What are some tips for starting and using this routine?
Here are some suggestions for deepening each step of the Feelings and Options routine:

- **Identify** Encourage students to write down all of the people who are involved (e.g., "The people involved are: Tim, Jason, and Tim's parents. The dilemma they are facing is that Jason feels overwhelmed by how much Tim is texting him, but Tim is having a hard time and needs a lot of support from his friends.")
- **Feel** Encourage students to take perspectives of how all characters involved are feeling. Prompt students to clearly label the feelings, and describe why they might feel that way. Ask: Are there any other ways the people in the situation might be feeling? How might additional people (e.g., friends not named in the dilemma) feel about the situation?"
- **Imagine** Have students imagine many ways of handling the situation, regardless of outcome. Then encourage them to select the dilemma that's most beneficial for everyone involved in the dilemma. Ask: Are there any downsides to the suggestion you came up with?
- **Say** After students write out the text, encourage them to consider whether they would really say the words they have come up with. Ask: Do you think the person(s) involved would really say that? Could you see yourself actually saying/sending a message like this, if you were the person in the dilemma? Why, or why not? What if the person instead said ___? What if person B didn't respond that way? What could person A do or say next?

As you wrap up, consider asking students: What's one thing you're taking away from Feelings and Options? This is a great way for students to identify and affirm the different thinking skills they practiced (e.g., perspective taking, considering different options, and anticipating tricky conversations).

This thinking routine was developed as part of the Digital Dilemmas project at Project Zero at the Harvard Graduate School of Education. Explore the full PZ Thinking Routine Toolbox at *pz.harvard.edu/thinking-routines*.

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project zero THINKING ROUTINES

Generate-Sort-Connect-Elaborate

A routine for organizing one's understanding of a topic through concept mapping.

Select a topic, concept, or issue for which you want to map your understanding.

- **Generate** a list of ideas and initial thoughts that come to mind when youthink about this particular topic/issue.
- **Sort** your ideas according to how central or tangential they are. Place central ideas near the center and more tangential ideas toward the outside of the page.
- **Connect** your ideas by drawing connecting lines between ideas that have something in common. Explain and write in a short sentence how the ideas are connected.
- **Elaborate** on any of the ideas/thoughts you have written so far by adding new ideas that expand, extend, or add to your initial ideas.

Continue generating, connecting, and elaborating new ideas until you feel you have a good representation of your understanding.

PURPOSE

What kind of thinking does this routine encourage?
This routine activates prior knowledge and helps to generate ideas about a topic. It also facilitates making connections among ideas. Concept maps help to uncover students' mental models of a topic in a non-linear way.

APPLICATION

When and where can I use it?
This routine can be useful as a pre-assessment before the beginning of a unit of study if students already have a lot of background information about the topic. Conversely, it can also be useful as a post or ongoing assessment to see whatstudents are remembering and how they are connecting ideas. Individual maps can be used as the basis for construction of a whole classroom map. Maps can also be done progressively, with students adding to their maps each week of the unit.

LAUNCH

What are some tips for starting and using this routine?

Depending on how much familiarity students have with concept maps, you may need to demonstrate making a concept map using this routine with the whole class. However, if students are relatively familiar with the idea of concept maps, you can launch right into the routine explaining that students will be making concept maps but in a structured way. Give time for students to complete each step of the routine before moving on to the next step. It isn't necessary that students generate an exhaustive list of all their ideas initially, but make sure they have time to generate a rich and varied list before moving on. Tell students that at any point they can add new ideas to their list and incorporate them into their map. If you are adding to a map over time, you might want to have students use a different color pencil each time they make additions. Explaining and discussing maps with partners helps students to consolidate their thinking and gain other perspectives.

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project zero THINKING ROUTINES

Headlines

A routine for capturing essence.

- Write a headline that captures the most important aspect of this topic/issue.
- How does your headline differ from what you would have said yesterday?

PURPOSE

What kind of thinking does this routine encourage?
This routine helps students identify the essence, or the core idea, of the topic/issue being discussed and reinforces taking notice of central themes as an essential tool in cultivating understanding.

APPLICATION

When and where can I use it?
This routine works especially well at the end of a class discussion or session when you want many or all students to succinctly summarize their thoughts. You can ask everyone for a headline that captures their impressions and key ideas about the topic/issue being explored. The routine helps students practice forming a concise synthesis and offers them an opportunity to listen to fellow students' thinking about the topic/ issue. As you support students in developing their headlines, explain that this is not about creating the best headline; rather, the goal is for students to hear different perspectives and surface nuances.

LAUNCH

What are some tips for starting and using this routine?
After students have had some learning experiences, ask the students to reflect individually about the core ideas of the topic/issue. For example, at the end of a class, you might say, "Think about all that we have been talking about today in class. If you were to write a headline for [topic/issue] right now that captured the most important aspect that should be remembered, what would it be?" Give them individual time to form their headlines. Then, invite the students to "Share your headline with your neighbor." You might close the class by asking, "Who heard a headline from someone else that they thought was particularly good at getting to the core of things?" (Note: The steps in this example are similar to those in the Think, Pair, Share routine.)

You may want to document students' headlines in order to revisit them at a later time or so that they can be updated periodically as students learn more about the topic/issue. In subsequent class sessions, referencing this documentation, you could ask a follow-up question to help students recognize or reflect on the changes in their thinking: "Compare your prior headlines with your current headlines: How has your headline changed? What new thoughts or ideas do you have now that you didn't have then?"

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project zero THINKING ROUTINES

Here Now / There Then

A routine for considering presentist attitudes and judgments.

1. Identify a controversial issue or fairness topic that has changed significantly over time and uncover student's basic knowledge about the topic.

   Column A: List **present** stances, values and judgments about the topic.

2. Ask kids to imagine they could travel back to a time when the attitudes about the fairness of this topic were different.

   Column B: List **past** stances, values and judgments about the topic.

3. Compare the past and present perspectives in Columns A and B. Why do you think things have changed? Why did people in the past not think the way we do today?

4. Close the discussion How could we find out more about the way people thought back then?

PURPOSE

What kind of thinking does this routine encourage?
The routine encourages students to consider past perspectives and develop a better understanding of how thinking changes over time and across cultures. It helps students acknowledge that we have strong stances regarding controversial issues, and that our stances are influenced by social and historical context. It is also helps to uncover stereotypical perceptions as well as ethnocentric and presentist judgments.

APPLICATION

When and where can I use it?
The routine works best when dealing with issues that at one point in time or in a different culture were considered controversial. It can be used with topics about which we have strong stances that are not necessarily shared by people from other cultures or people in the past. Examples of these topics might include: slavery, holocausts, genocide, human rights, women's rights, child labor, war, and so on. This routine works well when students have had some experience with the topic and have at least a basic knowledge of its historical development.

LAUNCH

What are some tips for starting and using this routine?

This routine works well as a whole class discussion. Use the idea of the time traveler to help students think about fairness issues and values that have changed significantly over time or place. When comparing past and present stances acknowledge that certain issues may not be controversial to us today. List how we think about it presently and ask students to step back and consider how people thought about the topic during another place and time. What was their reasoning? Make these ideas visible. Explore the possible reasons for our shifts in thinking about this topic. Why do we view it differently? How could we find out more information?

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Hotspots

A routine for noticing truth occasions.

Key Prompts:

1. Identify a topic or situation. Is this idea clearly true, or false, or somewhere between the two?
2. What makes it so uncertain? (or almost certainly true or false)
3. How important is it? What makes it important? (important or not so important)

PURPOSE

What kind of thinking does this routine encourage?
Spotting truth hotspots can be used on almost any topic or situation. It can be used to introduce a topic, to draw out students' initial thoughts. It can be used to review a topic, to look back at something students have studied, in the middle of a topic to take stock. It can be used to get students started on identifying projects or identifying issues for discussion in small groups or to launch a whole-class discussion.

APPLICATION

When and where can I use it?
A key part of thinking is spotting situations that need more thought, and where more thought is worthwhile. This spotting routine asks learners to spot "thinking hotspots" about truth within a topic or situation that might be worth more attention. It thus helps them to be more alert to truth hotspots in the future. Also, asking "What makes this idea this way?" draws from learners characteristics that make an idea more or less uncertain and more or less important. This greater awareness helps them to spot truth hotspots in the future.

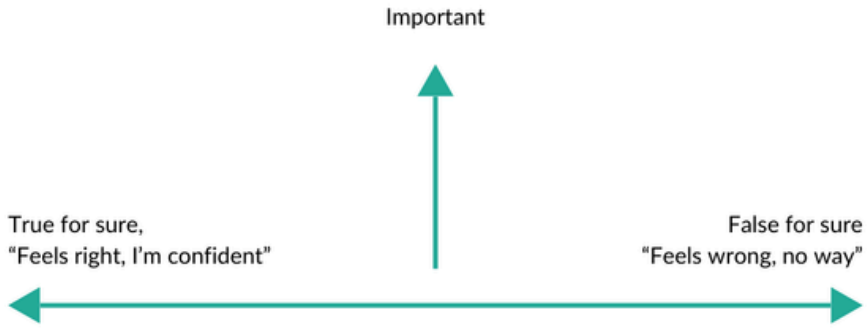
LAUNCH

What are some tips for starting and using this routine?
The spotting hotspots routine is best used for a topic or situation where students have some knowledge already. They may not have studied it formally, but at least they have some common knowledge. Otherwise, almost everything would come out "uncertain" and with little basis for judging its importance. Younger children may respond better to concrete situations, like a playground fight or an event in the news, than to abstract topics like nuclear energy. This routine makes thinking visible by helping students to see thinking opportunities — "thinking hotspots"—in situations. In particular, it helps students become more alert to situations where they might think more deeply about the truth of something.

Here are the key steps to the routine:

1. Teacher or student identifies a topic or situation.
2. Students identify ideas about the topic or situation as clearly TRUE, clearly FALSE, or uncertain and somewhere in the middle. And as more or less important to figure out.
3. Place ideas on a continuum. First, decide where to place the idea on the continuum between true and false. Then use a vertical axis to indicate importance, according to the student's judgment (see simple chart below). The teacher asks something like, "What makes this idea this way?" and draws out characteristics that put an idea "in the middle" rather than plainly true or false or make it important or not so important. The teacher does all this for several ideas from the class.
4. NOTE: Some students may reveal misinformation or misunderstandings at this stage. As with other thinking routines, while the students are thinking together it is not your role to correct them. Students may correct misinformation or misunderstandings themselves during the discussion or as they pursue a topic in the last step, or you may provide better information upon coming back to the topic later. Right now, you are functioning as a facilitator, not a source of information.
5. Teacher and students discuss disagreements about true-false and importance and place ideas on the chart, in more than one place if necessary. You do not have to resolve these disagreements, just acknowledge them. The goal is to raise consciousness of uncertainty and the reasons for it. Some disagreements may get resolved in the last step.
6. If the chart is lopsided, say with only some uncertain ideas in the middle or only important ideas, the teacher prompts to fill out the chart a little more. Example: "What are some ideas we are sure of?"
7. Teacher and students select "thinking hotspots" to investigate further, maybe right away or maybe later, perhaps using other routines.

Truth Hotspots Diagram



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project zero THINKING ROUTINES

How Else and Why?

A routine for cultivating a disposition to communicate across difference.

What I want to say is...
How else can I say this? And **why**?
How else can I say this? And **why**?
(Repeat question)

PURPOSE

What kind of thinking does this routine encourage?
This routine improves students' capacity to communicate with diverse audiences by nurturing understanding of their communicative choices, keeping in mind their intention, context, and audience.

APPLICATION

When and where can I use it?
This routine is broadly applicable to many aspects of the curriculum (e.g., when students are engaging with a story, historical event, or scientific finding). It is also useful at moments when students are asked to represent ideas or phenomena (e.g., when producing a graph, a poster design, or an interpretation of a work of art). This routine can also guide regular classroom discussions or informal interactions in and outside of school. Examples of provocations include but are not limited to film excerpts, students' own writings, classroom dialogue, and works of art.

LAUNCH

What are some tips for starting and using this routine?
Priming students' minds for the use of this routine may involve calling their attention to the importance of appropriate, effective, and respectful interactions and the value of reflecting on how we communicate with one another, especially with people whose lives and cultures are different from our own. Students will need an initial understanding of intention, audience, and situation as they develop skills for thinking about and engaging in complex communication. They also need to be aware of the variety of communicative means at their disposal (e.g., tone, body language).

Prior to starting the routine, you might want to explain that the routine involves an initial prompt, “What I want to say is …” that directs them to make a statement and explain their intention for the statement. The follow-up prompts ask students to examine their original statement and revise it based on their consideration of their own intention, their audience, and the situation or context. They will consider their reframing and revising with not only the verbal language they choose to revise but also their tone and body language (if the communication is taking place as a live exchange). You can choose how many times you want students to repeat the second prompt to further revise and reflect on their communication choices.

It is important for students to share with a partner or small group their responses to the “why” question of the follow-up prompt, and to use the ideas of intention, audience, situation, context, and delivery (tone and body language) in their explanation. Without doing so, the routine risks inviting students to repeat less effective forms of communication or reinforce communication misconceptions.

This thinking routine was developed as part of the ID Global and PZ Connect projects at Project Zero at the Harvard Graduate School of Education. Explore the full PZ Thinking Routine Toolbox at *pz.harvard.edu/thinking-routines*.

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project zero THINKING ROUTINES

I Used to Think... Now I Think...

A routine for exploring diverse perspectives to deepen understanding.

Think about what you have learned about the topic we have been studying and complete the following sentence stems:

- I used to think...
- Now I think...

PURPOSE

What kind of thinking does this routine encourage?
This routine helps students to reflect on their thinking about a topic or issue and explore how and why their thinking has changed. It helps consolidate new learning.

APPLICATION

When and where can I use it?
This routine can be used whenever students' initial thoughts, opinions, or beliefs are likely to have changed as a result of a learning experience. This may be after reading new information, watching a film, listening to a speaker, experiencing something new, having a class discussion, at the end of a unit of study, and so on.

LAUNCH

What are some tips for starting and using this routine?
Explain to students that the purpose of this activity is to help them reflect on their thinking about the topic and to identify how their ideas have changed over time. Here's a possible approach for using the routine:

"When we began our study of _____, you had some initial ideas about it. Take a minute to remember what ideas you previously had about _____. Write a few sentences using the sentence starter, "I used to think..."

"Now, think about how your ideas about _____ have changed as a result of what we've been studying/ doing/discussing. Again, in just a few sentences, write down what you now think about _____. Start your sentences with, "Now, I think..."

Have students share and explain their shifts in thinking. When you first begin using this routine, it is often helpful to do it as a whole group so that you can probe students' thinking to help them explain it in more depth. Once students become accustomed to explaining their thinking, students can share with one another in small groups and pairs.

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project zero THINKING ROUTINES

Imagine If...

A routine for imagining ways to improve objects or systems through creative redesign.

Choose an object or system and use the prompts.

- In what ways could it be made to be more **effective**?
- In what ways could it be made to be more **efficient**?
- In what ways could it be made to be more **ethical**?
- In what ways could it be made to be more **beautiful**?

PURPOSE

What kind of thinking does this routine encourage?
This routine helps students investigate objects and systems and how they might be redesigned by identifying new ideas and inviting divergent and convergent thinking.

APPLICATION

When and where can I use it?
This thinking routine can be used to explore the possibilities of improving, tinkering with, and/or tweaking any object or system. While it is important for students to generate ideas within a wideopen possibility space, it is helpful to place creative constraints on that possibility space. You may do this by limiting the variety of tools and materials students have access to, presenting certain functionality criteria, or identifying a particular population or user group.

LAUNCH

What are some tips for starting and using this routine?
TThis thinking routine asks students to imagine new ways to improve an object or system by looking at the possibility space around an object or system through four different lenses. While we find these four lenses helpful to consider, you and your students are encouraged to come up with others. When considering how to redesign or hack an object or system, it is exciting to see students generate a list of big, unbounded ideas, but it is also important for students to be sensitive to the design of their objects or systems. You might want to consider using one of the following thinking routines that help students think about objects and systems before doing this routine. These include: Parts, Purposes, Complexities; Parts, People, Interactions; or, Parts, Perspectives, and Me.

project zero THINKING ROUTINES

FACILITATION

This activity is recommended for the following learner age ranges:

Preschool
15-30 minutes
Heavily facilitated by peer/teacher/caregiver

Elementary School
15-30 minutes
Heavily facilitated by peer/teacher/caregiver

Middle School
15-30 minutes
Lightly facilitated by peer/teacher/caregiver

High School
15-30 minutes
Lightly facilitated by peer/teacher/caregiver

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project zero THINKING ROUTINES

Imagine If...
(For Younger Children)

A routine for finding opportunity.

Choose an object or system. Ask questions such as:

- How could you make it **more effective**? *(work better)*
- How could you make it **more efficient**? *(easier to use; faster to use)*
- How could you make it **more ethical**? *(better for the community; easier to share; safer; more fair)*
- How could you make it **more beautiful**?
- How could you make it _____?

PURPOSE

What kind of thinking does this routine encourage?
This routine encourages divergent thinking by prompting students to think of new possibilities for an object or system. It can also encourage convergent thinking by giving students a basis from which to narrow down their ideas so they can redesign or hack an object or a system. Ultimately, this thinking routine is about finding opportunity and pursuing new ideas.

APPLICATION

When and where can I use it?
You can use this thinking routine to explore any object or system. Though you can use this thinking routine on its own, we suggest you use it in combination with other Agency by Design thinking routines that encourage students to think more deeply about the object or system they are working with.

- This routine provides an opportunity for children to make their thinking visible. Children can show their ideas through drawings, role play, and/or writing.

- If you are using this thinking routine with young learners, consider having them think about an object or a system they have direct experience with. This way, all children can apply and build upon their prior knowledge in meaningful ways. We suggest you start with objects or systems in your classroom or community that students use regularly.
- Help children develop their language and literacy skills by facilitating class discussions and/orby encouraging them to brainstorm and share ideas with a partner.
- This thinking routine asks students to imagine how to make an object or system "more ethical." You will likely need to define "more ethical" for young learners by narrowing the scope of the term (for example, replacing "more ethical" with "better for the community"or "better for the environment"). When doing this, consider drawing upon your school community for guidance to ensure that you are focusing student thinking in ways that support the ethical behaviors that your school community is working to promote.
- To prepare children for using this thinking routine, consider using the Think, Feel, Care thinking routine. This will help children think about the desires and needs of people within the system, which will give them the background they need to imagine ways to improve the system for these people.

Note: This thinking routine is adapted from the Agency by Design "Imagine If..." thinking routine.

FACILITATION

This activity is recommended for the following learner age ranges:

Preschool
15-30 minutes
Heavily facilitated by peer/teacher/caregiver

Elementary School
15-30 minutes
Heavily facilitated by peer/teacher/caregiver

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project zero THINKING ROUTINES

Layers

A routine for analytically examining creative works through various frameworks.

Each layer consists of 4 possible elements to identify and examine in a creative work.

- **Narrative** The story, the back story, the hidden story, the message
- **Aesthetic** The appeal, the reward, the skill/mastery, the new/different/unusual
- **Mechanical** Technique, form, structure, methods
- **Dynamic** Surprise, tension, emotion, movement
- **Connections** To other works (in and out of the medium/genre), to history, to oneself, to the artist's other works or personal life

PURPOSE

What kind of thinking does this routine encourage?
The routine provides learners with a structure for looking analytically at creative works through a variety of different "layers" or frameworks.

APPLICATION

When and where can I use it?
Through using this routine, students learn that the process of analysis involves identifying many parts - or "layers" - of a creative work (e.g. literature, dance, painting, etc.). In this routine, students select the layer that they want to analyze. Some layers may be more familiar to students than others. Selecting unfamiliar layers may help students expand and deepen their understanding of the work.

LAUNCH

What are some tips for starting and using this routine?
This routine works best after students have had the opportunity to look closely at a creative work and to fully notice what is there. Students can then select a layer from the list to use in their analysis. This analysis can be done individually, with a partner, or in a whole group. Initially you may want to introduce one layer at a time with the whole class so that students have some collective experience using the layers. The analysis might be done verbally so that students hear and can build on others' ideas and contributions.

Here are some examples for how you can use this routine:

- **To identify prominent and hidden qualities.** Ask students, What layer immediately speaks to you? What makes you say that? Which layers seem more distant? What makes you say that?
- **To compare and contrast.** Ask students to contrast two works to see how they relate to one another: Where do you see connections as well as differences in terms of the layers?
- **To consider an individual element from one or more of the layers.** Ask students to pick one of the four elements from one or more of the five layers to explore the work.
- **To develop questions.** Ask students to use the layers and their elements to develop questions they want to ask an expert about the work of art.

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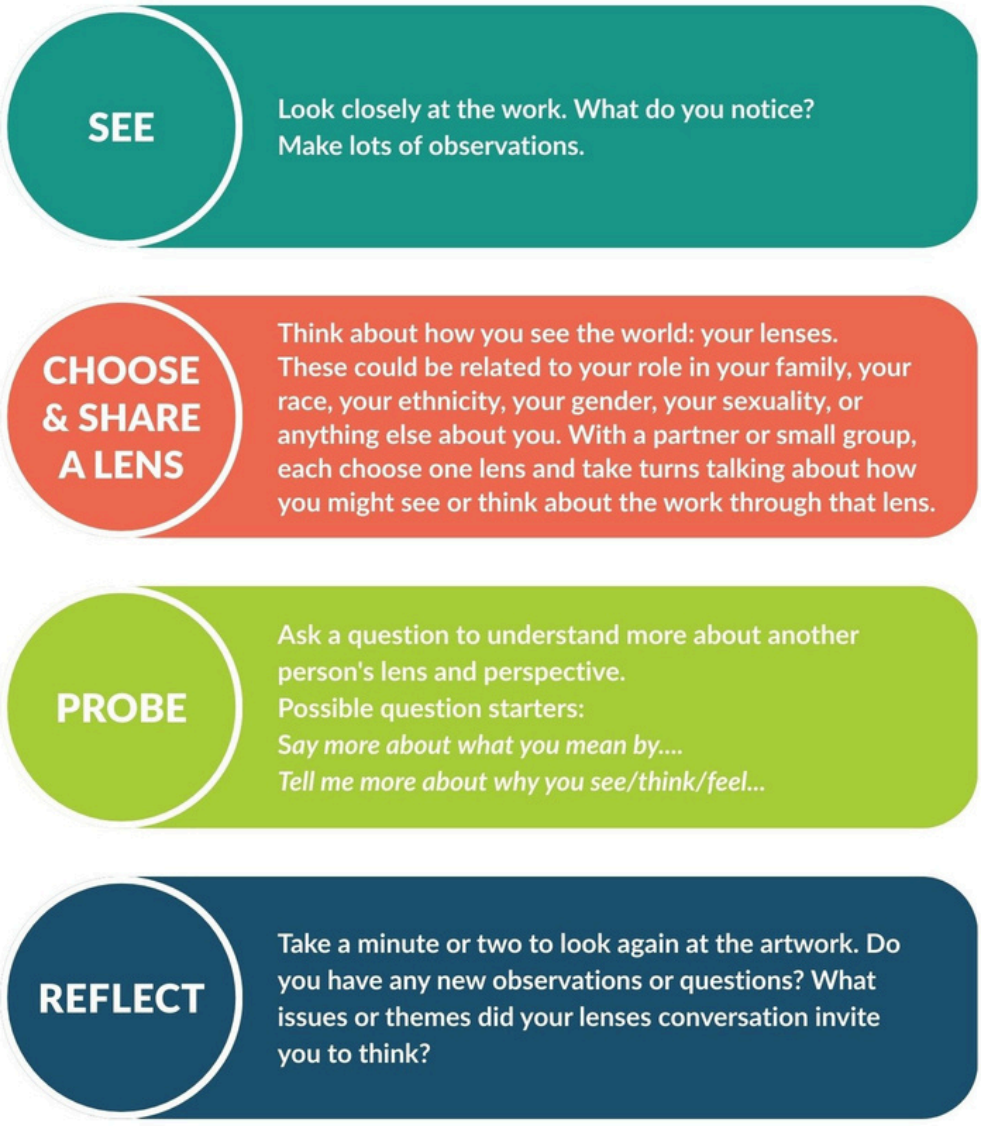
project zero THINKING ROUTINES

Lenses for Dialogue

A routine for looking through lenses and exchanging perspectives.



project zero THINKING ROUTINES

FACILITATOR NOTES LENSES FOR DIALOGUE

Before you begin

Choose an artwork that might speak to one or more civic themes—themes related to how we live together and how we aspire to live together. Civic topics may include issues such as climate and environmental change, migration, economic and other inequalities, gender and sexuality issues, race/racism, or any theme that addresses values, social conventions, power dynamics, institutions and systems that shape every level of community life, from the local to the global.

A word about lenses: The purpose of this routine is to look at an artwork with a distinct lens in mind. A lens is something you look through. Our lenses are affected by our identities or backgrounds. For example: gender, race, ethnicity, age, culture, place, occupation, family role, and more. These identities affect how we see and engage with the larger world. They also shape what we notice and think about when looking at an artwork. This routine invites students to be intentional about looking through distinct lenses as they explore an artwork.

Facilitating the Routine

Class time needed for this routine can range from 10-20 minutes or longer, depending on your facilitation choices. The routine is designed to support individual looking, thinking, dialogue with others, and then reflection. The discussion-oriented steps in the routine (Choose and Share a Lensand Probe) can be done in pairs or small groups.

**See Dialogue Moves for further ideas for deepening discussion.*
**See Viewing Moves for ideas to slow down and deepen looking at the artwork.*

This thinking routine was developed as part of the Arts as Civic Commons project at Project Zero at the Harvard Graduate School of Education. Explore the full PZ Thinking Routine Toolbox at *pz.harvard.edu/thinking-routines*.

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project zero THINKING ROUTINES

Listening: Ten Times Two

A routine for making careful observations about music.

1. Listen to a piece of music quietly. Let your ears wander and take in as much as possible.
2. List 10 words or phrases about any aspect of what you hear
3. Repeat steps one and two. Listen again and try to add 10 more words or phrases to your list

PURPOSE

What kind of thinking does this routine encourage?
The routine helps students slow down and make careful observations about music. It asks students to think about words or phrases to describe the work and encourages students to push beyond first listen or basic description.

APPLICATION

When and where can I use it?
The routine will work with any kind of music. Use "Ten Times Two" when you introduce a new piece of music to get students thinking carefully about it before having a discussion or using another routine. You can also use the Ten times Two routine after an in depth discussion about at a piece of music to both push forward and summarize some of the ideas and observations that were made during the conversation. The routine is useful before a writing activity. It gets students thinking about descriptive language and helps students make observations about the music.

LAUNCH

What are some tips for starting and using this routine?
If students are new to this routine, start with short excerpts of music. You can invite students to close eyes if that is comfortable for them or soften their gaze and just listen the first time. Then, during the second listen, they might write or draw in response to what they are hearing. If a list of ten things seems like a lot at first, begin with an invitation to list 3-5 things that students notice, then they can pair with a partner to compare lists.

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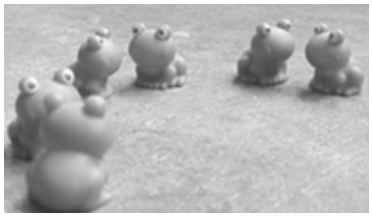
project zero THINKING ROUTINES

Looking: Ten Times Two

A routine for making careful observations about images, visual art, or objects.



1. Look at the image quietly for at least 30 seconds, letting your eyes wander
2. List 10 words or phrases about any aspect of the image
3. Repeat steps one and two. Look at the image again and try to add 10 more words or phrases to your list.

PURPOSE

What kind of thinking does this routine encourage?
This routine helps students slow down and make careful, detailed observations by encouraging them to push beyond first impressions and obvious features.

APPLICATION

When and where can I use it?
The routine can be used with any kind of visual art or non-art images or objects. The routine can be used on its own or to deepen the observation step of another routine. It can be especially useful before a writing activity because it helps students develop descriptive language.

LAUNCH

What are some tips for starting and using this routine?
You can invite students to do a first look with little guidance and encourage them simply to make a broad inventory of what they see. You might invite some guided looking during either the first or second "looking" opportunities by asking students to notice colors, shapes, and lines (borrowing from the thinking routine, "Colors, Shapes, Lines"). You can have pairs of students share their lists after both looking sessions to notice what their partners saw. If a list of ten observations seems too long to start with, shorten the task so students list or draw just 3-5 things they notice each time.

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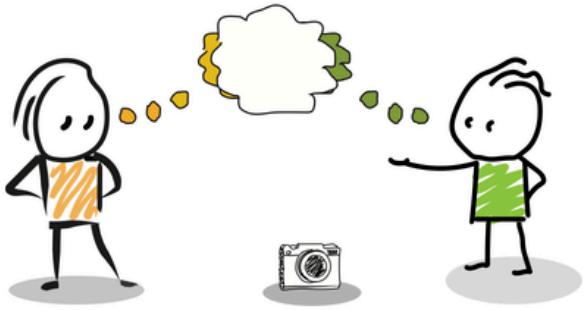
project zero THINKING ROUTINES

My Perspective, Other Perspectives (For Exploring Complexity)



Choose an object or system and ask:

My perspective

Have you ever used this? Why did you use it?
What do you like about it? Why?
What do you dislike about it? Why?
How would your life be different if you couldn't use this?

Other perspectives

Who else uses this? Why do they use it?
What do you think they like about it? Why?
What do you think they dislike about it? Why?
How would their life be different if they couldn't use this?

PURPOSE

What kind of thinking does this routine encourage?
The routine helps students explore complexity by encouraging them to consider that people may have different experiences with and perspectives on the same object or system. This routine also encourages students to think more carefully about how the object or system impacts their lives and the lives of others.

APPLICATION

When and where can I use it?

You can use this thinking routine to explore any object or system. You may choose to introduce the entire routine at once, or you may choose to break it down into parts over multiple occasions.

- This routine provides an opportunity for children to make their thinking visible. Children can show their thinking through drawings, photos, role play, and/or writing.
- If you are using this thinking routine with young learners, consider having them think about an object or a system they have direct experience with. This way, all children can apply and build upon their prior knowledge in meaningful ways. We suggest you start with objects or systems in your classroom or community that students use regularly.
- When possible, encourage students to gather information about other people's perspectives. For example, students might conduct interviews of people who use the object or system, observe people using the object or system, or use the internet or books to research specific perspectives.
- Consider using this thinking routine along with the Think, Feel, Care thinking routine to help children practice taking other people's perspectives.

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Name, Describe, Act

A routine for the looking closely, noticing, developing memory and analyzing.

Choose an image, object, or event to examine closely or select an object, event, or topic to analyze and:

- **Name** Recalling from memory, name and make a list of all the parts or features you can remember. These will most likely be nouns, things you can point to and name.
- **Describe** For each of the things you have named, add a description. What adjectives would you add to the nouns you have listed?
- **Act** For each of the things you named, tell how they act. What are they doing? What is their function? How do they add or contribute to the whole?

PURPOSE

What kind of thinking does this routine encourage?
This routine can be used to enhance close looking, develop descriptive language, and develop working memory. Depending on the stimulus/context, it can also be used to facilitate analysis of a topic.

APPLICATION

When and where can I use it?
This routine is used to help students notice and describe an image, object, or observed event in ever increasing layers of detail. For young children or for students learning another language the routine can also help to build up language proficiency.

LAUNCH

What are some tips for starting and using this routine?
To facilitate looking closely and develop working memory, have students look silently at an image object, or event for 1-2 minutes. Instruct them to notice as much as they can. Remove the image from view and ask students to name as many things as they can that they recall, focusing on just things they can name and touch, that is specific objects (i.e. “soldiers, guns, flames, etc.” rather than “a fight” or “war”). Doing this in writing helps students keep track of items. However, it can be done orally.

Pair students up for 3-5 minutes of “Describing.” The first person names an item off their list while their partner chooses a word to describe that item. Roles are reversed and the second person then names a new item while their partner describes it. This process continues until all items are named and described. If one person names something that their partner didn’t notice, then the person describes it him or herself. An alternative approach would be to have students describe everything on their list with a single word that captures its key qualities

For the “Act” component of the routine, you want students to focus on verbs that capture an action. This segment will also take 3-5 minutes. You might want to tell students that once a verb has been used it can’t be used again. This will force students to be more imaginative and not merely keep saying objects are “sitting.” An alternative approach that young children sometimes enjoy is to pick an object from the image and then silent walk around the room “acting” as if they were that object.

Conclude by returning to the image to view. Often students’ interest and curiosity are piqued by this routine and it can be a great opportunity to introduce background on your selected image that might set the stage for further study.

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project zero THINKING ROUTINES

Options Diamond

A routine for exploring the tensions of decision making.

1. Identify a couple of obvious options. Usually there are trade-offs or tensions between them that make the decision hard: Choose one and you get X but lose Y; choose the other and you lose X but get Y.
2. Make a diamond diagram, putting at the left and right corners the one or two main tradeoffs (the X's and Y's) pulling in opposite directions (see example in PDF).
3. Now have students brainstorm one to three solutions for each corner of the diamond.
   - Left side: Go with that trade-off
   - Right side: Go with that trade-off
   - Bottom: Compromise between them
   - Top: Clever solutions that combine the seeming opposites and get the best of both
4. Ask: What have we learned about the situation from finding these options? This is a way of understanding the situation better.

PURPOSE

What kind of thinking does this routine encourage?
This routine fosters creative thinking. It helps to explore decision making situations where a trade-off makes it hard to find a really good option. It focuses on resolving opposites. Sometimes, but not always, there are options that partly bring the opposites somewhat together. All this is also relevant to understanding. It helps in understanding situations even when you are not the real decision maker.

APPLICATION

When and where can I use it?
The options diamond helps with personal or classroom decision making when different factors pull strongly in opposite directions. It's also a useful way of exploring and understanding such situations in the news, history, or literature or science or medical policy, etc. For example, US President Harry Truman in deciding to drop the atomic bomb on Nagasaki and Hiroshima struggled with this trade-off: Kill many thousands of Japanese but shorten the war versus Let the war and its casualties continue. He chose to use the bomb. But what compromise options were there? And were there any options that might combine the opposites and end the war quickly without killing thousands of Japanese?

LAUNCH

What are some tips for starting and using this routine?

Remember, the left and right points of the diamond are not options themselves. They are the gains and losses that pull in opposite directions. You then write options near the left and right corners that go with one pull or the other; then the lower corner gets compromise options and the top corner gets any options that partly combine the opposites.

In many classroom situations the point is to use creative thinking to understand the situation better. Step 3 is the payoff and a final choice among the options may not matter. You can decide whether to go on to another routine for choosing among the options. Or you can just take a quick vote on some of the likely options. If you want, you can do this before step 3, to give students a little more to discuss in step 3.

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project zero THINKING ROUTINES

Options Explosion

A routine for creative decision making.

1. List the obvious options. There would not be a decision unless there were at least two or three obvious options.
2. Now brainstorm all sorts of different options to find the "hidden" options. Often there are hidden options that are the real best choices. Be imaginative! Piggyback on ideas already up, combine ideas to get new ones, look for ideas of a very different kind, imagine you are in different roles and suggest ideas from the perspective of those roles, etc.
3. Ask: What have we learned about the situation from finding these options? This is a way of understanding the situation better.

You may want to go on to a routine for comparing and choosing among options.

PURPOSE

What kind of thinking does this routine encourage?
This routine fosters creative thinking. It helps explore "hidden" options in a decision making situation. Often people don't make good decisions because they miss the hidden options. It is also relevant to understanding. It helps in building an understanding of decision-making situations even when you are not the real decision maker.

APPLICATION

When and where can I use it?
Students can use it for personal decision making or you and students can use it for classroom decision making. Also, you can use it with students as a way of exploring and understanding important decisions in the news or history or literature or science policy or medical policy, etc. You can ask students to make the decision personal by role playing, imagining that they were in the situation.

LAUNCH

What are some tips for starting and using this routine?
Emphasize that maybe there are good hidden options, maybe not – we have to find out by looking. Put the ideas on the blackboard or have students write them on Post-its and stick them up. Use an explosionlike diagram with radiating lines instead of a list if you want to emphasize the spirit, but a list is okay too. Remember, crazy ideas are okay – they are just part of the mix and they may lead to something else by piggybacking. In many classroom situations the point is to use creative thinking to understand the situation better, as in step 3. You don't need a final decision. You can decide whether it's a good idea to go on to another routine for choosing among the options. Or you can just take a quick vote on some of the likely options. If you want, you can do this before step 3, to give students a little more to discuss in step 3.

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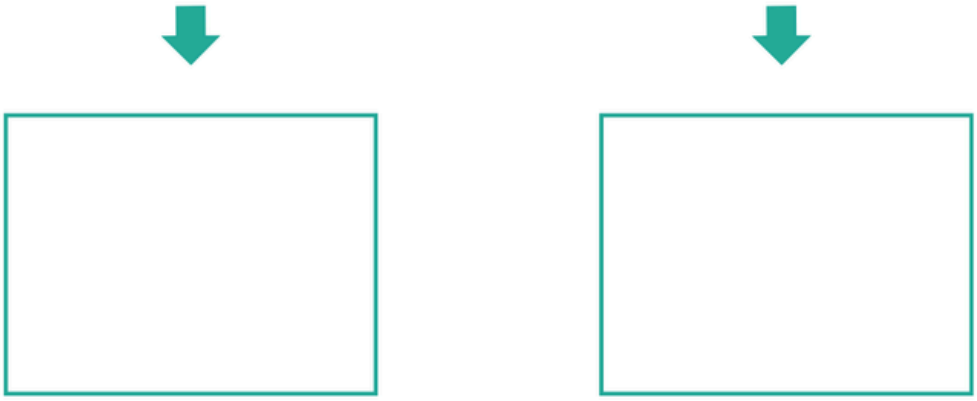
project zero THINKING ROUTINES

Outside In

A routine for connecting personal interests to topics in school.

1. List some of your hobbies or interests here . . . . . & . . . . . List some of your school topics or subjects here.



2. Connect a line between one of your hobbies/interests and a school topic. Along the line, write down a possible connection between the two. Do the same thing again with another interest and topic (the interest can connect to the same topic, or a different one).

3. Imagine what you could do to investigate or expand each of the connections further.

Connection #1: Possible investigation or expansion

Connection #2: Possible investigation or expansion

LAUNCH

What are some tips for starting and using this routine?

How can I help students make outside-in connections (and should I worry if the connections seem superficial)? At the broadest level, a connection is any type of relationship or association between two or more things. There is plenty of room for creativity, because connections can be metaphorical as well as literal. One way to help students find connections is to suggest that they look for certain types of connections. For example, they can look for connections related to form (e.g., how things are shaped physically or conceptually, what they look like, what things are made out of or what materials they use or involve); structure (e.g., how things are organized, how they work as a system), stories (e.g., the stories things tell or are told about them, or how things evolved over time,); defining ideas (e.g., main themes, definitions, key concepts); learning (e.g., how people learn about them, the learning challenges they involve); and culture (e.g., the customs and traditions people associate with them, the significance they have for groups of people). Don't worry if students make connections that at first seem broad or superficial. Most connections get more interesting once you spend some time thinking about them, and the 'imagine' step of the routine gives students an opportunity to dig into a connection and explore its depth. For example, a student might make a connection between her dance lessons and science class in school by saying that dance involves the human body and the human body is a topic in science. On the face of it, this may seem like quite a broad connection. But, as the answer to the next question illustrates, with a bit of expansion it can become quite interesting.

How can I help students imagine how to investigate the connections they make? A good way to start is to encourage students to brainstorm several questions about their connections. For instance, to continue with the foregoing example about dance, science, and the human body, here are some further questions students could ask. How are different systems of the body involved in dance—for instance the muscular system or the skeletal system? How is dance itself like a body system? What would it be like to create a dance inspired by the firing of neurons in the brain, or by the circulatory system? How does dance affect human health?

What should students do after they use the routine? Should they follow up on their ideas? There are two possible options. One of course is to encourage students to continue to investigate the ideas they came up with. They could do this as a special project, or as part of the regular curriculum. But even if there's no follow-up, just having students seek and expand on connections is worthwhile in itself. Why? Because knowledge likes to be activated, and students will find it interesting simply to reach for connections. Also, using the routine regularly helps students get in the habit of making connections. Recall the principle mentioned in the introduction to the Portable Knowledge materials: What you learn is what you do. Students who make connections frequently as part of the learning process are much more likely to continue that connection making later on. Moreover, sometimes school-based learning is seen as unconnected to students' lives outside of school. But learning happens everywhere. By getting in the habit of making connections 'outside-in,' students begin to see learning as what it truly is—a lifelong, lifewide endeavor.

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project zero THINKING ROUTINES

Parts, People, Interactions

A routine for examining systems by identifying their parts, the people involved, and how they interact.

Identify a system and ask the following questions.

- What are the **parts** of the system?
- Who are the **people** connected to the system?
- How do the people in the system **interact** with each other and with the parts of the system?
- How does a change in one element of the system affect the various parts and people connected to the system?

PURPOSE

What kind of thinking does this routine encourage?
This routine supports students to look closely at a system, identify its parts, recognize who participates in it, either directly or indirectly, and consider the consequences of changes in the system.

APPLICATION

When and where can I use it?
This thinking routine can be used to explore any system and helps stimulate curiosity, raise questions, surface areas for further inquiry, and introduce systems thinking. Students have the opportunity to notice that a change in one aspect of the system may have both intended and unintended effects on another aspect of the system. When considering the parts, people, and interactions within a system, students begin to notice the multitude of subsystems within systems.

LAUNCH

What are some tips for starting and using this routine?
Before beginning this routine, it may be helpful to introduce students to what a system is and develop criteria for what defines a system. Definitions are helpful, but we've found that concrete examples work best. In order to engage in this thinking routine, your students will have to identify a system to explore. One way to do this is to have your students situate an object within a broader system. Encourage your students to name the systems they would like to explore. You can then ask your students if their system meets the criteria for a system you identified. Systems are made up of subsystems and are themselves parts of broader systems. In order to avoid going down the rabbit hole of "everything is connected to everything," it may be helpful for you to encourage your students to define the boundaries of their system. Working in groups, it is helpful for students to first make a list of all of the parts and people involved in a system, and then to map out their system on chart paper to make the interactions between all of the parts and people in their system visible.

FACILITATION

This activity is recommended for the following learner age ranges:

- Preschool
  - 15-30 minutes
  - Heavily facilitated by peer/teacher/caregiver
- Elementary School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver
- Middle School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver
- High School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver

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project zero THINKING ROUTINES

Parts, People, Interactions (For Exploring Complexity)

A routine for exploring systems.

Identify a system and ask the following questions.

- What are the **parts** of the system?
- Who are the **people** connected to the system?
- How do the people in the system **interact** with each other and with the parts of the system?
- How does a change in one element of the system affect the various parts and people connected to the system?

PURPOSE

What kind of thinking does this routine encourage?
This routine helps students slow down and look closely at systems, with special emphasis on the various ways people engage with or are implicated in systems. Like the other routines that begin with identifying parts, the routine encourages students to make careful observations. It also encourages them to identify various kinds of causal interactions, and to consider the numerous ways that people affect and are effected by systems.

APPLICATION

When and where can I use it?
This thinking routine can be used to explore the human side of almost any kind of system. There are systems at work in virtually every aspect of our lives—environmental, governance, financial, workflow, logistical, distribution, mechanical, medicine, etc. The routine can be used on any kind of system, large or small, in which people figure in some way.

LAUNCH

What are some tips for starting and using this routine?
Before introducing the routine, it can be helpful to discuss the concept of systems with students. A good way to begin is to ask students what they think the word 'system' means, and encourage them to brainstorm several ideas and examples. A common definition of a system is: A group of interacting or interrelated elements that form a complex whole. While this definition is accurate, it isn't very illuminating unless it is accompanied by concrete examples, e.g., subway systems, town recycling systems, the lunch line system at school, etc. Since systems are everywhere, it can also be helpful to discuss what isn't a system. For example, a random heap of stuff, such as you might find in the back of a closet, isn't typically a system.

A good way to identify a system on which to use this routine is to begin with a concrete object or activity and then situate it within a broader system. For example, a postage stamp can be situated within a broader postal system and a bicycle helmet can be situated within a broader transportation system. A walk to school can be situated in a broader system of transportation. Brushing one's teeth can be situated in a broader system of health maintenance.

Systems are made up of subsystems, and are themselves parts of broader systems. In order to avoid going down the rabbit hole of everything is connected to everything, it may be helpful to encourage students to define the boundaries of their system. For example, if you are exploring the system of making an apple pie, you might limit it one particular pie; if you are exploring a recycling system, you might limit it to a geographical area.

How can students' thinking be made visible while using this routine? Working in groups, it is helpful for students to begin by making a list of all of the parts and people involved in a system, and then to map out their system on chart paper to make the interactions between all of the parts and people in their system visible.

FACILITATION

This activity is recommended for the following learner age ranges:

- Preschool
  - 15-30 minutes
  - Heavily facilitated by peer/teacher/caregiver
- Elementary School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver
- Middle School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver
- High School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver

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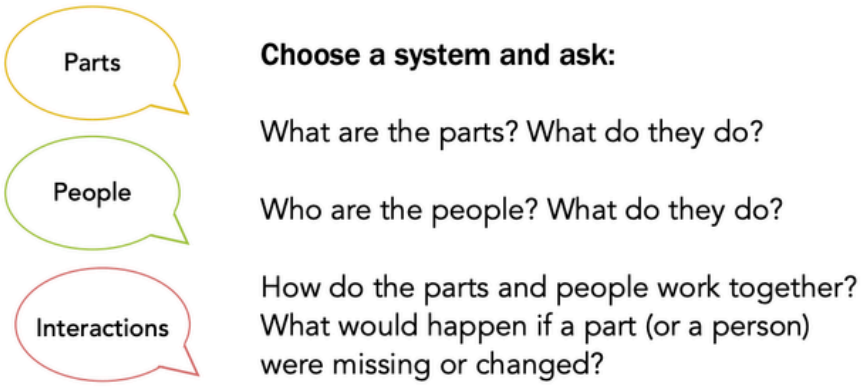
project zero THINKING ROUTINES

Parts, People, Interactions (For Younger Children)



PURPOSE

What kind of thinking does this routine encourage?
This routine encourages learners to slow down and look closely at a system. It helps them notice that there are different people who participate in the system and that they participate in different ways. It also encourages students to explore how one change in a system can impact the rest of the system. This thinking routine can help foster curiosity as children notice details, ask questions, make connections, and identify topics for future inquiry. It also helps children practice systems thinking.

project zero THINKING ROUTINES

APPLICATION

When and where can I use it?
You can use this thinking routine to explore any system. You may choose to introduce the entire routine at once, or you may choose to break it down into parts over multiple occasions.

- If you are using this thinking routine with young learners, consider having them explore a system they have direct experience with. This way, all children can apply and build upon their prior knowledge in meaningful ways. We suggest you start with systems in your classroom or community that students use regularly.
- Systems are made up of subsystems, and are themselves parts of broader systems. In order to avoid going down the rabbit hole of "everything is connected to everything," it may be helpful for you to encourage your students to define the boundaries of their system.
- This routine provides an opportunity for children to make their thinking visible. Children can show their thinking through drawings, visual mapping, photos, role play, and/or writing.
- You may realize, as you engage with this thinking routine, that the system you have selected is more complicated or abstract than you originally thought. This is okay. Help students look closely at the parts that they are interested in, and feel free to help them seek more information by asking each other, by using books or the internet, or by consulting with community members.It is okay if students have unanswered questions about the system. Consider documenting these questions to revisit if the opportunity presents itself later on. Note: This thinking routine is adapted from the Agency by Design Parts, People, Interactions thinking routine.

FACILITATION

This activity is recommended for the following learner age ranges:

- Preschool
  - 15-30 minutes
  - Heavily facilitated by peer/teacher/caregiver
- Elementary School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver

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project zero THINKING ROUTINES

Parts, Perspectives, Me

A routine for exploring the complexity of objects and systems.

Choose an object or system and ask:

- What are its **parts**?
  - What are its various pieces or components?
- What **perspectives** can you look at it from?
  - Different users, makers; different physical perspectives.
- How are ***you*** involved?
  - What connections do you have? What assumptions, interests, or personal circumstances shape the way you see it?

PURPOSE

What kind of thinking does this routine encourage?
This routine helps students explore complexity by encouraging them to look closely at details, to consider various viewpoints, users, and stakeholders, and to reflect on their own connections.

APPLICATION

When and where can I use it?
This routine can be used to explore virtually any object or system. It works particularly well with objects that have many parts–and can be taken apart—as well as systems that have various roles and users connected to them (e.g., systems involved in managing or providing resources, social systems, organizational systems, transportation systems, or governance systems).

LAUNCH

What are some tips for starting and using this routine?
The three elements of this routine can be introduced all at once, but each of the steps encourages a different kind of thinking, so it is helpful to distinguish the steps from one another and to give each step time to unfold. For the parts step, give students plenty of time to look at or otherwise experience the topic in detail. If the object is physically present, students can sketch it or make a diagram. If appropriate, they can take it apart (taking things apart—from doorknobs to old household appliances to toys—is a particularly powerful way to look closely at the parts of something).

For the perspectives step, encourage students to imagine different physical viewpoints if appropriate. Encourage them to think broadly about how different people interact with or are connected to the object or system including considering who is involved in making it, who is affected by it, and who cares about it. When possible, encourage students to gather information about other perspectives, for example by interviewing people or doing research. For the your involvement step, encourage students to consider the different ways the object or system influences their lives or the lives of other people. Encourage them to consider any feelings, assumptions, connections, beliefs, attitudes, or associations they have with it. As with the other routines that begin with the naming of parts, students can make their thinking visible by creating lists, sketches, and diagrams.

FACILITATION

This activity is recommended for the following learner age ranges:

- Elementary School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver
- Middle School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver
- High School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver

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project zero THINKING ROUTINES

Parts, Perspectives, Me (For Exploring Complexity)

A routine for exploring the complexity of objects and systems.

Choose an object or system and ask:

- What are its **parts**?
  - What are its various pieces or components?
- What **perspectives** can you look at it from?
  - Different users, makers; different physical perspectives.
- How are ***you*** involved?
  - What connections do you have? What assumptions, interests, or personal circumstances shape the way you see it?

PURPOSE

What kind of thinking does this routine encourage?
The routine helps students explore complexity by encouraging them to look closely at the details of something, considering its various viewpoints, users, and stakeholders, and reflecting on their own connections and involvement with it.

APPLICATION

When and where can I use it?
The routine can be used to explore virtually any object or system. It works particularly well with objects that have many parts – and can be taken apart—as well systems that have various roles and users connected to them, such as systems involved in managing or providing resources, social systems, organizational systems, transportation systems or governance systems. The three elements of this routine can be introduced all at once, but each of the steps encourages a different kind of thinking, so it is helpful to distinguish the steps from one another, and to give each step time to unfold.

LAUNCH

What are some tips for starting and using this routine?
What are some tips for the 'parts' step? Give students plenty of time to look at or otherwise experience the topic in detail. If the object is physically present, students can sketch it, or make a diagram. If appropriate, they can take it apart. (Taking things apart—from doorknobs to old household appliances to toys--is a particularly powerful way to look closely at the parts of something).

What are some tips for the 'perspectives' step? Encourage students to imagine different physical viewpoints if appropriate, for example by zooming in, zooming out, or taking a bird's eye view. Encourage them to think broadly about how different people interact or are connected to the object or system: who is involved in making it, who is affected by it, who cares about it? When possible, encourage students to gather information about other perspectives, for example by interviewing people or doing research.

What are some tips for the 'How are you involved?" step? Encourage students to consider the different ways the object or system touches their lives or the lives of people they know. Encourage them to consider any feelings, assumptions, connections, beliefs, attitudes or associations they have with it.

How can students' thinking be made visible while using this routine? As with the other routines that begin with the naming of parts, students can make their thinking visible by creating lists, sketches, and diagrams.

FACILITATION

This activity is recommended for the following learner age ranges:

- Elementary School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver
- Middle School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver
- High School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver

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project zero THINKING ROUTINES

Parts, Purposes, Complexities

A routine for looking closely at systems.

Choose an object or system and use the following prompts.

- What are its **parts**?
  - What are its pieces or components?
- What are its **purposes**?
  - What are the purposes of each of these parts?
- What are its **complexities**?
  - How is it complicated in its parts and purposes, the relationship between the two, or in?

PURPOSE

What kind of thinking does this routine encourage?
This routine sparks curiosity by encouraging students to observe carefully, looking beyond the obvious features of an object or system, to surface questions and identify areas for further inquiry.

APPLICATION

When and where can I use it?
This thinking routine can be used to explore any object or system. The routine provides an opportunity to make students' thinking visible as they create lists, maps, and drawings of the parts, purposes, and complexities of various objects and systems.

LAUNCH

What are some tips for starting and using this routine?
You may introduce the three elements of this routine all at once, or you may want to introduce the three elements of the routine one at a time. If an object students are working with is present and/or physically visible, students might not need a lot of background knowledge. However, if students are working with a system (for example, a system of government), it may be helpful for students to have background knowledge or to have an opportunity to reflect on their experiences interacting with that particular system.

To take this routine to the next level, after students have considered the parts, purposes, and complexities of an object as it is, you may consider having students take apart the objects they are working with—and then continue to identify the parts, purposes, and complexities they notice, documenting their noticings on their maps or drawing by using a different color pencil, pen, or marker. You may consider swapping out the word "complexities" for more accessible terms, such as *puzzles* or *questions*.

FACILITATION

This activity is recommended for the following learner age ranges:

- Preschool
  - 15-30 minutes
  - Heavily facilitated by peer/teacher/caregiver
- Elementary School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver
- Middle School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver
- High School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver

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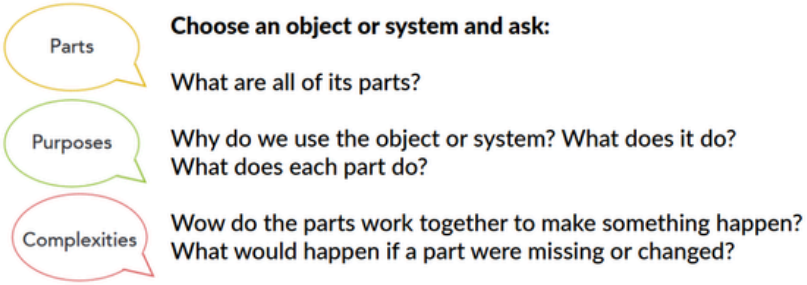
project zero THINKING ROUTINES

Parts, Purposes, Complexities (For Younger Children)

Looking closely.



PURPOSE

What kind of thinking does this routine encourage?
This routine encourages learners to slow down and make careful, detailed observations as they look beyond the obvious features of an object or system and think about how it works. This thinking routine can help foster curiosity as children notice details, ask questions, make connections, and identify topics for future inquiry.

APPLICATION

When and where can I use it?

You can use this thinking routine to explore any object or system. You may choose to introduce the entire routine at once, or you may choose to break it down into parts over multiple occasions.

- This routine provides an opportunity for children to make their thinking visible. Children can showtheir thinking through drawings, photos, role play, and/or writing.
- If you are using this thinking routine with young learners, consider having them explore an objector a system they have direct experience with. This way, all children can apply and build upontheir prior knowledge in meaningful ways. We suggest you start with objects or systems in yourclassroom or community that students use regularly.
- Help children develop their language and literacy skills by encouraging them to work togetherand to discuss what they notice with a partner. Consider creating a class word wall of descriptivelanguage that the class can use and add to over the course of the school year.
- To encourage children to deepen their understanding of the parts, purposes, and complexities ofthe object under study, have them take apart the object. Then, repeat the thinking routine whilelooking closely at the separate parts.
- You may realize, as you engage with this thinking routine, that the system you have selected ismore complicated or abstract than you originally thought. This is okay. Help students look closelyat the parts that they are interested in, and feel free to help them seek more information by askingeach other, by using books or the internet, or by consulting with community members. It is okayif students have unanswered questions about the system. Consider documenting these questionsto revisit if the opportunity presents itself later on.

FACILITATION

This activity is recommended for the following learner age ranges:

- Preschool
  - 15-30 minutes
  - Heavily facilitated by peer/teacher/caregiver
- Elementary School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver

This thinking routine was originally developed by the Artful Thinking project, and adapted by the Agency *by* Design project at Project Zero, Harvard Graduate School of Education. Explore the full PZ Thinking Routine Toolbox at *pz.harvard.edu/thinking-routines*.

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project zero THINKING ROUTINES

Parts, Purposes, Complexities (For Exploring Complexity)

A routine for looking closely at systems.

Choose an object or system and use the following prompts.

- What are its **parts**?
  - What are its pieces or components?
- What are its **purposes**?
  - What are the purposes of each of these parts?
- What are its **complexities**?
  - How is it complicated in its parts and purposes, the relationship between the two, or in?

PURPOSE

What kind of thinking does this routine encourage?
The routine helps students slow down and make careful, detailed observations by encouraging them to look beyond the obvious features of an object or system. The routine stimulates curiosity, raises questions, and surfaces areas for further inquiry. The routine is one of three PZ Thinking Routines that begins with the naming of parts. Naming parts is an important way of exploring complexity because it reveals how objects and systems have numerous interacting components.

APPLICATION

When and where can I use it?
The routine can be used to explore virtually any object or system. It works particularly well with things that students can directly observe or diagram. If you use the routine with an object or system from the natural world, as opposed to a human-designed object or system, you may want to substitute the word 'function' for 'purpose. This is because humanmade objects and systems are made with intentional purposes in mind, whereas in the natural world, form and function co-evolve without requiring explicit intentionality.

How much background knowledge do students need in order to use this routine? If the topic is an object that is physically or visible, students may not need a lot of background knowledge. However, if students are working with a system that extends beyond what they can observe at the moment—like democracy or recycling—it may be helpful for students to have some background knowledge, or to give them an opportunity to reflect on their experiences interacting with that particular system.

LAUNCH

What are some tips for starting and using this routine?

Should the routine be used all at once, or step-by-step? The three elements of this routine can be introduced all at once, but it is often helpful to begin by taking extra time with the parts step, in order to give students plenty of time to make detailed observations. (If an object is physically present, students can be encouraged to draw its parts, as a way of looking closely.) Once students have taken time with the parts step, the next two steps—purposes and complexities— often flow together naturally.

How can students' thinking be made visible while using this routine? The routine provides an opportunity to make students' thinking visible through creating lists, maps, and drawings of the parts, purposes, complexities of various objects and systems. Often students discover linkages between the three steps of the routines and their diagrams can including lots of connecting lines and annotations—and even look messy at times. This is generally a good thing, as it is an indication that students are discovering the genuine messiness of complexity.

Can this routine be used with very young students? Yes, though generally for students in preK-4 you may want to lead the discussion and make one collective chart or diagram. Also, you may consider swapping out the word complexities for more accessible terms, such as *puzzles* or *questions*.

FACILITATION

This activity is recommended for the following learner age ranges:

- Preschool
  - 15-30 minutes
  - Heavily facilitated by peer/teacher/caregiver
- Elementary School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver
- Middle School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver
- High School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver

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project zero THINKING ROUTINES

Pass the Parcel

A routine for fostering intrinsic motivation for deep learning and understanding.

Invite students to think about a goal they have and imagine it as a treasure inside a parcel that is wrapped in many layers. With each layer they unwrap, they get closer to their treasure:

As you **REFLECT** on what has been learned about . . . (current or past topics), think about connections with (this topic) . . . and your goal. With these connections, you have unwrapped the first layer of the parcel.

IDENTIFY significant connections that you would like to explore further and think about why they are significant to you. With each significant connection you identify, you unwrap another layer.

EXPLORE all ways you can to understand what it is about these connections that can help you unwrap the parcel and reach the prize. Each exploration removes another layer.

What will you do to **ACT** on all you have learned, so the last wrapping paper will come loose and your treasure is in reach?

PURPOSE

What kind of thinking does this routine encourage?
To heighten awareness that what is learned in and out of school are often connected with each other. These connections can enable moving forward to achieve goals. This fosters intrinsic motivation. It provokes curiosity, wonder and delight in finding the unexpected. Exploring the connections can broaden and deepen understanding.

LAUNCH

What are some tips for starting and using this routine?
What is meant by goals? Short term goals, career goals, health and wellbeing, sporting, social. See examples on the following page.

Why Pass the Parcel? Pass the Parcel is a children's party game where, through a process of unwrapping layer after layer of paper, the child who unwraps the last layer receives the prize hidden deep inside. The exact form varies, including passing the parcel around a circle of children as music is playing, and, when the music stops, the child holding the parcel unwraps a layer, and so it continues. This thinking routine uses Pass the Parcel as a metaphor. As learning and understanding broaden and deepen, the layers are removed; core knowledge, understanding and constructive ways forward, are the prize. If students do not know the game, you can explain it briefly.

How could Pass the Parcel be used in a classroom? A parcel isn't necessary. Teachers can select topics directly from the curriculum, current events, or concepts such as freedom, democracy, peace, measurement etc. (see next page). Students can also suggest topics. Students commence progression through the steps quietly with reflection, then with discussions and recording the responses either individually, in a group, or as a whole class, sharing what they think they know about the topic and connections they are making.

After this, encourage students to continue their explorations any ways they think will be particularly helpful e.g. time in the library for research, drawing concept maps or diagrams of how things are connected or effect each other, list questions you would like to ask, places you could go to find out more… At a later stage invite students to share experiences related to the topic and what they have explored. The last step can be composed of discussions and formulating action plans to reach goals.

How and why invite students to suggest something they would like to explore? When students suggest topics, there is immediate engagement and shared ownership of the learning and it can evoke intrinsic motivation toward seeing how these topics connect to the curriculum.

How might you follow up this thinking routine to maximize the portability of knowledge? When students have had time to explore the topic using their own goals as a focus in their explorations, perhaps they could adopt the more traditional version of Pass the Parcel to share their new learning about the topic. They might list key points they have found out about the topic and wrap the list up as a 'treasure', providing a brief overview of their findings; and continue wrapping up this treasure, on each layer pasting a question that they had been asking on their learning journey; and then invite other students in the class to play the game. The lucky student unwrapping that final layer would instantly have the results of this exploration of the topic. This way individuals or small groups could share their experiences and new knowledge of the topic seen through different perspectives and build on their learning and experiences.

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project zero THINKING ROUTINES

Peel the Fruit

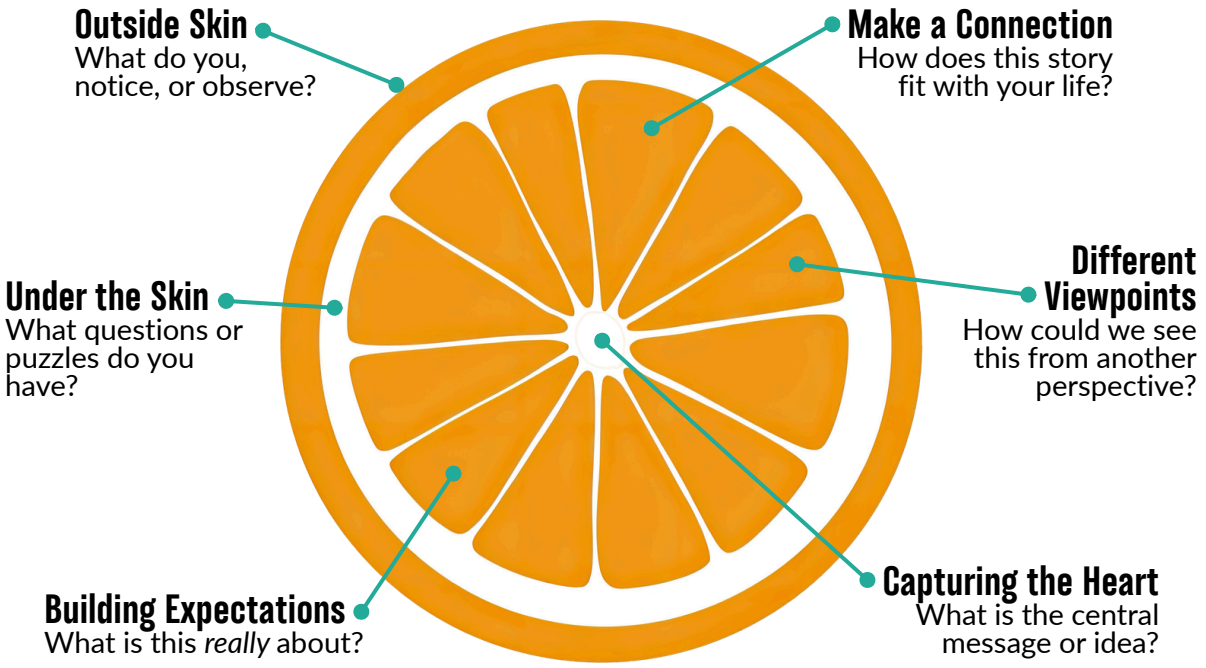
A routine for tracking and guiding understanding.

1. **Put some version of the map up** in a convenient location or give learners copies. See example below and notes about different ways of using the map.

2. Briefly state that the group will be **tracking progress and planning** with the map from time to time. Note how the map uses the metaphor of "peeling the fruit." getting familiar with the surface of something, seeking puzzles and mysteries to investigate, and pursuing these in various ways to arrive at core understandings.

3. Refer to the map to **choose next steps and mark progress** from time to time during the exploration of a topic (no need to do everything every time). Use it as a way of thinking about what routines to use or simply what kind of conversation or other activity to have.

4. When the map is used collectively by a class, you may want to invite students to **put up post-its on the map** over time to mark insights associated with any of the map elements.



project zero THINKING ROUTINES

PURPOSE

What kind of thinking does this routine encourage?
We often want to develop learners' understanding of a complex topic over days or weeks. This map can help. It's not a routine but a way of planning and tracking over time the exploration of a topic. It can help in choosing good routines too.This routine supports reasoning by having students form claims, explanations, or interpretations and support them with evidence. Questioning claims helps students see reasoning as an ongoing process.

APPLICATION

When and where can I use it?
Whenever there's a topic that calls for a broad and rich understanding and learners have enough time to look at it in different ways – anything from a single long lesson to several lessons or a unit. You can use it with students collectively, to help them maintain a bird's eye view of progress through a topic and to make with them good choices about what to do next. You can use it yourself, to plan topics and to track progress. You can also give copies to students for their individual self-management in pursuing a general class topic or individual projects.

LAUNCH

What are some tips for starting and using this routine?
Explain that the map is for tracking and guiding the exploration of the topic. Explain the metaphor briefly. Invite learners to help chart progress by using the map.

You can create a giant version of the map to put on the wall of a classroom (see diagram below), or just put labels up for the categories if it's easier to organize on the wall, or personalize the process in some other way. If you're tracking two or three topics at the same time or multiple groups you might: have two or three wall maps, color code paths on a single map, give learners page-size copies to track their own progress, or invent something else. Whatever works! The main idea is to make visible the developing understanding to mark progress and choose next steps.

It usually makes sense to start with the "skin" and go to "getting under the skin" with mysteries and then on from there to "substance" and toward the "core." You need not use all of the "substance" approaches—whatever fits—and there's no fixed order. You can go back to something and add at any time!

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project zero THINKING ROUTINES

Portable Surprise

A routine for finding patterns in a topic and similar patterns in very different situations.

(pattern = any generalized relationship across many cases, often with variations, e.g. good guy/bad guy in literature, transformative inventions in history, food webs in ecologies)

1. Brainstorm a few big general patterns that showed up in the topic. Point to ones that surprised you, like a rabbit popping out of a magician's hat. Pick one to focus on that seems especially interesting.

2. Brainstorm a few other situations, very different, where the same pattern might show up. Pick one to focus on that seems especially interesting.

3. Compare and contrast the pattern in the initial topic with the pattern in the new situation. What are the key similarities? Differences? What similarities and differences seem most important?

PURPOSE

What kind of thinking does this routine encourage?
Abstracting important surprising patterns from a topic and also finding somewhat similar patterns in very different situations and comparing.

How does this help build portable knowledge? The thinking routine makes more portable some of the ideas about the topic itself. And also, the thinking routine cultivates the general habit of seeking portable knowledge.

APPLICATION

When and where can I use it?

Turning to examples mentioned earlier, one can see in global warming a pattern of confusing controversy that also showed up around cigarettes and today around genetically engineered crops. One can extend the concept of food webs in a natural ecology to ask about the food webs of humans today in our complex societies, or prehistorically, or the food webs of pets, or even the "food webs" in an analogical sense in patterns of consumption around product marketing and sales. Or, for a new example, if students are studying nuclear weapons, one can ask what were the "game changer" weapons of other eras, for instance the bow and arrow, and how were the impacts the same and different in a broad sense. Or, to generalize even further, one could explore "game changer" products (like the iPhone) or ideas (like democracy) in the same spirit. How is the pattern of "game changing" similar and different to that of weapons?

LAUNCH

What are some tips for starting and using this routine?

What if students have trouble extracting patterns? Emphasize the idea of surprise. Draw from the students what surprised some of them, at least a little bit. Then what pattern made it surprising? Help and hint a little as you need to.

What if the students have trouble finding other situations that might be similar? Again, help and hint a little as you need to.

What if students want to compare and contrast with a new situation that actually isn't a very good analogy? It doesn't have to be a really good analogy. Remember, step 3 invites similarities and differences and an assessment of their importance. You can easily ask at the end: As it turns out, is this a really good analogy or only a so-so analogy? Even so-so analogies can be illuminating, the contrasts as revealing as the similarities!

Can we discuss more than one pattern and connection? By all means, if time allows. One way to get more ideas in play is to have students apply Portable Surprise in small groups, each group picking a favorite analogy to develop. At the end, the groups share a few headlines about their discoveries.

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project zero THINKING ROUTINES

Projecting Across Distance

A routine for considering how a topic, event, or issue impacts not only one's own community and country, but also other communities and countries far away.

Pick a topic, event, or issue that might be approached differently in different parts of the world or even your own country, such as food security, population growth, or marriage practices. Then, consider it from the following perspectives.

How is the topic, event, or issue playing out or viewed in...

- your community?
- another city or town in your country?
- a country east or west of your country, where people may think differently about the theme, event, or issue?
- a country north or south of your country, where people may think differently about the theme, event, or issue?

What might account for the similarities and/or differences **between** and **within** the communities and countries?

PURPOSE

What kind of thinking does this routine encourage?

This routine encourages learners to take a broader, multi-perspectival view of a topic, event, or issue. While there are other thinking routines that also encourage learners to take diverse perspectives, this routine differentiates itself by inviting learners to specifically compare and contrast perspectives with firstly another community in their country, followed by countries that are geographically different from their own country. In doing so, learners come to understand that:

- Not everyone in the same country or even in the same city, holds the same views, and even if they hold broadly similar views, it's the shades of differences that may be of greatest interest.
- People in countries that seem to be like theirs may not always think the same way, and that in fact it could be the case that people who are most geographically distant that may share similar views as them.
- It is important to gather a diversity of perspectives on a topic, event, or issue if one is to try to understand it as fully as possible, or to try to walk in someone else's shoes for a day.

APPLICATION

When and where can I use it?

An important thing to keep in mind is that learners may not be able to immediately identify communities and countries where the selected topic, event, or issue is approached differently. In that case, you may want to do any of the following:

- Have ready a set of communities and countries that you put forward for learners to explore.
- Invite students to pool what they already happen to know, taking advantage of the diversity among them. Then, assign them to research (e.g. conduct Internet searches; interview adults like parents) some points from the pool of knowledge and bring back their findings.

LAUNCH

What are some tips for starting and using this routine?

As this routine involves discussing different communities and countries, it will be important to keep a lookout for learners who make sweeping assumptions or broad generalizations about other cultures, peoples, and places. This could lead to misunderstandings and stereotyping of other cultures, peoples, and places, which need to be addressed quickly. Some ways to do so are to follow up with "What makes you say that," introduce a counter example to highlight the discrepancy in their thinking, etc., or ask the question: *Do you think everyone in that country/community is likely to think the same way?* At all times, it's crucial not to let the misunderstanding or stereotype take root.

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project zero THINKING ROUTINES

Projecting Across Time

A routine for understanding a topic (e.g. ideas, phenomena, problems) across a broad span of time reaching backward into the past, and forward into the future.

Pick a topic (e.g. tourism; computers; global migration patterns).
Then, consider it using the following prompts:

Map what you think or already know. What do you know about the topic?

Reach back in time. How has the topic played out in different forms / contexts / places over the last 10 years? The last 100 years? The last 1000 years?

Reach forward in time. How do you think the topic will play out 10 years into the future? 100 years? 1000 years?

Map how your thinking about the topic has changed. How do you view the topic now?

PURPOSE

What kind of thinking does this routine encourage?
This routine encourages learners to view topics (e.g. ideas, phenomena and problems) from the vantage of a broader timescale, than what they have learned from the media, or experienced in their own lifetime. Often, ideas (e.g. childhood; debt; happiness), phenomena (e.g. the Internet; migration), or problems (e.g. lack of access to clean water; terrorism), can look very different depending on how far back in history or how far forward into the future we look.

For instance, the idea of “childhood” can seem relatively unproblematic when learners explore it through their own lived experience, or even by going two or three generations back to how their parents or grandparents experienced it. Imagine inviting them to go even farther back in time, to before the 17th century when “childhood” as an idea did not exist and children worked in the fields from a young age. How would that impact the way learners understand what being a child means? Similarly, inviting learners to project forward in time to understand the idea of “childhood” requires them to consider the conditions that make “childhood” possible or even viable, as well as how what they have come to accept as “childhood” may change dramatically.

LAUNCH

What are some tips for starting and using this routine?

What are some tips for the step "Map what you think or already know?" This step is intended to invite learners to make visible what they think about the topic. A good way to begin the routine is to have learners map this step on a large chart in one color, and then each of the next steps in a different color. That way, at a glance, they can see how their thinking develops with each step.

What are some tips for the step "Reach back in time?" Here, it may be necessary to circumscribe the problem space for your learners: how far back do you want them to go? Is there a particular context(s) you want them to pay attention to? If so, you may want to provide resources that will help them explore the topic, e.g. a carefully curated set of websites, articles, books, etc. Where feasible, this step also offers the opportunity to teach learners about historical understanding and evidence, e.g. how do we know what we know about the past? Whose past is it, and who decides? What evidence has been offered for the conclusions?

What are some tips for the step "Reach forward in time?" This step should encourage a certain amount of imagination and fantasy (especially the farther into the future one reaches for), but not fantastical imaginings that are simply meant to be outrageous. A good follow-up is "What makes you say that?" to prompt learners to provide thoughtful reasons to support their responses. Also, learners may choose to hedge in their response—"Maybe X, maybe Y, depending"—which is perfectly reasonable.

What about some other time frames? It's important that the routine is adapted for your learners, context, and purpose, so if the time frame of 10/100/1000 does not work well (e.g. 10 years is too short for any observable change), feel free to choose other time frames like 20/100/1000 or even 20/200/2000.

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project zero THINKING ROUTINES

Red Light, Yellow Light

A routine focusing students on signs of puzzles of truth.

1. **Identify a source or range of experiences to investigate**, e.g. the editorial page, a political speech, a pop science source, rumors on the playground.
2. **Students look there for "red lights" and "yellow lights,"** specific moments with signs of a possible puzzle of truth, like sweeping generalizations, blatant self-interest.
3. **Round up students' observations.** Make a list of specific points marked R for red or Y for yellow with the sign (see sample chart). Also, ask students to identify "red zones" and "yellow zones," whole areas that tend to be full of red or yellow lights. Write them on the board in circles.
4. **Ask**: What have we learned about particular signs that there could be a problem of truth? What have we learned about zones to watch out for?

Red light, Yellow light only identifies potential issues of truth. You may want to go on to some other truth routines to dig into a couple of the issues.

PURPOSE

What kind of thinking does this routine encourage?
In the general clutter of everyday life, moments that need deeper thinking tend to be invisible. Students have to learn to see them. This routine focuses students on signs of puzzles of truth, and also on typical red zones and yellow zones where such puzzles are common. To build up this sensitivity, use the routine often in deliberately different ways.

APPLICATION

When and where can I use it?
Wherever there might be interesting puzzles of truth: a text that might have questionable claims, the daily paper, TV news, political speeches, a mystery story, a math proof that might have weaknesses, playground activities and conversations, home life, pop science, potentially risky behaviors, self-critique of something one has written, etc. For settings outside of school, students can keep logs over a day to a week. Typical red zones are the editorial pages of newspapers, political speeches, playground arguments, because so many red lights occur within them.

The source should be large enough to take some time, like a chapter or keeping track of rumors for a few days. That way, students have to keep alert in a sustained way, which practices their skills of noticing puzzles of truth.

LAUNCH

What are some tips for starting and using this routine?
Explain that "red lights" are specific moments with signs of a possible puzzle of truth, signs like sweeping statements, one-sided arguments, obvious self-interest, etc. See the sample chart for others. Yellow lights are milder versions of the same thing.

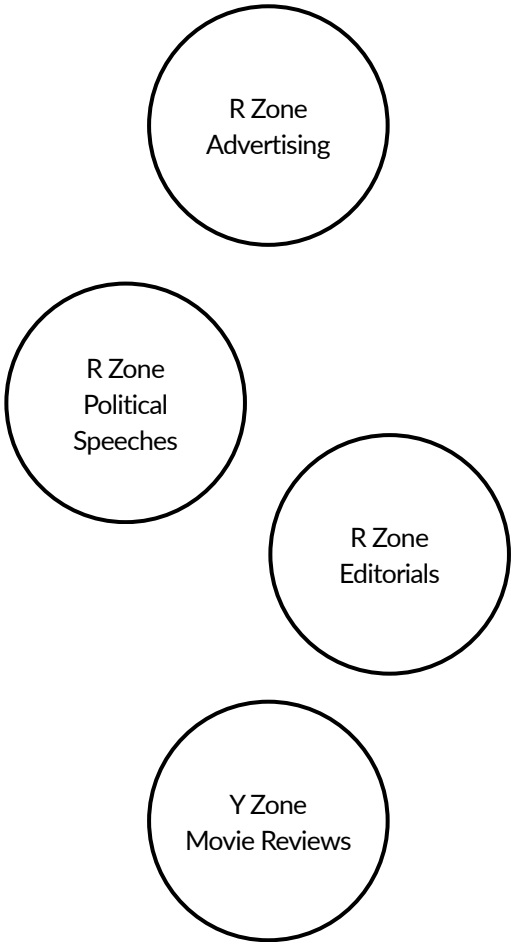
Naturally students may disagree on what's red vs. yellow vs. green in particular cases. Have students explain the signs and their judgments briefly, but mainly the routine is for detecting potential puzzles of truth. The real way to investigate a couple of the more important red or yellow lights is to dig further into the issue with another truth routine.

List of some signs to start with. Students can add to this.

Red Lights/Yellow lights for problems of truth

- Sweeping generalization
- One-sided arguments
- Bold claim, no argument
- Blatant self-interest
- Extreme conviction
- No obvious expertise
- Angry claims
- Feelings: seems implausible, uncertain, tentative
- Plainly an opinion

Students investigating a newspaper might find examples (abbreviated) like these:



R The only honorable way out is to win on the battlefield (political, extreme statement, no argument)
Y The majority of people agree… (evidence?)
R I'm sick and tired of the way…. (editorial, angry claim)
Y The senator expressed his judgment that… (tentative)
R You can save more now than ever before… (ad, blatant self-interest)
Y Thousands of people flock to these kinds of self-medication (re the medications, lack of expertise)
Y Both teenagers and young adults will like this film (opinion)

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project zero THINKING ROUTINES

Reflect, Connect, Project

A Routine to Reflect Back and Think of Future Directions for Learning and Making.

Choose a Making Move you've been working with, or have seen in action, and consider...

Reflect
What stood out to you as you used this Making Move, or watched a classmate or friend use it? Why do you think this Making Move might be important for learning?

Connect
How could this Making Move connect to things you are learning or doing outside this class or learning experience?

Project
How might you use this Making Move differently in the future? What's another Making Move that you're inspired to use as your work goes forward?

PURPOSE

What kind of thinking does this routine encourage?
This thinking routine helps learners to think in the past, present, and future, viewing their making in the context of a long-term and broad trajectory of learning. It is meant to cultivate an ongoing reflective practice in the classroom or other maker-centered learning context.

APPLICATION

When and where can I use it?
This routine is meant to be used within classrooms and other learning environments that have an active and ongoing dialogue around the Agency by Design Making Moves. It can be used when thinking about any of the Making Moves. Learners might use this routine to reflect on their own use of a Making Move, or to reflect on a peer's use of a Making Move. The routine can be used on its own, or in combination with another routine or other tools related to the Agency by Design framework. Here are some ideas and considerations for putting this thinking routine into practice:

APPLICATION

When and where can I use it?

This routine is meant to be used within classrooms and other learning environments that have an active and ongoing dialogue around the Agency by Design Making Moves. It can be used when thinking about any of the Making Moves. Learners might use this routine to reflect on their own use of a Making Move, or to reflect on a peer's use of a Making Move. The routine can be used on its own, or in combination with another routine or other tools related to the Agency by Design framework. Here are some ideas and considerations for putting this thinking routine into practice:

- The three elements of this routine can be introduced to learners all at once or one at a time.
- You may find that in the beginning, it is helpful to distinguish for learners the difference between the use of the word project as a noun (as in a multi-stage task that unfolds over time) and its use as a verb (as in look forward), which is the third part of this routine. This, and other language in this routine and in the Making Moves, should always be adapted to best fit the needs of your learners and your context.
- As with all routines, it is helpful to use the language of the routine early and often in order to help learners internalize the mental moves it encourages.
- You might use the language of this routine to make students' thinking visible by incorporating it into concept map prompts, group conversation structures, personal journal-writing, etc.
- You could hang this routine up in the learning space in order to encourage the language of the routine to be used in teaching, learning, and reflection on an ongoing basis.

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project zero THINKING ROUTINES

Reporter's Notebook

A routine for separating fact and feeling.

Identify **a situation, story, or dilemma** for discussion.

Name the **facts and events** of the situation.
(What evidence confirms these?)

Identify the **thoughts and feelings** of the characters/participants in the situation.
(What evidence confirms these?)

Make your **best judgment** about the situation, based on this information.

PURPOSE

What kind of thinking does this routine encourage?
This routine helps students to distinguish among facts, thoughts, and judgments and to organize ideas and feelings, encouraging perspective-taking.

APPLICATION

When and where can I use it?
Consider using the Reporter's Notebook routine in many situations: when discussing imagined or real moral dilemmas or topics from history, literature, or science; after reading a chapter or watching a video or performance; or, when inviting students to think about actual events from their own lives. This routine is most useful "mid-investigation," after some information about a given situation has already been shared and explored. Maybe the topic is growing increasingly more complex as students' learning advances, or perhaps the topic has generated disagreements among students. Perhaps students are mistaking opinions (either in the texts being examined or in the students' discussions) for facts. Using this routine can help students deepen their understanding of a topic and clarify for themselves the roots of the complexity or confusion.

LAUNCH

What are some tips for starting and using this routine?
This routine is best introduced in a whole class discussion. Once students understand the routine, they can work independently or in small groups. Ask students to imagine they are newspaper reporters who need to differentiate the facts of a given event from the thoughts and feelings of those involved in the event. The stance of a reporter helps students to clarify issues, including points of agreement and disagreement, by giving them the opportunity to step back from their own perspectives or initial understandings of a given situation. Invite the students to create a reporter's notebook grid by drawing a 2x2 grid on a piece of paper. Over the two columns on the top, students write "Clear" and "Need to Check." On the side, they write "Facts & Events" and "Thoughts & Feelings." Then students consider the event or issue under discussion and list responses in the appropriate portion of the grid. As you listen to students' conversations, encourage them

project zero THINKING ROUTINES

REPORTER'S NOTEBOOK

	CLEAR	NEED TO CHECK
FACTS & EVENTS What happened?		
THOUGHTS & FEELINGS How did characters think or feel about it?		

My Best Judgement:

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Same and Different

A routine for considering and questioning dichotomies.

Choose a debate, incident, or object in which opposing views are clearly apparent, or images that look different but are grouped together.

Notice
- Often judgments are made at first glance. What was the first impression you had about this?

Perspective Taking
- From what other points of view could this be perceived? What would one say from those points of view?

Same and Different
- What are the similarities? Differences? How is this case the same and different at the same time?

PURPOSE

What kind of thinking does this routine encourage?
This routine helps learners go beyond the surface of similarities and differences. Complex matters are often presented and accepted as is. It's not until they are questioned and explored more thoroughly and deeply and from different angles that they can be better understood and more thoughtful opinions and decisions made.

APPLICATION

When and where can I use it?
This routine can assist students of all ages to consider dichotomies that are commonly present in everyday life. Same and different are constant puzzles. Different countries have different currencies, different homes, schools and languages, yet they all share the same or similar purposes. Words can sound exactly the same as another, yet be spelled quite differently, and have very different meanings. People can be grouped together or separated because of one or more of the following: religion, skin color, education, socio-economic status, background, country of birth, etc.

LAUNCH

What are some tips for starting and using this routine?
How can we assist our students in realizing their thinking has changed over time and not feel guarded about admitting they have changed their mind? Give examples of how highly regarded scientists changed their views about the impact of sugar in diet as they learned more about the way bodies function and respond to different foods and combinations of foods. You can also model how your ideas have changed and how and why this happened and assure students that their thoughts are valued, not judged, and provide time for reflection. The thinking routine I Used to Think … Now I Think … can assist with this.

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project zero THINKING ROUTINES

Same, Different, Connect, Engage

A routine to nurture empathetic perspective taking and bridge building.

Same In what ways might this person and you be similar?
Different In what ways might the person and you be different?
Connect In what ways might the person and you be connected as human beings?
Engage What would you like to ask, say, or do with the person if you had the chance?

Same In what ways might this person and you be similar?	**Different** In what ways might the person and you be **different**?
Connect In what ways might the person and you be connected as human beings?	**Engage** What would you like to ask, say, or do with the person if you had the chance?

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project zero THINKING ROUTINES

See, Feel, Think, Wonder

A routine for exploring works of art and other interesting things.

See What do you **see**?
Feel What **feelings** emerge for you as you look at this piece?
Think What does this piece make you **think** about?
Wonder What do you **wonder** about this piece?

PURPOSE

What kind of socio-emotional learning and thinking does this routine encourage?
This routine encourages students to make careful observations and thoughtful interpretations. It provides opportunity to connect and share motions and it helps stimulate curiosity and sets the stage for inquiry. Use this routine when you want students to think carefully about why something looks the way it does or is the way it is. Use the routine at the beginning of a new unit to motivate student interest or try it with an object that connects to a topic during the unit of study. Consider using the routine with an interesting object near the end of a unit to encourage students to further apply their knowledge and ideas.

APPLICATION

When and where can I use it?
Ask students to make an observations about an object—it could be an artwork, image, artifact, or topic—and follow up with what they think might be going on, what they think this observations might be and what feelings emerge when looking. Encourage students to back up their interpretation with reasons. Ask students to think about what this makes them wonder about the object or topic. The routine works well in a group discussion but in some cases you may want to ask students to try the routine individually on paper or in their heads before sharing out as a class. Student responses to the routine can be written down and recorded so that a class chart of observations, interpretations, and wonderings are listed for all to see and return to during the course of study.

(Excerpts retrieved from the See Think Wonder Thinking Routine)

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project zero THINKING ROUTINES

See, Wonder, Connect x2

A routine for looking closely and making connections to deepen understanding.

Explore an image, object, or work of art:

1. Look closely: What do you notice? Make many observations.
2. What questions do you have? What do you wonder about?
3. How could this connect to subjects you study in school?
4. How could this connect to your personal interests or hobbies?

PURPOSE

What kind of thinking does this routine encourage?
The goal of the routine is to help students appreciate how looking closely at something can expand one's understanding of it, which in turn can provide a basis for making connections to other things. The routine encourages students to explore the interconnectedness of knowledge, and to understand that if they intentionally look for connections, they can find them. And because this routine is supposed to be used 'routinely,' the eventual goal is for students to internalize the habit of looking for connections.

APPLICATION

When and where can I use it?
It works well toward the beginning of a topic of study because it begins by drawing on visual information students can readily see, and thus tends to be immediately engaging. Also, using the routine early in a topic of study encourages students to frame their learning as transfer-oriented from the outset. That said, the routine is a variation of the Visible Thinking routine, See | Wonder | Connect, which is a highly versatile routine that can be used at virtually any point in a learning trajectory.

LAUNCH

What are some tips for starting and using this routine?
What kind of image or object should the routine be used with? The routine encourages students to look deeply and go beyond the obvious, so it works with a wide variety of images and objects—even items whose surface appearance is plain. Everyday objects, images in the media, book covers, works of art, all work well. Often teachers choose a physical object or image that directly relates to a topic—a ruler, a map, a leaf, a portrait, a historical artifact or photograph. But don't be afraid to experiment: Let yourself be surprised by what students will be able to uncover, even when you can't envision ahead of time where students' observations will lead them.

Are there specific ways to encourage the See step? The main thing is to provide a timeframe that encourages students to prolong their looking, but doesn't feel too long. For example, give students a full minute to look quietly, noticing as many details as they can, before asking students to share their observations. Also, encourage students to elaborate by asking them to describe a particular observation in more detail. It also helps to simply prompt them to keep looking by periodically asking, "What else do you see?"

Are there specific ways to encourage students the Wonder step? One important thing you can do is to encourage students to ask different kinds of questions. For example, they might ask questions about what an item is made of, how it is made, who made it or how it came to be, who it is important to, how it be different, and what role the item plays in larger systems.

Are there specific ways to encourage the Connect steps? There is no getting around the fact that making connections can be challenging. Don't be afraid to encourage students to stretch to make connections: Does the object or image visually resemble something else they have seen? Is it structurally similar to other things they know about? Is its purpose similar? Was it made in a way that reminds them of other things? Does it connect to events they know about or can imagine? Does it have a history that connects to other things? Are there ideas or assumptions behind the object that connect to other contexts? Is it part of a larger system that connects to other things?

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project zero THINKING ROUTINES

See, Think, Make, Discuss

A routine for thinking about civic art through making.

Prepare
Before you begin this routine, you will need to gather some materials to make something. Paper and pencil are more than enough, but you might also want to consider other materials, perhaps even things that don't seem like art materials. What about that piece of junk mail you were going to recycle or that can of seltzer whose contents you just emptied into a cup? Feel free to use whatever you have available, even if it seems unconventional.

SEE Look closely at the work. What do you notice? Make lots of observations.

THINK How might this artwork connect to ideas or questions about civic life? Brainstorm several thoughts.

Civic life = how we live together as a community, from the local to the global.

MAKE Choose an idea or question about civic life that came to mind during the THINK step, and explore your thoughts about it by making or drawing something.

DISCUSS Talk about what you made and the civic idea or question you explored.

project zero THINKING ROUTINES

BEFORE YOU BEGIN

Choosing an Artwork: Before beginning the routine, you will want to select a work of art. Any artwork that can be read through a civic lens works well, whether that reading is aligned with the artist's intention or is an interpretation you bring to the work. Artworks that could be read through multiple civic lenses, rather than explicitly focusing on one topic, leave a bit more room for interpretation, which may be more generative for student discussions.

Preparing Materials: You will also need to gather some art-making materials before starting the routine. These materials can truly be whatever is at hand. Simple drawing materials are more than enough. You can also consider more unconventional making materials, like aluminum foil, string, tape, recycled materials, or anything else that is lying around. The process of making is more important than the product that is created, so expensive or high quality art materials are not essential. Furthermore, we recommend limiting students to using only one to three types of materials, as creativity is often sparked through limitations.

FACILITATING THE ROUTINE

The purpose of this routine is to help students think critically through making. Art making allows for a multiplicity of meanings to be shared and explored at once, evoking rather than defining interpretations. Making also offers many opportunities to think and express oneself in ways that are often difficult or impossible to articulate in words. You might consider using this routine when you want students to explore emotions around a civic topic or when engaging with a topic that can be difficult to discuss or put into words. If your students need more support during the See step, consider implementing a Viewing Move; extra support for the Think and Discuss steps can be found in the use of a Dialogue Move.

Supporting Making: Keep in mind that making can look many different ways, be done individually or collaboratively, and take vastly different amounts of time. Feel free to leave the making prompt as open ended or as specific as fits your needs. We recommend limiting the time spent on making - you might be surprised with what can be created in only 5 minutes! In fact, limiting the amount of time and materials spent on making can lower the stakes and help students to be more free and take more risks with their expression. To further help lower the anxiety that can often come with sharing artworks, students need not share what they've made (although, of course, they are welcome to). Students can focus on sharing their making process and any new thoughts or insights they came to through making rather than the product of their making.

Here are just a few ideas of what making could look like: a drawing, a collage, a small sculpture made of recycled materials, a plan for a larger artwork, a remix of the original artwork, a poem or song lyrics, a short play/skit/dance.

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project zero THINKING ROUTINES

See, Think, Me, We

A routine for connecting to the bigger picture.

1. **SEE:** Look closely at the work. what do you notice? Make lots of observations.

2. **THINK:** What thoughts do you have about the work?

3. **ME:** What connections can you make between you and the work?

4. **WE:** How might the work be connected to bigger stories— about the world and our place in it?

BEFORE YOU BEGIN

Choose an artwork or image. This routine works well with a wide variety of works, so feel free to be experimental or adventurous in your choice. If you typically use classrooms norms for respectful discussion, you may want to refer to them before you begin: The routine invites learners to make personal connections, so it's especially important to establish an atmosphere of trust and care.

FACILITATING THE ROUTINE

The **SEE** step encourages close looking. Ask learners to fully describe what they see, and to hold off making interpretations or giving opinions until the THINK step. If you like, deepen the SEE step by using a strategy or two from the Viewing Moves.

The **THINK** step encourages learners to share thoughts about the work. All thoughts are welcome, but you can give some direction to the step by asking questions such as: What's going on in the work? What might it mean? What makes you say that?

The **ME** step asks learners to make personal connections, so it's a moment when a safe and trusting atmosphere is especially important. You may want to model this step by sharing your own personal response first. If you're working with a large group, it can be helpful to do this step in pairs or trios. You can also add support to the discussion by using strategies from the Dialogue Moves, such as NAME or STORYTELLING.

By asking for "bigger stories," the **WE** step invites learners to reach for connections beyond themselves. One way to help them do this is to ask them to consider how the personal connections they identified in the ME step might connect to larger themes of human experience. This step can be challenging for students. As with the ME step, it can be helpful for you, the facilitator, to model a response by sharing your own reflections.

This thinking routine was developed as part of the Arts as Civic Commons project at Project Zero at the Harvard Graduate School of Education. Explore the full PZ Thinking Routine Toolbox at *pz.harvard.edu/thinking-routines*.

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project zero THINKING ROUTINES

See, Think, Wonder

A routine to make careful observations and thoughtful interpretations.

- What do you see?
- What do you think about that?
- What does it make you wonder?

PURPOSE

What kind of thinking does this routine encourage?
This routine encourages students to make careful observations and thoughtful interpretations. It helps stimulate curiosity and sets the stage for inquiry.

APPLICATION

When and where can I use it?
Use this routine when you want students to think carefully about why something looks the way it does or is the way it is. Use the routine with a relevant object (such as an artwork, image, artifact, chart, video, etc.) at the beginning of a new unit to motivate student interest, or try it with an object that connects to a topic during the unit of study. Consider using the routine with an interesting object near the end of a unit to encourage students to further apply their knowledge and ideas.

LAUNCH

What are some tips for starting and using this routine?
Once you present the object to your students, give them time to observe it. It may be useful to explain that they are first going to describe exactly what they see, not what they think they see. In the second step when students describe what they think about what they're seeing, you could ask them follow-up questions like, "What else is going on here?" or "What do you see that makes you say that?" These questions help move students away from giving unsupported opinions encouraging them instead to use evidence to explain their thoughts. In the third step, help students articulate what they are wondering by asking them what questions remain for them.

The routine generally works well in a group discussion. You may want to document the students' responses and post them in a place where all students can see them to encourage future consideration. When doing this as a group, you may want to ask students to try the routine quietly on their own first (perhaps documenting their own thinking in writing) before discussing in a group.

This thinking routine was developed as part of the Visible Thinking project at Project Zero at the Harvard Graduate School of Education. Explore the full PZ Thinking Routine Toolbox at *pz.harvard.edu/thinking-routines*.

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project zero THINKING ROUTINES

Seek to See

A routine to nurture a disposition toward proactive empathic perspective taking, de-stigmatization, and recognition of dignity

Take some time to look closely at this story (with an image, video, or text) drawing on the information you have, explore the following ways of seeking to see.

I seek to see . . .

- **Multiple Feelings** What might be this person's various feelings in this situation?
- **Strength** What might be this persons strengths, cultural richness and power?
- **Connections** What might be some ways in which we connect as human beings?
- **Human Dignity** What words would I choose to honor your humanity and make you shine?

Take a moment to reflect about your experience seeking to see. Did you notice any shifts in your thinking, perspectives, or feelings? Did anything surprise you? What questions do you have?

PURPOSE

What kind of thinking does this routine encourage?
TWe often see inequality through the lens of distribution of resources. This routine addresses a different kind of inequality: Recognition gaps defined as "the disparity in worth and cultural membership between groups in a society." Recognition gaps situating inequality in the realm of culture, and amenable to influence through education.
Recognition of worth and belonging have intrinsic value because human dignity and social justice do as well. Additionally, recognition—the opposite of stigmatization and discrimination—is associated with mental health and wellbeing.

Seek to See is designed to direct our gaze onto another human being with a deliberate intention to see complexity, connections and human dignity.

APPLICATION

When and where can I use it?
Provocations for this routine can provide an opportunity to understand a subject that remains typically invisible or a target for stigmatization. Provocations ideally will provide background and wider contextual information regarding the person being explored. Provocations should provide a richness of multiple colors, with possible positives as well as negatives responses in terms of the feelings, emotions and connections present surrounding the provocation being explored.

If the provocation features issues of violence, discrimination, or other difficult issues, it is important to think about opportunities for students to consider their agency and reasons for hope associated with the circumstance being explored, so students don't solely end in a place of fear or anger.

LAUNCH

What are some tips for starting and using this routine?
One might want to pay close attention to the range of **emotions** offered by a provocation, as well as nuance with which students describe emotions, opportunities to uncover complexity and the way that students bring their own emotions in terms of the emotional states of the lives they're learning about.

In exploring at **strengths**, we should seek to see a variety of sources of strengths ranging from the personal character driven values, to the cultural roots and contributions of a person, to their capacity to bridge cultures.

For the **connections** stage, we may seek to observe the kinds of connections students find to bridge differences (socioeconomic, cultural, etc.) while also highlighting the importance of understanding the self and the other as holding complex identities.

In seeking to see **human dignity**, we should seek provocations that present opportunities to see the whole person, their story and their potential. In the process we should seek to move towards a proactive view bringing a critical eye towards stigmatization and considering actions towards de-stigmatization that they may propose organically in the process.
In certain moments, pairing this routine with the question "What makes you say that?", can help deepen the conversation and help students uncover their reasoning and evidence related to a given provocation.

This thinking routine was developed as part of the ID Global, Reimagining Migration project at Project Zero at the Harvard Graduate School of Education. Explore the full PZ Thinking Routine Toolbox at *pz.harvard.edu/thinking-routines*.

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project zero THINKING ROUTINES

Slow Complexity Capture (Find, Capture, Explain, Wonder)

A routine for slowing down to look closely at complexity.

Find Find an object or scene that captures your eye. In a word or phrase, say what it is.

Capture Take some time to look carefully at your item. Capture it by slowly making a drawing of it or taking pictures of it from different angles or perspectives. Don't worry about creating a 'good' or realistic drawing or picture. The goal is to use drawing or picture-taking to help you look closely and notice details. Spend at least 5-10 minutes observing through drawing, looking, and/or photographing.

Explain After you have visually captured your item, write a paragraph (or tell a friend) about how it is complex

Wonder What new ideas and questions do you have about your item?

PURPOSE

What kind of thinking does this routine encourage?
The routine helps students slow down and use the medium of drawing or photography as a tool for looking closely. By taking time to make extended observations, students will begin to see the complexity inherent in the design, composition and constitution of even everyday objects and scenes.

APPLICATION

When and where can I use it?
The routine mainly emphasizes the complexity of parts and interactions, but it may also help student see other kinds of complexity.

LAUNCH

What are some tips for starting and using this routine?
If students are uncomfortable with drawing, emphasize that the purpose of the routine is to help students make careful observations, not to produce polished work. If students don't know where to start, offer some simple strategies or techniques. For example, create a grid of 4 or 6 squares and ask students to fill in each square with a quick sketch. Or have students make a contour drawing by drawing the outlines of something without lifting their pencil from the paper—and perhaps by looking only at the object, not at the drawing or their hand.

The routine can also be varied. Students can capture an object or item from several different perspectives (as in the picture above); they can focus in on a single aspect of something and capture it in careful detail; they can make a rough sketch of an entire scene; they can make multiple sketches of the same scene. Additionally, students can vary their work configurations. For example, they can work in pairs, or small groups in which each person creates a capture of the same item from a different angle or perspective. Then they share their captures with one another, discuss them, and together describe the way in which their item is complex.

When responding to student work, emphasize the observational qualities of students' drawing, rather than their technical qualities. Support students' explanations of complexity by using language and ideas from the 'Ways Things Can be Complex' page. Support their wonderings by appreciating the many different kinds of questions they have, and invite them to explain how observing through drawing/photographing helped them to ask new questions.

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project zero THINKING ROUTINES

Step In, Step Out, Step Back

A routine for nurturing a disposition to take social/cultural perspectives responsibly.

Step In: What do you think this person might feel, believe, know, or experience?
Step Out: What would you like or need to learn to understand this person's perspective better?
Step Back: What do you notice about your own perspective and what it takes to take somebody else's?

PURPOSE

What kind of thinking does this routine encourage?
A routine to nurture perspective taking, recognizing that understanding others is an ongoing process that reveals as much about ourselves as it does about the people we seek to understand.

APPLICATION

When and where can I use it?
This routine can be used with a broad range of topics from examining the perspectives of people in a story, a historical event, or a contemporary news article, to considering non-human perspectives such as species in an ecosystem, to collective perspectives such as interest groups in a given conflict. You may choose an image, video, story, or classroom incident to ground students' thinking.

LAUNCH

What are some tips for starting and using this routine?
When you start the routine, remind students that in the first prompt, "step in," they are reasoning with the information they have, which is always limited. You may point to the speculative nature of their interpretations. In the second prompt, "step out," you invite learners to see that there is more to understanding another person than the first impression they as individuals construct. As they share their views, students may detect stereotypes in their own initial thinking and feel uneasy about "having been wrong" in their initial thinking. It is important to normalize the fact that we all have first impressions of others and others have them of us, and that this routine is designed to help us all get better at understanding other persons' perspectives beyond someone's own initial assumptions. In the final prompt, "step back," students may explore how prior knowledge, cultural, and/or linguistic perspectives can inform or obscure their interpretation(s). This routine lends itself to small groups. You may invite students to write their responses to each question individually on separate small pieces of paper (sticky squares can be helpful but aren't necessary) first and then share.

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project zero THINKING ROUTINES

Step Inside (Perceive, Know About, Care About)

A routine for getting inside viewpoints.

Three core questions guide students in this routine:

1. What can the person or thing perceive?
2. What might the person or thing know about or believe?
3. What might the person or thing care about?

PURPOSE

What kind of thinking does this routine encourage?
This routine helps students to explore different perspectives and viewpoints as they try to imagine things, events, problems, or issues differently. In some cases, this can lead to a more creative understanding of what is being studied. For instance, imagining oneself as the numerator in a fraction. In other settings, exploring different viewpoints can open up possibilities for further creative exploration. For example, following this activity, a student might write a poem from the perspective of a soldier's sword left on the battlefield.

APPLICATION

When and where can I use it?
This routine asks students to step inside the role of a character or object—from a picture they are looking at, a story they have read, an element in a work of art, a historical event being discussed, and so on—and to imagine themselves inside that point of view. Students are asked to then speak or write from that chosen point of view. This routine works well when you want students to open up their thinking and look at things differently. It can be used as an initial kind of problem-solving brainstorm that opens up a topic, issue, or item. It can also be used to help make abstract concepts, pictures, or events come more to life for students.

LAUNCH

What are some tips for starting and using this routine?
In getting started with the routine, the teacher might invite students to look at an image and ask them to generate a list of the various perspectives or points of view embodied in that picture. Students then choose a particular point of view to embody or talk from, saying what they perceive, know about, and care about. Sometimes students might state their perspective before talking. Other times, they may not and then the class could guess from which perspective they are speaking.

In their speaking and writing, students may well go beyond these starter questions. Encourage them to take on the character of the thing they have chosen and talk about what they are experiencing. Students can improvise a brief spoken or written monologue, taking on this point of view, or students can work in pairs with each student asking questions that help their partner stay in character and draw out his or her point of view.

This routine is adapted from Art Works for Schools: A Curriculum for Teaching Thinking In and Through the Arts (2002) DeCordova Museum and Sculpture Park, the President and Fellows of Harvard College and the Underground Railway Theater

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Project Zero adapted this routine from from Art Works for Schools: A Curriculum for Teaching Thinking In and Through the Arts (2002) DeCordova Museum and Sculpture Park, the President and Fellows of Harvard College and the Underground Railway Theater.

Sticking Points

A routine for mapping messy controversies.

Choose a big issue and look at these four facets...

Facts: What facts do people differ on? What facts do they agree on?

Values: What values do people differ on? What values do they agree on?

Interests: What practical interests (e.g., investments, land, group loyalty) do people differ on? What practical interests do they share?

Policies: What policies (i.e., general actions to take) do people differ on? What policies do they agree on?

PURPOSE

What kind of thinking does this routine encourage?
It's to help learners understand today's complicated world by digging into the big controversies so common today. Straightforward agreement is often difficult, but learners can get a grip on complicated issues through appreciating the sources of different viewpoints. Over time, students can come to some insights about controversies in general.

APPLICATION

When and where can I use it?
ILead a class discussion. Students might get ready with small group discussions beforehand. As students talk, make their thinking visible with a concept map on the whiteboard, or have your students post Post-its. Facts-Values-InterestsPolicies is a good order but use any order for your purposes. Expect to spend 20 minutes to an hour or more, depending on how deep you want the class to go. Sticking Points can apply to interpersonal conflicts and local community and school controversies. The routine can be used on past controversies and issues as well as those that are current. And, the routine can be used in many disciplines including history, social studies, science and the arts.

LAUNCH

What are some tips for starting and using this routine?

- *What if my students don't know that much about the issue?* You could have an initial conversation size that up, and make an assignment to investigate different viewpoints.

- *Controversies are about disagreement. Why does the routine also ask where people agree?* Because advocates tend to foreground the disagreements, often it seems there is disagreement on everything! It's good for learners to appreciate where the agreements are, because they are a starting point for resolving, or at least reducing, the controversy.

- *What about investigating how different groups have different viewpoints?* You could ask students to form clusters and chart the Facts-Values-Interests-Policies for different groups. A question prompt could be: How does the pattern of Facts-Values- Interests-Policies differ by different groups (e.g. scientists, people from different political parties, business people, workers, different nationalities, etc.)?

- *Does the class or group have to agree on a resolution of a controversy?* No, that would be asking too much. The main idea of using this routine is to help learners understand the controversy and appreciate different sides. However, ways forward can be explored using these prompts for follow-up conversations: How could progress be made without necessarily completely agreeing (e.g. people might try to settle some facts, find some common interests, arrive at a compromise policy)?

This thinking routine was developed as part of the PZ Connect project at Project Zero at the Harvard Graduate School of Education. Explore the full PZ Thinking Routine Toolbox at *pz.harvard.edu/thinking-routines*.

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project zero THINKING ROUTINES

Stop, Look, Listen

A routine for clarifying claims and sources.

The routine follows a simple 3-step structure:

Stop: Be clear about the claim. Define your question from your list of facts and uncertainties.

Look: Find your sources. Where will you look? Consider obvious and non-obvious places.

Listen: Hear what the sources tell you with an open mind. Is it possible for your source to be biased and how does it affect your information?

PURPOSE

What kind of thinking does this routine encourage?
The Stop, Look, Listen routine helps students investigate truth claims and issues related to truth. It allows students to stand back and think about ways to obtain information when trying to find out about the truth of something. Students are encouraged to think critically about sources. This process helps students appreciate the deeper complexity of truth situations by addressing issues of bias and objectivity.

APPLICATION

When and where can I use it?
This routine invites students to think carefully about the process of initiating truth investigations. It can be used in any situation in which students need to find out more about a truth claim. It helps students step back and take a questioning stance in order to clarify a claim. Use the routine when you want students to be open minded and to think broadly about sources of information.

LAUNCH

What are some tips for starting and using this routine?
Begin by helping students to pin down a claim about a topic. Students may have a good idea about a question or claim they would like to investigate. Help them take stock of what they know by creating a list of facts and uncertainties around their claim. Students may need to redefine or restate their claim.

Once a clear claim has been identified, ask students what they can do to investigate it. Brainstorm source by encouraging students to think broadly about different kinds of information available to them. Consider having students make a mind map of sources that can provide information about their claim. Then, document students' ideas by creating a chart of identified sources, the perspectives of the source and potential biases they might represent. Display the chart on the wall and add comments as each source is investigated. Keep this chart accessible so students can return to it during the investigation of future truth claims. Track the instances or types of bias that students identify and use it as a way to further conversations about new situations in the classroom.

FACILITATION

This activity is recommended for the following learner age ranges:

- Elementary School
  - 30-60 minutes
  - Heavily facilitated by peer/teacher/caregiver
- Middle School
  - 30-60 minutes
  - Heavily facilitated by peer/teacher/caregiver
- High School
  - 30-60 minutes
  - Lightly facilitated by peer/teacher/caregiver

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project zero THINKING ROUTINES

Stories

A routine for uncovering accounts of complex issues.

Consider how accounts of issues, events, people, society, etc. are presented; what has been left out, and how might you want to present the account.

What is the **story** that is presented?
What is the account that is told?

What is the **untold** story?
What is left out in the account? What other angles are missing in the account?

What is ***your*** story?
What is the account that you think should be the one told?

PURPOSE

What kind of thinking does this routine encourage?
This thinking routine invites learners to (1) explore how accounts of issues, events, people, society, etc. are presented; (2) tease apart and rethink the various angles, dimensions and scope of accounts; (3) consider what has been left out in the account; and (4) take a stand on the kind of account that they would want to hear.

APPLICATION

When and where can I use it?
It is advisable to use the routine's prompts in the order they are presented, as the sequence is designed to support learners in uncovering how narratives are presented and how they could be re-thought. You may want to use additional prompts to target specific goals, such as:

- To move from a local to a global perspective: *"What is a **bigger** story that this story could be connected or related to?"*
- To think about what an individual's account could be: *"What is a **smaller** story that this story could be connected or related to?"*
- To consider how different disciplinary experts might explain or describe the issue, event, people, society, etc.: *"What is the **scientific/historical/literary/etc.** story?"*

You may also consider framing the prompt *"What is **your** story?"* as *"What is **his/her** story? What is **our** story?"* when you want learners to explore multiple perspectives on the issue, event, people, society, etc., and begin to craft a collective account.

LAUNCH

What are some tips for starting and using this routine?

- The prompt "What is the **story**?" invites learners to understand how an issue, event, people, society, etc., is presented. If the material used with the thinking routine presents more than one perspective, consider dividing the learners into groups and assigning them a specific perspective. That way, these different "stories" will flow naturally into the next prompt, where different angles and perspectives are brought into the mix.
- The prompt "What is the **untold** story?" invites learners to explore the multiple angles, perspectives and dimensions in complex issues, events, people, society, etc., as well as hypothesize the motivations for the account: whose account is this? What has been left out? What has been emphasized? What was made less significant? Who does this account advantage and disadvantage? In what way is it easy to not pay attention to? Unpack it for learners by asking them: what might people not want us to know about the issue, event, people, society, etc., or what do people not pay attention to or don't think about?
- The prompt "What is ***your*** story?" invites learners to bring their own perspective and aspirations into the topic or issue. Unpack the prompt by asking learners: How would you rewrite this story? What perspective would you take and why? How would the story ideally be like 'in a different world'? While it is not intended to be an invitation for learners to relate their own experiences, the prompt is still roomy enough for learners to use their lived experiences as a basis for what they hope to see happen with the topic or issue, or why their aspiration matters.

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project zero THINKING ROUTINES

Take a Stand

A routine for scaffolding perspective-taking.

Choose a digital dilemma to analyze. (If you are choosing a non-PZ developed digital dilemma, consider the important criteria for dilemmas to work well with this routine.)

Take a Stand
Individually, read or listen to the dilemma and consider your thoughts about it. What's your view on the dilemma? Make notes that help explain your perspective.

Stand Back
In small groups or as a class, share your individual perspective(s) about the dilemma and listen to the perspectives or your fellow students.

Look Again
Individually, look again at your original response. What had you not considered that other people brought up? Have you changed your perspective, and if so, what changed? How has your thinking shifted after hearing your classmates' perspectives— even if you haven't changed your mind?

Look Beyond
Finally, look beyond this specific dilemma or case. Consider: How does the dilemma remind you of other situations explored in class or that you've seen, heard about, or experienced?

Optional: **Complicate**
If using a dilemma in our compilation, consider the provided prompts that "complicate" the scenario. Or, if you used your own dilemma, consider how your response might change if certain details in the dilemma shifted (e.g., who was involved, the relationships between the people involved, when /where it happened, etc.) Discuss these questions in your group and share responses.

PURPOSE

What kind of thinking does this routine encourage?
This thinking routine scaffolds perspective taking especially in situations and dilemmas that lack clear right or wrong answers, especially those involving privacy, community, and civic life. By using this routine repeatedly, students develop the thinking dispositions to: 1) slow down and self-reflect, and 2) explore and engage with others' perspectives, 3) seek and evaluate evidence, and 4) envision options and potential impacts. The routine supports students to engage in the habit of considering/reconsidering their own and others' perspectives.

APPLICATION

What are some tips for starting and using this routine?
Choose a digital dilemma either from the PZ and Common Sense-developed dilemmas here, pull a relevant story from the news/current events, or develop your own dilemma that captures a tension you want to explore. Print copies of the "Take a Stand" handout or provide students with a link to the electronic version of the template. Feel free to revise, reword, or adjust this activity to make it work for your class. You know your students best! The four-step routine helps students think through the dilemma and reflect on their own views and values as they listen to perspectives that differ from their own.

This activity can be used anywhere educators see a connection to their learning goals, especially where students can discuss or debate an issue about digital life with each other. For example, in: advisory period, where students are learning social-emotional learning (SEL) or character education skills. library or media class, where students are learning about digital citizenship and technology.

Social studies, particularly related to timely civic and political issues, free speech, and constitutional protections (e.g., in U.S. contexts, the First Amendment). English language arts (ELA), particularly relevant to developing students' skills for argumentative or persuasive essay writing. student council or peer leadership, where students are exploring life decisions and being a good leader.

Students find more meaning in the Take a Stand activity—and take it more seriously—when teachers make explicit connections to class curriculum and learning goals. These connections might be easy to find when the dilemma relates to the class material (e.g., talking about the protest dilemma in a civics course). If not, name other connections to course topics and learning goals.

LAUNCH

When and where can I use it?
This activity can be done in varied amounts of time, and it often expands to fill whatever time teachers give it. You can get through the routine in 30 minutes, or allow the discussion to expand for a longer block. This activity works best with a whole class, whether in the classroom or using synchronous video instruction in distance learning.

As you plan for discussions of dilemmas, it's important to remember that these discussions will surface disagreements that provide important learning opportunities but also can trigger discomfort. It's valuable to know that you aren't looking for "right" or "wrong" answers. While acknowledging that no set of norms will keep the conversation safe for everyone, it's also helpful to establish or revisit classroom norms before you dive in.

Consider the following suggested classroom norms (informed by norms from Facing History and Ourselves):

- Listen for understanding. Try to understand what someone is saying before rushing to judgment.
- Make comments using "I" statements.
- Think with your head and your heart.
- Acknowledge the difference between intention and impact.

If one student's perspective on a dilemma is not shared by most of the class, be alert that they might feel understandably vulnerable in their stance. It can also be challenging if a student feels personally attacked by other students' perspectives. Think about your classroom climate and your student group when deciding how to modify the routine and/or which dilemmas to cover. If you're looking for more support, consider these two resources: Courageous Conversations, Cult of Pedagogy or Contracting, Facing History and Ourselves.

As you consider how to facilitate this thinking routine and the discussion, first, choose a digital dilemma that you want students to analyze. Be sure to read the Keys for Using Dilemmas to incorporate these strategies into the lesson. Make a copy of the Take a Stand student handout. Copy and paste the dilemma and the "complicated questions" into the worksheet. Use the teacher guidance below, which mirrors the sections in the student handout, to support students through each step of the thinking routine. Note that some steps are done individually and others are done as a class. To provide educators with additional tips for using the routine, we have created a teacher's guide to the "Take a Stand" handout.

This thinking routine was developed as part of the Digital Dilemmas project at Project Zero at the Harvard Graduate School of Education. Explore the full PZ Thinking Routine Toolbox at *pz.harvard.edu/thinking-routines*.

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project zero THINKING ROUTINES

Take Note

A routine for distilling ideas and identifying emerging questions.

After a lecture, film, reading, or discussion learners "take note" of **one** of the following:

- What is the most important point?
- What are you finding challenging, puzzling or difficult to understand?
- What question would you most like to discuss?
- What is something you found interesting?

PURPOSE

What kind of thinking does this routine encourage?
This routine can be used to enhance students' memory of and engagement with ideas by focusing on capturing the heart and distilling key issues and questions after a learning episode rather than in the midst of it. This allows them to participate fully knowing that there are times to consolidate their learning afterwards.

APPLICATION

When and where can I use it?
This routine can be used after any episode of learning, whether from a discussion, video, lecture, or reading. The routine may be done at the end of class or at regular intervals (every 10-15 minutes) during the class. It can also be done after an independent reading. Another alternative use is as an "exit ticket" strategy in which students make their note on an index card at the end of class and turn it in to teacher upon exiting the room.

LAUNCH

What are some tips for starting and using this routine?
Explaining the psychology behind this routine can help students learn how to learn. The routine is not simply an activity you are asking to students to do, but a tool designed to help us use out brains more effectively for learning. Explain to students that our learning and memory are enhanced by the regular distillation of key ideas and points as well as our identification of emerging questions and puzzles. Furthermore, the sharing of these ideas and questions supports the group's learning by facilitating continued exploration, discussion, and synthesis as well as providing the instructor with learning feedback. Encourage learners to engage actively in the learning episode without taking notes so as to be fully present. Some students who are used to spending their time in class may initially resist this call to engage. For these students, it may be useful to share some of the research on note taking and the value of engagement. You might also want to prepare notes in advance yourself, and tell students you will be sharing your notes of key points with them after class but the point now is to engage with the ideas and issues.

At regular intervals (if there is a lot of content) or at the end of the lesson, pass out index cards and ask each student to make note using any one of the above prompts. Have students record their thinking anonymously. This will take approximately 3-5 minutes. Whether done at intervals or at the end, there needs to be some kind of sharing of the notes. This could be done in a number of ways:

1. Have small groups share and discuss what they have written.
2. Have one group collect their index cards and pass them to another group. Upon receiving the new note cards, the cards are randomly distributed and each student reads and responds to the card they receive. Cards are then recollected and passed back to the group from which they came.
3. The teacher collects all note cards and redistributes them randomly. Students then read aloud the note card they receive. The teacher may document and organize the responses.
4. If using the exit ticket method, the teacher collects, reads, and summarizes the Take-Note cards as a form of formative assessment and begins the next class by sharing or in some way making use of what was shared.

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project zero THINKING ROUTINES

The 3 Whys

A routine for nurturing a disposition to discern the significance of a situation, topic, or issue keeping global, local, and personal connections in mind.

- Why might this (topic/question) matter to me?
- Why might it matter to people around me (family, friends, city, nation)?
- Why might it matter to the world?

PURPOSE

What kind of thinking does this routine encourage?
This routine nurtures a disposition to discern the significance of a situation, topic, or issue, keeping in mind global, local, and personal connections.

APPLICATION

When and where can I use it?
The routine can be applied to a broad range of topics, issues, and activities. You may use a powerful image, text, quote, video, or other materials to ground students' thinking. You may find this routine useful early in a unit after the initial introduction of a topic, when you want students to consider carefully why it might be worth investigating further. Teachers have also used this routine to expand on a given issue in order to help students become aware of that issue's far-ranging impact and consequences at the local and global levels. In other cases, the routine is used to create a personal connection to a topic that might initially seem remote.

LAUNCH

What are some tips for starting and using this routine?
Ensure that the students have clarity about the focal point of the analysis. For example, you might ask your students, "Why might understanding social inequality matter to our community?" as opposed to, "Why might this image matter?" Use the questions in the order proposed or in reverse order, beginning with whichever one you think will be the most accessible entry point. For instance, students might identify and articulate the purpose and significance of a story they are reading by first reflecting about why the story matters to them, and then moving on to the questions about their community and the world. In other cases, a teacher may seek to construct a more personal connection to a distant event, thus beginning with the world, then working inward. Invite students to work on one step at a time. If they try to work on all three questions at once, nuances and distinctions between the personal, local, and global may be lost. If time allows, compare and group students' thoughts to find shared motivations and rationales for learning the topic under study.

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project zero THINKING ROUTINES

The 4 Cs

A routine for structuring a text-based discussion.

Connections
What connections do you draw between the text and your own life or your other learning?

Challenge
What ideas, positions, or assumptions do you want to challenge or argue with in the text?

Concepts
What key concepts or ideas do you think are important and worth holding on to from the text?

Changes
What changes in attitudes, thinking, or action are suggested by the text, either for you or others?

PURPOSE

What kind of thinking does this routine encourage?
The routine provides learners with a structure for a text-based discussion built around making connections, asking questions, identifying key ideas, and considering application.

APPLICATION

When and where can I use it?
After reading a text, or even several different but related texts, small groups can use this routine to discuss the text and explore the application of ideas. The individual questions for each of the Cs may be adjusted to fit the needs of the group and the text being read. Sometimes you may want to focus on a just 2 or 3 Cs rather than all 4.

LAUNCH

What are some tips for starting and using this routine?
This routine works best with small groups. Before beginning discussion, give group members time to identify passages from the text that correspond to each of the Cs. These should be underlined or in some way identified so that they can later be shared with the group. In the group, discussion begins by one person sharing a connection that he or she made, reading the text passage and explaining the connection. Members of the group may add their comments on the offered passage. The next member of the group then shares his or her passage and the discussion continues in the matter until each member has shared a "connection." The group then moves on to the next "C," always making sure that the chosen text passage is first shared and then discussed. You may have groups record the text passages to document the discussions.

This thinking routine was developed as part of the Cultures of Thinking project at Project Zero at the Harvard Graduate School of Education. Explore the full PZ Thinking Routine Toolbox at *pz.harvard.edu/thinking-routines*.

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project zero THINKING ROUTINES

The Complexity Scale

A routine for unpacking complexity.

1. Say some broad things you know about a topic—observations, facts, ideas. Think of different kinds of things.
2. Place each statement somewhere on the complexity scale.

Simple ——————————————————————— **Complex**

3. Explain why you want to place it there.
4. Reflect: What new insights and questions do you have about the topic?

PURPOSE

What kind of thinking does this routine encourage?
This routine helps students build a more multi-dimensional mental model of a topic by identifying different aspects of the topic and considering their complexity. The benefit of the routine consists mainly in the reasoning students do in order to choose and explain their ratings. Of less importance is assigning each idea to the "right" place on the scale.

APPLICATION

When and where can I use it?

Use the Complexity Scale when you have been studying a topic for a while and students know something about it. But don't wait until the end of the topic to use the routine. The Complexity Scale is a good way to surface students' conceptions so you can get a sense of how they understand the topic in its entirety.

LAUNCH

What are some tips for starting and using this routine?

Put up a scale with simple on one end and complex on the other. Have students write statements on post it notes about what they know about a topic. Then, have them place statements along the continuum. Feel free to discuss placement. You can even place a statement in more than one spot on the line (sometimes things are simple in one way, but complex in another).

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project zero THINKING ROUTINES

The Explanation Game

A routine for exploring causal understanding.

The routine focuses first on identifying something interesting about an object or idea:

I *notice that...*

And then following that observation with the question:

Why is it that way? or *Why did it happen that way?*

PURPOSE

What kind of thinking does this routine encourage?
This is a routine for understanding why something is the way it is. This routine can get at either causal explanation or explanation in terms of purposes or both.

APPLICATION

When and where can I use it?
You can apply it to almost anything: a pencil, cell phones, forms of government, historical documents, or events. Students can work in pairs or groups of larger size, even as a whole class. The explanation game can also be used solo. The first time the routine is used, the teacher may need to take an active role in scaffolding the conversation and modeling how to ask questions of explanation and clarification of others. Over time, students can begin to emulate the conversational moves and questioning they have seen modeled.

LAUNCH

What are some tips for starting and using this routine?
Begin with something "on the table"—an object like a cup or a compass; a document like a poem, or a picture; a historical event; a scientific theory, etc. The first person (this might be the teacher initially) points out an interesting feature of the object: "I notice that... That's interesting. Why is it that way?" or "Why did it happen that way?" (or some similar why question). The other people in the group try to answer the question or at least to propose possible explanations and reasons. As these students share their ideas, the person asking the original question follows up by asking, "What makes you think so?" The group works together to build explanations rather than merely deferring to an outside source, the teacher, or a textbook to provide an answer.

Student questions and explanations become visible to the class as they are shared. Responses to the routine can also be written down and recorded so that there is a class list of evolving ideas. A few key issues or puzzles might then be chosen for further investigations. A conversation could also be recorded as a chart with four columns representing the key structures of the conversation:

1. The Observation that is initially made
2. The Question that comes out of that observation
3. The various Explanations/Hypotheses that the rest of group puts forth
4. The Reasons/Justifications that are given in support of the explanations

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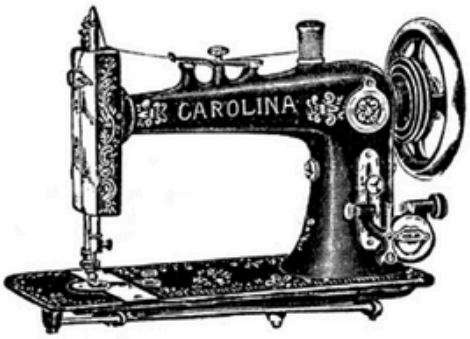
project zero THINKING ROUTINES

Think, Feel, Care

A routine to foster perspective taking within a system.



Step inside a system:

Choose a variety of people within a system and then step inside each person's point of view. As you think about what you know about the system, consider what each person might think, feel, and care about:

Think: How does this person understand this system and their role within it?

Feel: What is this person's emotional response to the system and to their position within it?

Care: What are this person's values, priorities, or motivations with regard to the system? What is important to this person?

PURPOSE

What kind of thinking does this routine encourage?
This routine encourages learners to consider the different and diverse perspectives held by the various people who interact within a particular system. The goal of this routine is to help learners understand that the variety of people who participate in a system think, feel, and care differently about things based on their positions in the system. This routine fosters perspective taking, raises questions, and surfaces areas for further inquiry.

APPLICATION

When and where can I use it?
This thinking routine can be used to explore the perspective of any person within a particular system. This routine can be used on its own, or in combination with another routine.

Here are some ideas and considerations for putting this thinking routine into practice:

- Working individually or in small groups, it may be helpful for you to have students sketch out a small monologue or scene that contains some of the different people who participate in a particular system. They can then assume the role of various people in their system, and act out the scene, with each student portraying a different person's perspective.

- Once students portray a person in their system in one way, ask them how they might portray the same person in an entirely different way. This will prompt your students to understand that even within particular groups of people, there is no one perspective, but rather an array of perspectives that different and unique people may hold.

- Learners should be encouraged to consider how what people think, feel, and care about may be in alignment within a particular system, or misaligned. When misalignments emerge, ask your students how these tensions are dealt with or negotiated within the system? Discussions about unequal power structures within a system may arise.

- While this routine asks learners to step inside the role of a character and to imagine how they may think, feel, and what they might care about from that point of view, it is important to remember that students can never really know and understand someone else's perspective. When engaging in this thinking routine, it is important for students to push beyond stereotypes and to try to imagine the lived experiences of particular people. Encourage your students to develop specific people to play (e.g., Julia, a migrant worker, John, a used car salesman, and Martin, a Republican senator) as opposed to types of people (e.g., a migrant worker, a used car salesman, and a Republican senator).

- When perspective taking, learners will likely draw on assumptions about the types of people represented in their system. As they do so, you may lead students in a discussion that addresses where these assumptions come from. You can encourage students to challenge their assumptions by asking them what they really know about someone else's perspective, and what they may need to do (e.g., conduct interviews, speak with a grandparent, etc.) in order to find out about someone else's perspective.

FACILITATION

This activity is recommended for the following learner age ranges:

- Preschool
  - 15-30 minutes
  - Heavily facilitated by peer/teacher/caregiver
- Elementary School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver
- Middle School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver
- High School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver

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project zero THINKING ROUTINES

Think, Feel, Care (For Younger Children)

Exploring complexity.

Choose a system that involves several people. Select one person from the system and ask:

What does this person do in the system? How do you know?

If you were this person, how would you feel? Why would you feel that way?

If you were this person, what would you care about? What would be important to you?

To help children think about different perspectives, choose a different person from the system and ask the questions again.

PURPOSE

What kind of thinking does this routine encourage?

This routine encourages students to consider the diverse perspectives that different people within a particular system may have based on their role in the system. This routine fosters perspective taking and can help children generate new questions and/or ideas about the system, how it works, and how it might be improved.

APPLICATION

When and where can I use it?

You can use this thinking routine to explore the perspectives of anyone within a system. You may choose to introduce the entire routine at once, or you may choose to break it down into parts over multiple occasions. Though you can use this thinking routine on its own, we suggest you use it in combination with other Agency by Design thinking routines that encourage students to look closely and carefully explore the parts and purposes of the system they are working with.

- If you are using this thinking routine with young learners, consider having them think about asystem they have direct experience with. This way, all children can apply and build upon theirprior knowledge in meaningful ways. We suggest you start with systems in your classroom or community that students use regularly.
- You may also use this routine to explore the interactions presented in a fictional story. Children can explore the different perspectives of the characters based on their different roles in the story.
- Consider having children engage in this routine through role play, with each student portraying adifferent person in the system. Once students portray someone one way, have them try to portraythe same person in a different way. This can help children understand that, even within particular-groups of people, there is no one set perspective, but rather an array of possible perspectives.
- Students will likely draw on their assumptions about the various people in their system. Considerdiscussing where these assumptions come from. You can encourage students to challenge theirassumptions by asking them what they really know about someone else's perspective, and whatthey could do in order to find out more about someone else's perspective. For example, theycould conduct interviews with someone in that role, seek out more information using the internetor books, or even find ways to experience that role themselves.
- Encourage students to notice that not everyone in a system feels the same way or cares about thesame things. Have them talk about this. Why might different people have different perspectives?What problems might arise because of this? What might be some benefits of this?

FACILITATION

This activity is recommended for the following learner age ranges:

- Preschool
  - 15-30 minutes
  - Heavily facilitated by peer/teacher/caregiver
- Elementary, Middle, and High School
  - 15-30 minutes
  - Lightly facilitated by peer/teacher/caregiver

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project zero THINKING ROUTINES

Think, Pair, Share

A routine for active reasoning.

Pose a question to students. Give students a few minutes to **think**.
Invite students to **pair** with a nearby student to **share** their thoughts.

PURPOSE

What kind of thinking does this routine encourage?
This routine promotes understanding through active reasoning and explanation. Because students listen to and share ideas with others, it also encourages students to understand multiple perspectives.

APPLICATION

When and where can I use it?
This routine can be used when it would help for students to process their thinking aloud with another student. For example, you may ask students to Think, Pair, Share before starting a science experiment, in the middle of solving a math problem, after reading a passage of a book, etc. Sharing can also be done in small groups.

LAUNCH

What are some tips for starting and using this routine?
When first introducing the routine, you may want to remind students to take turns, listen carefully, and ask clarifying questions of each other. One way to encourage students to listen actively to each other is to tell them that when they have completed their conversations, you will ask some students to explain their partner's thinking. Encourage students to make their thinking visible by asking them to write or draw their ideas before and/or after sharing (perhaps in a journal). You may also want to document students' ideas and display them in the learning space. Sometimes it is useful to have pairs or groups summarize their ideas for the whole class.

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project zero THINKING ROUTINES

Think, Puzzle, Explore

A routine that sets the stage for deeper inquiry.

- What do you think you know about this topic?
- What questions or puzzles do you have about this topic?
- How might you explore your puzzles about this topic?

PURPOSE

What kind of thinking does this routine encourage?
This routine activates prior knowledge, generates ideas and curiosity, and prepares students for deeper inquiry.

APPLICATION

When and where can I use it?
This routine works especially well when introducing a new topic, concept, or theme in the classroom. It helps students recall what they already think they know about the topic and then invites them to identify puzzling questions or areas of interest to pursue. This routine can help you gauge students' current understanding of a topic and inform your subsequent lesson planning. You might return to this thinking routine throughout the study of a topic to surface changes in student conceptions.

LAUNCH

What are some tips for starting and using this routine?
With each question in the routine, give students adequate time to think about and identify their ideas. You may want to have students write down their individual responses to the routine before sharing them with others. Students could document their responses in a journal.

The routine can be used to engage a class at the beginning of a new topic or unit. At first, students may list seemingly simplistic ideas and questions. It also is common for students to have misconceptions about the topic. As you're documenting students' responses, be sure to list these initial conceptions so all ideas are available for reconsideration after further study. Encourage students to think about things that are truly puzzling or interesting to them.

At times, you may want to use some of the questions in the routine by themselves. For example, you can use the think section of the routine at the end of a unit as a reflective tool. You may also want to use the puzzle section of the routine to remind students that there will always be more to learn about a topic and that learning is a continual process.

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Thinking with Images

A routine for uncovering relational patterns.

Pick a topic/issue/problem. Then consider the topic/issue/problem in light of the following prompts:

How is this *like*...? How is this *not like*...?

PURPOSE

What kind of thinking does this routine encourage?
This thinking routine invites learners to use analogical thinking to reason about topics/issues/problems by examining the relations that hold within topic/issue/problem and each image.

APPLICATION

When and where can I use it?
This routine sets up the conditions for learners to (1) identify systematic correspondences between the topic/issue/problem and each image; (2) draw inferences about each image in relation to the topic/issue/problem; (3) evaluate the inferences for how they might be adapted to fit the qualities of each image; and (4) consider new ways of thinking about the topic/issue/problem using each image.

LAUNCH

What are some tips for starting and using this routine?

- While learners do not need any background knowledge to use this routine, you may want to allow learners some time to think about what they are seeing in each image. For instance, invite learners to quietly jot down their impressions of each image before providing the topic/issue/problem for them to explore. You may choose to have learners work with all four images (this usually supports them in seeing more and more complexity as they begin to compare and contrast the various images), or have them gravitate to the images that call their attention (this supports learners' agency as well as provides information on the images that work well for specific topics/issues/problems), or work with one image at a time (this allows learners to build on preceding ideas as well as systematically develop their thinking).

- How else can this routine be used? You may consider inviting learners to design or search for other images that they feel represent or describe their thinking about the topic/issue/problem. Also, having learners reflect on which images work better for the topic and which ones less so is a good way of clarifying and deepening their thinking. Another idea is to invite learners to respond to the images with other images of their choice. This nonverbal way of building on ideas can overcome learners' apprehensions about language, as well as open up a novel way of communicating ideas.

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project zero THINKING ROUTINES

True for Who?

A routine for exploring truth claims from different perspectives.

1. **Discuss.** What kind of situation was the claim made in? Who made it? What were people's interests and goals? What was at stake?

2. **Brainstorm.** Make a list of all the different points of view you could look at this claim from.

3. **Dramatize.** Choose a viewpoint to embody and imagine the stance a person from this viewpoint would be likely to take. Would he or she think the claim is true? False? Uncertain? Why? Go around in a circle and dramatically speak from the viewpoint. Say:

    - *My viewpoint is...*
    - *I think this claim is true/false/uncertain because...*
    - *What would convince me to change my mind is...*

4. **Stand back.** Step outside of the circle of viewpoints and take everything into account: What is your conclusion or stance? What new ideas or questions do you have?

PURPOSE

What kind of thinking does this routine encourage?
The True for Who routine helps students cast a wide net for facts and arguments by imagining how an issue looks from different points of view. The routine also helps students see how different viewpoints and situations might influence the stances people are likely to take.

APPLICATION

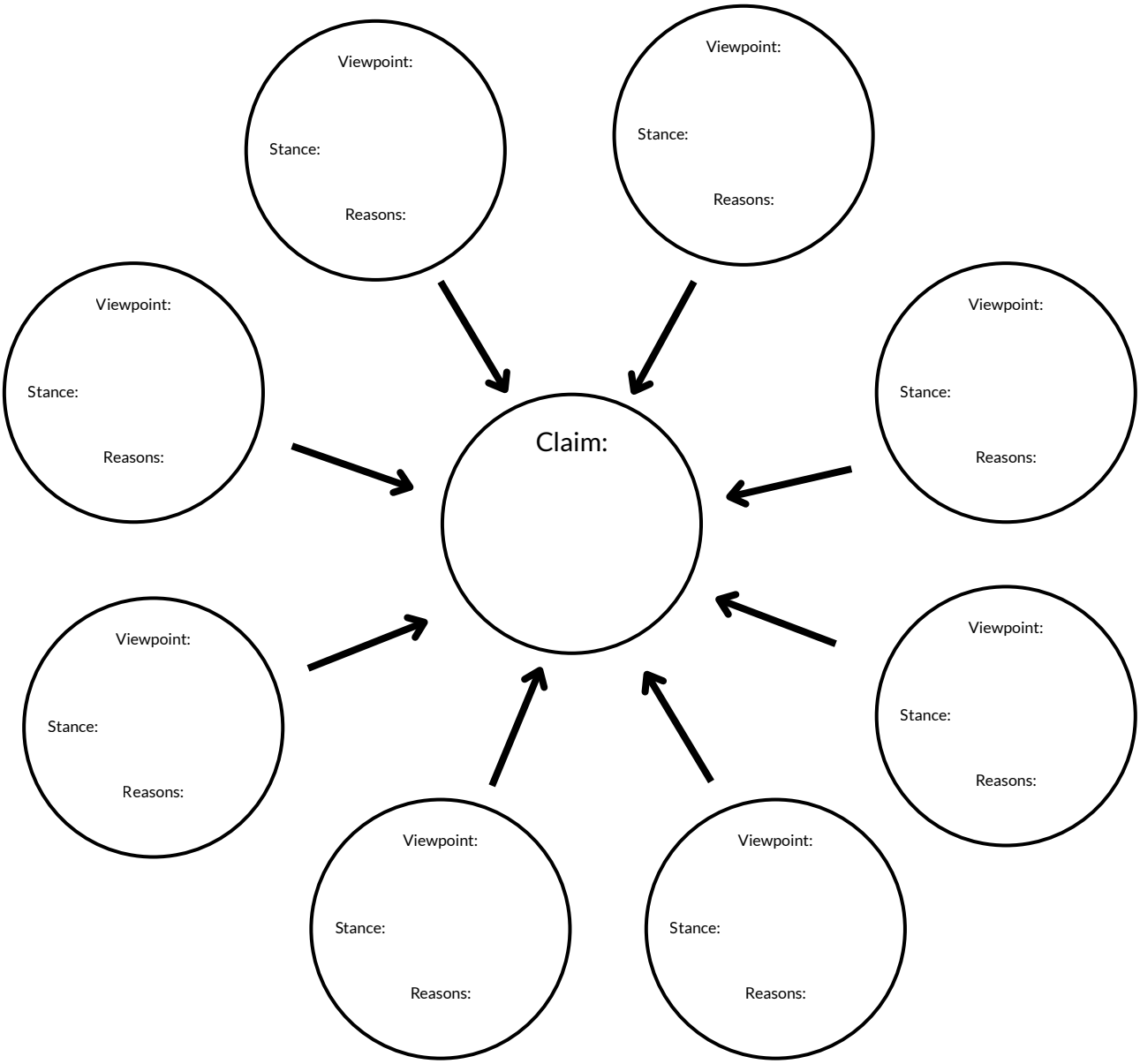
When and where can I use it?
What we think is true often depends on what we see and care about from our own perspective. Like the Circle of Viewpoints routine, this routine helps students consider the roles of context and perspective in shaping what people believe. It can be used at any point in the process of puzzling about truth, once the truth-claim has been clarified.

LAUNCH

What are some tips for starting and using this routine?
Begin the discussion by clarifying a claim and imagining various perspectives on the topic. After the brainstorm, ask each student to choose one of these viewpoints to embody. Give them time to prepare to speak about the topic from that perspective and to elaborate on the viewpoint using the three sentence stems to structure what he or she says.

Taking turns, students can go around the circle and speak briefly about their chosen viewpoint. The circle of viewpoints can be graphically documented on the board or on a poster using the formatted sheet on the next page. After many different viewpoints are dramatized, ask students to step out of their roleplaying and reflect on the issue. What do they think about the claim now? What are some questions about the claim now?

project zero THINKING ROUTINES



What is your conclusion or stance?

What new ideas or questions do you have?

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Tug for Truth

A routine for exploring tensions of truth.

1. Identify a question of truth—a controversial claim that something is true or false—where you know there is some evidence on both sides that students can bring forward.

2. Ask students if they have an opinion about it (it's okay not to have one).

3. Draw a tug of war diagram on the board (or tape a piece of rope on the wall and use Post-its to make it more dramatic). Explain that students can add two kinds of things. One is evidence — tugs in the Yes, True direction or the No, False direction. The other thing to add is a question about the tug of war itself, a question that asks for more information or about "what if" we tried this or we tried that, what would the results be?

4. Finish the lesson by asking students what new ideas they have about the question of truth. Can we decide now? Do some people lean one way and some the other? Is the best answer in a "gray area"—most of the time true but not always, or half the time? How could we settle it if we had to?

PURPOSE

What kind of thinking does this routine encourage?
It encourages students to reason carefully about the "pull" of various factors that are relevant to a question of truth. It also helps them appreciate the deeper complexity of matters of truth that can appear black and white on the surface.

APPLICATION

When and where can I use it?
It can be used in any situation in which questions of truth come up, and there is evidence to be considered, evidence from common knowledge or from information resources like newspapers or encyclopedias or the Internet. Questions of truth can come from school subjects or everyday life. Newspaper headlines are full of claims from politicians and others that can be evaluated. Science brings many issues like whether genetically engineered foods are dangerous or how old the universe is. History comes with endless controversies, for instance about who really started a particular war or what everyday life was like at various times in the past. Many works of literature create suspense by offering only bits and pieces of information until the end: Before the end, can you the reader figure out who the culprit really is or what secret from the past the heroine is hiding?

LAUNCH

What are some tips for starting and using this routine?

This is a routine that builds on children's familiarity with the game of Tug of War to help them understand the complex forces that "tug" at either side of a question of truth (there is also a Tug of War for fairness dilemmas with the same basic structure). The routine uses a rope or a diagram to represent pulls toward true or false in evaluating a claim. The tug of war is between True and False. Help students think about the various factors that tug at one side of the rope or the other, as well as other considerations related to the issue. A natural follow-up to the activity is to have students investigate facts related to the questions written above the Tug of War.

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project zero THINKING ROUTINES

Tug of War

A routine for exploring the complexity of fairness dilemmas.

1. Present a fairness dilemma.

2. Identify the factors that "pull" at each side of the dilemma. These are the two sides of the tug of war.

3. Ask students to think of "tugs", or reasons why they support a certain side of the dilemma. Ask them to try to think of reasons on the other side of the dilemma as well.

4. Generate "what if?" questions to explore the topic further.

PURPOSE

What kind of thinking does this routine encourage?
This routine builds on children's familiarity with the game of tug of war to help them understand the complex forces that "tug" at either side of a fairness dilemma. It encourages students to reason carefully about the "pull" of various factors that are relevant to a dilemma of fairness. It also helps them appreciate the deeper complexity of fairness situations that can appear black and white on the surface.

APPLICATION

When and where can I use it?
This routine can be used in any situation where the fairness dilemma seems to have two obvious and contrasting ways of being resolved. Dilemmas can come from school subjects or everyday life: testing of medicine on animals, adding people to a game once it has started, censoring a book in a library, and so on.

LAUNCH

What are some tips for starting and using this routine?
The routine works well as a whole class activity. Present the dilemma to the class. Draw or place a rope with the two ends representing the opposing sides of the dilemma and ask students to think about what side of the dilemma they would be on and why. Students can write their justifications on post-it notes. Encourage students to think of other reasons or "tugs" for both sides of the dilemma, and then have students add their post-it notes to the rope. Stand back and ask students to generate "What if's:" questions, issues, factors or concerns that might need to be explored further to resolve the issue. Write and post these above the rope. Finish the lesson by asking students to reflect on the activity. What new ideas they have about the dilemma? Do they still feel the same way about it? Have they made up minds or changed their minds?

The display of the tugs and What if's? on the rope helps to make students' thinking visible. Most importantly, their ideas are displayed in a way that shows their interconnectedness. The collaborative thinking process of the group as a whole is represented through the "action" of the tug of war. This is a key point about making thinking visible: It shows the dynamic interaction of people's thoughts in a context of a shared inquiry. Documenting thinking and making it visible in the classroom can facilitate this interaction in order to make the inquiry richer.

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Unveiling Stories

A routine for revealing multiple layers of meaning.

What's the story?

What is the human story?

What is the world story?

What is the new story?

What is the untold story?

PURPOSE

What kind of thinking does this routine encourage?
This routine helps students investigate the layers of meaning in images and texts and explore the human condition, systemic issues, and the power and limitations of representations in the news.

APPLICATION

When and where can I use it?
This routine can be used when introducing and/or exploring topics involving issues of local, national, and global importance. It can be useful in the arts, geography, literature, and history, especially when supporting students' disposition to look beyond the surface of a atopic, story, or image in order to uncover and explore complexity.

LAUNCH

What are some tips for starting and using this routine?
Depending on your goals, you may consider selecting just a few of the routine's questions.. You may also consider modifying the order in which the questions are introduced. In using this routine with your students, you may see "the story" interpreted in one of the following ways: 1) "the story" told by the article, image, or material that they read, or 2) "the story" proposed to explain or contextualize the event depicted, i.e. "the human story that led to the contamination of the Mexican gulf begins with our dependence on fossil fuels."

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project zero THINKING ROUTINES

Values, Identities, Actions

A routine for exploring some rich civic aspects of works of art.

Take some time to look closely at the work. What do you notice? Make lots of observations.

VALUES

What values does this work invite us to think about?
(Values are kinds of things that people value – fairness, justice, safety, respect, traditions, a nation or group a person belongs to, creativity, anything like that.)
Dig a little deeper with any of these questions: Are they your values? Others' values? Whose? Does the work affirm or challenge or raise puzzles about these values?

IDENTITIES

Who is this work speaking about? And who is this work trying to speak to? (Not necessarily the same people!)
Dig a little deeper with any of these questions:
Is anyone left out of the story that should be in it? Do you fit in or not so much for this story? Why?

ACTIONS

What actions might this work encourage?
(Actions could include doing something concrete, refraining from doing something, just learning more, etc.)
Dig a little deeper with any of these questions:
Who's actions – yours, others, what others? Why?

Look back at the work now. Spend a little time.

Do you get any ideas that might add to Values, Identities, or Action?

VIA is a relatively straightforward routine for exploring some rich civic aspects of works of art. It is quite easy to facilitate in most settings. Here are a few tips to make the experience a success.

BEFORE YOU BEGIN

A key aspect of a good VIA session concerns choice of the work or works. A fertile work for VIA has one or more fairly easily discernible civic themes, where matters of Values, Identities, and Actions invite comment.

Often, such works feature a deliberate civic statement of some sort ... but not always. The work might have been fashioned for other reasons altogether and still happen to invite civic connections. In other words, VIA is not necessarily an attempt to interpret the work as intended by an artist but an attempt to see the work through a civic lens.

You might select works in advance that you feel would support a good conversation. But also, it often plays well simply to say a few words about what makes for a good VIA conversation and turn participants loose in a gallery, or with a book of art, etc., to make their own choices. If students are having difficulty saying something about the art, you might use any of the Viewing Moves or Dialogue Moves to support their discussion.

FACILITATING THE ROUTINE

Often no direct facilitation is necessary. You can distribute the routine on paper and ask participants individually or in small groups to go through the steps. For a large group talking together, or perhaps for people trying VIA for the first time, it may make sense to lead the participants trough the process. This can be done with a handout or orally.

The top question under each of V, I, and A is the key question. Be sure participants get to that one. As to the “dig a little deeper” questions, there is no need to cover every single one. That’s a lot of conversation! You can pick and choose according to what explorations look most promising for the work of art in question and the participants.

You can make choices with the age of the participants in mind. For younger participants feel free to focus on participants questions and rephrase them if needed. However, be positive minded about what younger participants can tackle. They often have more to say than one might imagine about digging deeper questions. Participants using VIA individually or in small groups often explore VIA more richly if they expect a general conversation involving all participants at the end, where people briefly and informally report in headlines. You can facilitate that conversation.

This thinking routine was developed as part of the Arts as Civic Commons project at Project Zero at the Harvard Graduate School of Education. Explore the full PZ Thinking Routine Toolbox at *pz.harvard.edu/thinking-routines*.

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project zero THINKING ROUTINES

Walk the Week

A routine for connecting school topics to everyday life to broaden understanding beyond the classroom.

This thinking routine is aimed at heightening sensitivity of the many connections there are in our everyday lives with the topics we are studying at school, broadening our understanding of these topics and thinking far beyond them being only school subjects.

Focus on a topic you are learning about:

- **Log** some instances as ideas about this topic came up this week.
- How do these ideas **connect** with what you are learning about this topic?
- How can you **apply** what you have learned in this topic to other parts of your life?

PURPOSE

What kind of thinking does this routine encourage?
Too often, subjects and topics are learned at school to meet curriculum and assessment requirements and what happens outside of school is another part of life. When students can see connections between what they are learning at school with the world outside of school and vice versa, it can foster intrinsic motivation in understanding what they're learning.

LAUNCH

What are some tips for starting and using this routine?
How could I help my students see how this thinking routine can be of value to them? Choosing something that students can immediately relate to and modeling how it has helped can provide a concrete example.

For example, consider the science topic Energy, discussed as a property of objects that can be transferred to other objects, with explorations of the energy we use to keep warm or cool, read at night or on a cloudy day. During the week, what could students notice that connects to this? Could it be something to do with the new sleek windmills in fields just outside of town? On the news, there was talk about hydroelectricity. On the bus, I heard someone talking about solar energy.

Looking at Mum turning up the thermostat, I thought about our old log fire that we could watch as we warmed up in front of it. Advertisements in shop windows showed pictures of different foods, describing them as energy foods. The basketball coach talks about building up our energy. How do all these different things connect to what we are learning? Maybe thinking more about this will help me be fit, energetic and comfortable despite the weather.

Why is the Log step included as part of the structure of this routine? Away from school, it's easy to miss things that connect to a current school topic. The request to "log" some connections brings the topic to the forefront of the mind and heightens sensitivity.

When this happens more than once with the same topic, but through different sources, what may at first seem to be a novel experience evolves toward an instinctive routine: looking for possible connections. Logging, whether in a notebook or digital device, as the ideas come to mind or soon after, also builds a visual memory of something noticed that can be retained and referred to.

When the week is over? Invite students to share the instances they logged and the connections they made and how they have or could apply what they learned. After observations and experiences are shared and discussed, enable time for reflections on the topic, perhaps incorporating the thinking routine "I used to think...And now I think...

What is the broader impact of walk the week? In today's busy world, the week is often neatly parsed into separate parts. School days are usually structured around the timetable. After-school time gets shaped around family commitments, homework and extracurricular activities, weekends around family matters, social engagements, time and sporting or club meetings or practices. Getting into the habit of holding onto current school topics while going about the rest of your week can broaden understandings and surface examples of how what's being learned applies to day to day living.

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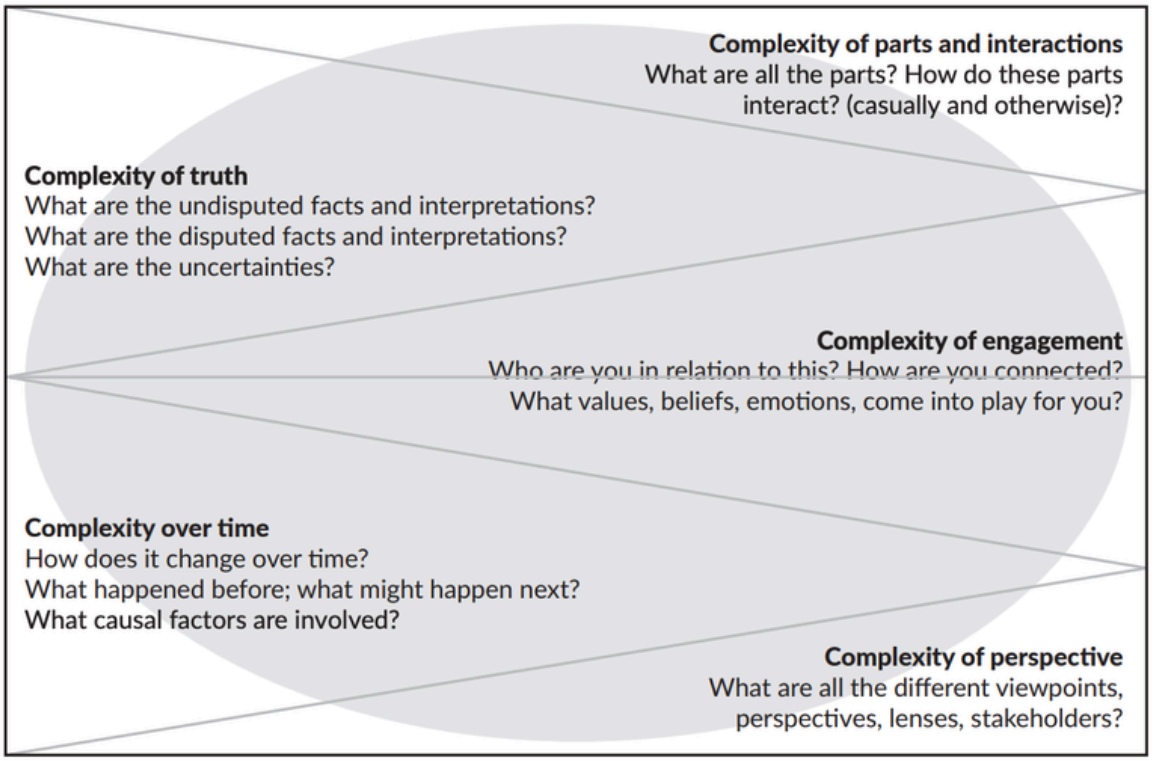
project zero THINKING ROUTINES

Ways Things Can Be Complex

A guide for organizing one's understanding of a topic through concept mapping.



Things can be complex in many ways. In fact, that is often what makes them complex. The diagram on the next page identifies several different types of complexity. Each type is accompanied by a few questions that help point students toward the kind of complexity in question. Not all the categories apply equally well to a single topic. For example, a bicycle can be viewed as a complex object: it has many interacting parts, it may change over time, and so on. But it isn't really useful to consider the complexity of truth related to a bicycle. A bicycle just is. That is unless you are examining the environmental friendliness of bicycle manufacturing, in which case truth may well be an issue.

The diagram can be useful in several ways:

- Use it to plan an introductory lesson on complexity. Choose a topic or object. Tell or show it to students, and show them the Ways Things can be Complex chart. Ask students to consider each of the categories and explore which kinds of complexity can be uncovered in the topic.
- Use the chart to expand your own ideas about complexity.
- Use it to help you think about which types of complexity you particularly want to emphasize in relation to particular topics.
- Use the chart to identify and gauge the quality of students' thinking: Which kinds of complexity are they exploring? How deeply are they exploring them?

This thinking routine was developed by Project Zero at the Harvard Graduate School of Education as part of the Visible Thinking Initiative – Exploring Complexity project with generous support from Independent Schools Victoria (AU). Explore the full PZ Thinking Routine Toolbox at *pz.harvard.edu/thinking-routines*.

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What Can Be

A routine for considering how complexities unfold.

Choose something (there are endless possibilities, from objects to systems, human-made or natural, ideas or issues, people and societies or narratives) that affect us and has changed in many ways.

Review How did it get to be the way it is now? Who/what might have caused or influenced these changes?

Predict How else might this change in the future?

Imagine and Create Change always comes with challenges. If you could turn the challenges of these predicted changes into opportunities, what do you imagine could be? What could be created?

Consider how things that have been the same or similar for a long time are changing rapidly in our complex world and despite the unknown, how might we plan to positively shape changes ahead?

PURPOSE

What kind of thinking does this routine encourage?
The only constant in our complex world is change. Change can elicit uncertainty and fear, yet change can also provide new opportunities. This routine is designed to encourage exploration of possibilities and to build agency in imagining and creating how things could change and what can be.

APPLICATION

When and where can I use it?
Objects and situations can be accepted as is, status quo. But how else could they be? The "What Can Be" routine is designed to encourage the freedom to imagine possibilities. The creative process can involve solving a problem, creating an alternative, making something new or an exploration of options. These learning opportunities can include using found or available materials that could be discarded, and/or specifically selected materials, and time given for imagining how else they could be, to dream, plan and design. New or adapted objects and/or systems can be created individually, in groups or as whole class activities.

LAUNCH

What are some tips for starting and using this routine?

- The Review step (seeking to identify how today's objects, ideas, systems and narratives came about in the first place, and what caused changes along the way) can help develop understanding of why beliefs that have been held and objects created have changed. This in turn reveals how they might be developed further or new creations built on what has been learned from past experiences. Why are schools in so many ways the same as when our grandparents were at school? What changes could be made in schools for a better fit for growing up in our changing world?

- Students may need to be encouraged to take risks with this thinking routine experience. History provides many examples of people trying something differently and persevering when things went awry. A fun and engaging way to encourage students to think about what can be could be to look at accidental discoveries.

- *How can predictions be made when so much is unknown?* The word "predictions" encourages safe risk taking. It is not expressing certainty of what will be, rather forecasts, likelihoods and projections, and therefore worth exploring what could happen with the aim of providing positive options ahead, or, if predictions are not so positive, giving time to seek alternative ways ahead.

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What Makes You Say That? (Adapted)

A routine for observation, description, explanation-building, and evidence-based reasoning.

- What's going on?
- What do you see that makes you say that?

PURPOSE

What kind of thinking does this routine encourage?
This routine cultivates observation, description, explanation-building, and evidence-based reasoning. Because students share their interpretations, they are encouraged to see multiple perspectives.

APPLICATION

When and where can I use it?
Use this routine when you want students to look closely at something and uncover their reasoning about the way it works, how it came to be, or why it is the way it is. For example, you can use this routine when students are looking at works of art or historical artifacts, reading poetry, making scientific observations and hypotheses, or delving into broad conceptual topics like culture, change, or creativity. It also is useful for gathering information on students' prior understandings when introducing a new topic.

LAUNCH

What are some tips for starting and using this routine?
To use this routine effectively, it is important to pay attention to the flow of classroom discussion, considering where in that flow the routine could help students expound on their reasoning. For example, listen for students giving opinions, making assumptions, or offering interpretations about a topic and consider inserting the question, What makes you say that? As you are planning lessons, you also may be able to anticipate the moments when using this routine will deepen the discussion and help students explore what's underneath their thoughts and assertions.

Asking *What makes you say that?* models for students the type of thinking that you want them to develop. Over time, students may internalize the routine and begin to support their interpretations with evidence without being asked.

Document student responses in a way that maintains the flow of the conversation. Consider recording class discussions, keeping an ongoing list of explanations, or asking students to document their thinking through sketches, drawings, models, or writing. This documentation can be displayed and revisited in the classroom, helping you as well as the students to see the development of their thinking.

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Who Am I?

A routine to explore the complexity of identity.

Think about who you are and then about someone else. Consider how you have become who you are, where you belong and what that can mean in our changing world.

Explore
Who am I? How has my identity developed?

Connect
I am connected to my parents, their parents and my brother and sister and I'm in the basketball team. Who else and what else am I connected to?

Identify
If I wanted others to know who I am, what would identify me? Do we have more than one identity?

Belong
Where do I think I belong? Do I have a sense of belonging to more than one group, more than one place?

PURPOSE

What kind of thinking does this routine encourage?
It is not unusual for people, systems, objects or ideas to be judged or given labels without others really knowing much about them. This routine encourages students to reserve judgment, take time to find out more about what they see and/or hear, and explore more deeply and broadly other people, and develop greater understanding of similarities and differences.

APPLICATION

When and where can I use it?
Identity can be a sensitive matter in some contexts. As a teacher, you make decisions all the time about what will serve your students and your context well. If in your judgment, this routine will not serve your context well, don't use it! Or maybe parts of it would serve well, or some adaptation. This does not mean we should never take up sensitive matters in our classes; arguably we should from time to time. But it's always a judgment call. Other questions that can foster further understanding of the ways an individual's identity is developed include: Does where you, or where your parents were born influence your identity? Does the place you live, your school, your friends shape your identity in certain ways? What about your religion and/or skin color? What do you think has shaped your identity?

LAUNCH

What are some tips for starting and using this routine?
The routine can be introduced and incorporated in one lesson, or in one or more steps over time. The process can be planned or shaped in response to discussions it elicits, and depending on the purpose and context in which it is being utilized. It can be broken into steps in whichever order is most relevant and time frame that is effective, sometimes involving revisiting steps over time.

- How can similarities, not only differences, be brought into discussions about identity? Students in the same school can wear different clothes, eat different foods and celebrate different festivals. Often their identities are shaped by their differences, and generalizations are often made that group them with others sharing those attributes. Invite students to find similarities they share. Invite them to look for similarities among students who learn differently to others, among those who come from different family structures, or those who make very different choices in how they spend time away from school, e.g. playing sport, going shopping, playing the saxophone, studying, meeting friends, painting, making things, spending time in hospital, staying alone or with family. Often the many similarities they discover they share are unexpected.

- What are some tips for the “identify” step? You could invite students to role play, introducing themselves to each other as if meeting for the first time, and ask each other questions that would help them get to know each other better. Discuss the multiplicity of identity. Who do people think you are? Can the same person be a sister, a daughter, a student, a swimmer, a friend? Where do you think you belong? Is a sense of belonging important?

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project zero THINKING ROUTINES

Word-Phrase-Sentence

A routine for capturing the essence of a text.

As an individual, review a text and then select a:

- **Word** that captured your attention or struck you as powerful
- **Phrase** that moved, engaged, or provoked you
- **Sentence** that was meaningful to you, that you felt captures the core idea of the text

As a group, discuss and record your choices. Begin by sharing your words, then phrases, then sentences. Explain why you made the selections you did. Looking at your group's collective choices of words, phrases, and sentences, reflect on the conversation by identifying:

- What themes emerge?
- What implications or predictions can be drawn?
- Were there aspects of the text not captured in your choices?

PURPOSE

What kind of thinking does this routine encourage?
This routine provides structure for a conversation. It can be used as both a discourse and as a thinking routine.

APPLICATION

When and where can I use it?
This routine helps learners engage with and make meaning from text with a particular focus on capturing the essence of the text or "what speaks to you." However, the power and promise of this routine lies in the discussion of why a particular word, a single phrase, and a sentence stood out for each individual in the group as a catalyst for discussions.

LAUNCH

What are some tips for starting and using this routine?

1. Set up. Give time to read selected texts in advance unless text is short and can be read on spot. Encourage active reading and highlighting. Not necessary to read with word-phrase-sentence protocol in mind.

2. Select a word-phrase-sentence. Ensure what is selected is meaningful to the reader. Note: some teachers like to change the order to sentence-phrase-word and this is okay.

3. Share selections. In groups of 4-6 people, ask learners to share and record their choices, explaining why they selected them. Sharing and discussion should occur in rounds, so the discussion is facilitated. First participant shares a word and explains why she chose it, inviting others to comment and discuss. The words are recorded and then the next person shares, records, and discusses until everyone has their turn. The group then moves to phrases and finally to sentences.

4. Invite reflection on the conversation. Each group looks at its documented responses. They identify common themes that emerge from these responses and then the implications and/or predictions they suggested. Finally the group identifies any aspects of the text that were not represented in their choice of sentences, phrases, and words.

5. Share the thinking. Post documentation from all the groups. Allow time to look at the sentences, phrases, and words chosen and the themes and implications drawn. Invite each group member to reflect briefly on his or her current understanding of the text and how using the routine contributed to his or her understand understanding of it.

It is not essential to complete all three steps to achieve the purpose, sentence, and word.

This thinking routine was adapted from the Text Rendering Experience from the School Reform Initiative by the Cultures of Thinking project at Project Zero, Harvard Graduate School of Education. Explore the full PZ Thinking Routine Toolbox at *pz.harvard.edu/thinking-routines*.

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project zero THINKING ROUTINES

The Story Routine: Main, Side, Hidden

After closely examining the source document or material, **respond to the prompts and explain your thinking**:

- What is the **main** or central story being depicted or documented?
- What is the **side** story (or stories) happening on the sidelines or around the edges? This might not necessarily involve the main characters.
- What is a **hidden** story—that is, a story that may be obscured, unspoken, or happening below the surface that we aren't readily aware of initially?

BACKGROUND

This thinking routine was originally developed for Densho, a Seattle-based organization dedicated to preserving, educating, and sharing the story of the World War II-era incarceration of Japanese Americans. Densho educators were looking for new ways to extend learners' understanding beyond the central story of the historical events and to get learners thinking about the deeper issues and implications of those events both for that time period and for today. This kind of thinking was particularly important because many of the documents preserved from that period are propaganda materials that were used to justify the internment to the American people and the world.

PURPOSE

What kind of thinking does this routine encourage?

This routine helps students analyze events and explore documents in more depth by constructing a set of narratives around those events. Learners begin with the main narrative to capture the core or central story being told. Then, by looking for side stories, students are encouraged to identify additional points of view that are present, but which are not the main focus. Looking beneath the surface of events to uncover the hidden story, learners are invited to explore the complexities of the narrative or the situation.

APPLICATION

When and where can I use it?

Source material should embody depth and some degree of complexity. It might be a primary source document from an historical archive, a complex work of art, a news event or photograph, events from a story or novel, a case study, a data set, or even a social problem or situation.

LAUNCH

What are some tips for starting and using this routine?

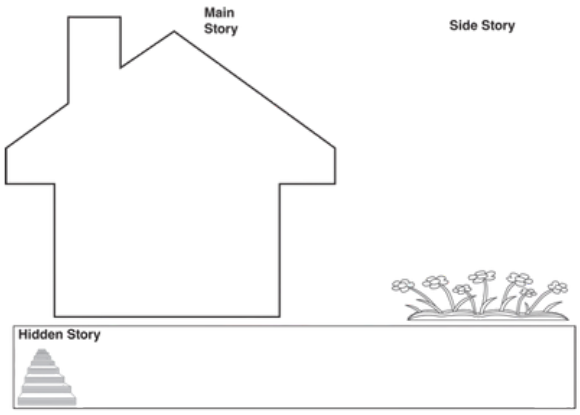
1. **Set up**. Introduce the source material and invite students to examine it carefully. This could be done individually or in pairs. This slow looking (Tishman 2017) shouldn't be rushed, and students should be explicitly encouraged to take time to notice things that aren't immediately apparent.

2. **Main.** Invite students (working individually or in pairs) to identify the main story that captures the central ideas that are most apparent.

3. **Side.** Prompt students to identify possible side stories. What else is going on here around the edges? Who are the other actors that might be contributing to the main story but are not the major players?

4. **Hidden.** Finally, ask students to consider the hidden or untold story. What is not immediately apparent but might nonetheless be important in understanding what is actually happening? What might be obscured or left out, either intentionally or unintentionally?

5. **Share Thinking.** Convene a whole class discussion. Often there is agreement regarding the main story, and so this doesn't typically need much discussion. A class will often generate a wide variety of side stories, and students' understanding of the situation can be enhanced by hearing these. As students share, ask them for elaboration, explanation, and evidence. Discuss the hidden stories. It is not uncommon for some students' side stories to be others' hidden stories, and this is fine. You might follow up by asking, “So what is that making you wonder about this situation?” or “Why do you think that story has been hidden?”



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